MA147 Mathematical Methods and Modelling 1

Ferran Brosa Planella

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Preface

These are the lecture notes for the first-year Maths module MA147 Mathematical Methods and Modelling 1 taught at the University of Warwick. These notes are available both as a PDF and a static website (which should be suitable for screen-reading devices), you can access both on Moodle. The source files are available on GitHub. I might update the notes as we go, fixing typos and improving explanations. You can keep track of those changes in the CHANGELOG. Further material is also available on Moodle.

These lectures notes, which aim to be self–contained, are based on the notes created in 2022 by Dr Björn Stinner, Dr Emma Davis and Dr Shreyas Mandre. For the academic year 2025-26, I have made significant changes to the content and structure of the notes, and I have also added new material on the topics of systems of differential equations.

For further reading, the following books are recommended:

- Robinson, James C. An Introduction to Ordinary Differential Equations. Cambridge University Press, 2004.
- Witelski, B. and Bowen, M., Methods of Mathematical Modelling: Continuous Systems and Differential Equations. Springer, 2015.
- Logan, David. A First Course in Differential Equations. Springer, 2015.
- Holmes, Mark H. Introduction to the Foundations of Applied Mathematics. Springer, 2019.

Aims and structure

The main aim of this module is to introduce the basic concepts of (ordinary) differential equations and difference equations and their solutions.

By the end of this module, you should be able to:

- Analyse, solve and interpret:
  - Differential equations (first order and second order).
  - Difference equations (first order and second order).
  - First order systems of differential equations and difference equations.
- Understand the modelling cycle and be able to formulate and analyse simple models.

- Use scaling, non-dimensionalisation and linear stability techniques to reveal and understand the main underlying dynamics/driving factors.

Throughout, we will use applied examples from a range of fields (often the sciences) to motivate and apply the methods covered. However, **no prior knowledge of science is required** to do well in this module or understand the content. In any instance where a fundamental principle, law or rule from science is used, it will be given to you and you will not be expected to memorise it or know its derivation or origin.

Module structure

These notes have six chapters:

- **Introduction & notation:** we will start by introducing the notation used throughout the notes and the module, as well as some general advice on how to use these notes.
- **Differential equations:** here we will initially learn how to classify differential equations and recap methods of solving first order equations covered in A-Level (or equivalent). We will see a more generalised method and look at some techniques for analysing the behaviour equations that we cannot necessarily solve. We will then extend our methods to second order differential equations, with a particular focus on the simplified linear case where we have constant coefficients, some of which may also be familiar from school.
- **Difference equations:** in this chapter we move away from things about continuous time and introduce difference equations, where our models are constructed from discrete sequences rather than continuous functions. Difference equations are easier to analyse computationally, so we will first observe how Euler's method can be used to discretise continuous equations. We will then cover topics from previous chapters applied to difference equations and a first introduction to the concept of 'chaos'.
- **Modelling:** in this chapter we will discuss the modelling cycle and how to formulate models and associated mathematical models using differential equations. This chapter contains very little examinable content and will be aligned with Week 6 of the term, which is breathing week.
- **Dimensional analysis:** in which we will introduce the concepts of dimensional analysis. We will see that dimensional analysis is a useful tool that can be used to help us construct, validate and simplify models, as well to create generalised problems that can be used to investigate wider ranges of scenarios.
- **Systems:** here we will consider systems of multiple equations, first for discrete difference equations and then for continuous differential equations. We will use matrices to solve these systems, as well as explore alternative methods of analysing and visualising solutions.

As these lecture notes have gone under a major editing process, they will doubtlessly contain errors and typos. There are also probably sections that could be better explained. If you spot anything or you have any suggestions, please do let me know via email.

Dr Ferran Brosa Planella, Summer 2026

1 Introduction & Notation

1.1 How to use these notes

These notes are comprehensive and cover some of the items we are going to discuss in the course with more mathematical rigour than you will be required to remember. This module is focused on learning and applying methods rather than memorisation of proofs or definitions and, as such, these notes have been structured to try and create some clarity about what concepts you are expected to **know** and what you are expected to **understand**, which are two very different things.

Any content, formulae or methods that you are expected to **know** and remember, will be surrounded with a box:

> **!** Important
>
> E.g. this content is assessed - remembering it will be beneficial for the exam.

Examples of this include:

- Definitions you may be asked to state in an exam, e.g. what is a forward problem.
- Content you may be expected to use, e.g. the core fundamental dimensions.
- Methods you may be expected to apply, e.g. nondimensionalisation, Euler's method.
- Formulae you may be expected to apply, e.g. solving first order differential equations.

All other content, unless otherwise specified, you are expected to **understand**. This is content that you do not need to memorise, but may have similarities or overlap with assessed content.

Examples of this include:

- Applied modelling examples: you would be given the model in the exam (or the information required to construct it), e.g. logistic population growth model.
- Examples applying methods from the course, e.g. solution to a specific first order differential equation.
- Derivations: understanding these will help you remember the results they demonstrate, e.g. derivation of stability of stationary points.

1.2 Notation

1.2.1 Variable naming conventions

Variables are kind of placeholders and used to express the quantities involved in our model. The letters used are not particularly important, for instance,

$$f(x) = x^2$$

means the same thing as

$$g(y) = y^2.$$

However, there are a few conventions:

- The variable t often represents time. Alternatives, used for changes of variable can include s, τ or modifications to t such as $\tilde{t}$.
- x, y, z are often used for spatial variables.
- Derivatives with respect to time are often denoted using dots:

$$\frac{\mathrm{d}}{\mathrm{d}t}z(t) = \dot{z}(t), \quad \frac{\mathrm{d}^2}{\mathrm{d}t^2}z(t) = \ddot{z}(t), \quad \ldots.$$

- Derivatives with respect to space are often denoted using dashes:

$$\frac{\mathrm{d}}{\mathrm{d}x}y(x) = y'(x), \quad \frac{\mathrm{d}^2}{\mathrm{d}x^2}y(x) = y''(x), \quad \ldots.$$

- The initial value of a variable is often described using a subscript 0, e.g. $x(0) = x_0$.

1.2.2 Functions

Mathematical functions will play a central role in this module. A function is a mapping from one set to another, e.g. $f : \mathbb{R} \to \mathbb{R}$ is a function that maps real numbers to real numbers. This means that the function f takes an input $x \in \mathbb{R}$ (remember that calling it x is an arbitrary choice) and produces an output $f(x) \in \mathbb{R}$. In some cases this might seem overkill to write, but it is always important to keep in mind the range of values the input and output of a function can take to avoid studying unnecessary cases.

We will also encounter functions that take two inputs (or produce two outputs). For example, imagine a function f that depends on both time t and space x and returns two values (e.g. the concentration of two chemicals u and v). Then we would write $f : [0, \infty) \times \mathbb{R} \to [0, \infty) \times [0, \infty)$, which means that f takes two inputs $t \in [0, \infty)$ and $x \in \mathbb{R}$ and returns two outputs $u \in [0, \infty)$ and $v \in [0, \infty)$. Here $\times$ is the Cartesian product between two sets, which is the set of all possible pairs of elements from the two sets. For simplicity, when we have the product between two instances of the same set, e.g. $\mathbb{R}$, we write $\mathbb{R}^2$ instead of $\mathbb{R} \times \mathbb{R}$.

1.2.3 Vectors and matrices

There is also specific notation to represent that we are talking about vectors and matrices. For vectors we often use **bold** font to imply something is a vector, e.g.

$$\boldsymbol{v} = (v_1, v_2).$$

However, when writing by hand it is easier to use underlined symbols, e.g.

$$\underline{v} = (v_1, v_2).$$

We will represent matrices using **capital letters**, with the component entries indicated by the lowercase equivalent with subscripts, e.g.

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

1.2.4 Mathematical operators

Knowledge of key mathematical operators will be assumed, these include:

- $\sum_a^b$ represents the discrete sum of items indexed from a to b (the symbol is a capital sigma).
- $\int_a^b$ represents the continuous integral from a to b.
- $\prod_a^b$ represents the the discrete product of items indexed from a to b (this is a capital pi).
- $a \in B$ represents that a belongs to or is contained in B, for example $x \in \mathbb{R}$ implies x is a real number.
- $\lim_{t \to T}$ is the limit of some value that depends on t as t tends towards some value T, e.g. $\lim_{t \to 0}$.
- $\approx$ means "approximately equal to" (usually due to approximations either numerical or analytical).
- $A \implies B$ means "if A is true then B is true" and NOT "A is true, therefore B is true".
- $A \iff B$ means "A is true **if and only if** B is true". Remember that this also works the other way around: "B is true **if and only if** A is true".
- $A := B$ means "A is defined to be B" or "A is equal to B by definition".

2 Differential Equations

Aims of the chapter

By the end of this chapter, you should be able to:

- Explain what an ordinary differential equation is and classify it.
- Solve simple first order differential equations.
- Analyse the behaviour of first order autonomous equations.
- Solve linear second order differential equations.

Because of their fundamental importance and success in many scientific fields, differential equations have become a field within the mathematical sciences and are an integral part of training in mathematics. One fundamental question is whether a differential equation has solutions and, if so, whether the solution is unique. We will see some counterexamples showing that this is not always the case but have to delay deeper results on this questions because they require sophisticated analytical tools that we don't have to our disposition yet.

If one can show that there is a solution, a natural question is whether we can find an explicit formula for it. We will study some techniques that enable us to do so for some simple differential equations. However, in general, this is not the case, and for this reason computational methods have been developed that approximate the solution. We may occasionally make use of the computer but the in-depth study of such methods is left to future modules.

Even if we cannot find an explicit solution formula, we still may be able to make some qualitative predictions about the behaviour of the solution. For time dependent problems this comprises the long-term behaviour, for instance, whether the solution grows, or remains bounded, or oscillates, or converges to an equilibrium state. We will also look a bit into so-called stability properties, i.e., how this qualitative behaviour changes if data or parameters in the differential equation change. Such questions will also be picked up in follow-up modules.

2.1 General concepts

General scalar differential equations can be written in the form

$$\mathcal{F}\left(x, y(x), y'(x), y''(x), \ldots, y^{(k-1)}, y^{(k)}\right) = 0 \tag{2.1}$$

for some given function $\mathcal{F} : (\alpha, \beta) \times \mathbb{R}^{k+1} \to \mathbb{R}$ with an interval $(\alpha, \beta) \subset \mathbb{R}$. We here use the notation

$$y^{(l)}(x) = \frac{d^l}{\mathrm{d}x^l} y(x)$$

for the l-th derivative of a function $y : (\alpha, \beta) \to \mathbb{R}$. In the general form (Equation 2.1) we make the implicit assumption that the highest derivative $y^{(k)}$ occurs in the equation (otherwise we could consider a function $\mathcal{F}$ that does not depend on this argument). We will focus on equations where we can solve the equation for $y^{(k)}$ and write it in the form

$$y^{(k)}(x) = \phi\left(x, y(x), \ldots, y^{(k-1)}(x)\right) \tag{2.2}$$

for some function $\phi : (\alpha, \beta) \times \mathbb{R}^k \to \mathbb{R}$.

Our equation is expressed in terms of the *variables* x and y. We distinguish:

- *Independent variables* can be varied independently of any other variables. In Equation 2.5, x is an independent variable.
- *Dependent variables* are unknown quantities that depend on independent variables. In Equation 2.5, y is a dependent variable, which depends on x.

The equation may also depend on parameters, which are constant quantities that are fixed for a given problem but can be varied between problems. For example, if we wrote an equation for a weight on a spring, the mass of the weight would be a parameter.

The variables are kind of placeholders and used to express the differential equation. We could have used other letters, for instance,

$$z^{(k)}(\theta) = \phi\left(\theta, z(\theta), \ldots, z^{(k-1)}(\theta)\right) \quad \text{and} \quad x^{(k)}(t) = \phi\left(t, x(t), \ldots, x^{(k-1)}(t)\right)$$

express the same differential equation Equation 2.2. Note that x stands for the dependent variable in the last reformulation. This concept of a variable is known from other areas such as the formulation of algorithms or integration: given an integrable function $g : (a, b) \to \mathbb{R}$, the expressions

$$\int_a^b g(x)\mathrm{d}x, \quad \int_a^b g(y)\mathrm{d}y, \quad \int_a^b g(\theta)\mathrm{d}\theta$$

all mean the same, x, y, or θ stand for a variable that has no effect on the integral.

In the literature, the variable t is often stands for time and x, y, z are spatial variables. Recall from Chapter 1 that we will sometimes use dots and dashes to denote derivatives.

! Important

Definition 2.1. For a differential equation of the form Equation 2.1 we define:

1. The *order* of a differential equation is the order of its highest derivative.
2. The differential equation is *autonomous* if $\mathcal{F}$ does not depend on the independent variable, so $\mathcal{F} : \mathbb{R}^{k+1} \to \mathbb{R}$, and the differential equation reads
$$\mathcal{F}\left(y(x), y'(x), \ldots, y^{(k)}\right) = 0.$$
3. The differential equation is *homogeneous* if setting the dependent variable and all its derivatives to zero yields zero, i.e.
$$s(x) := \mathcal{F}(x, 0, 0, \ldots, 0) = 0.$$
Otherwise, the differential equation is *inhomogeneous*.
4. The differential equation is *linear* if $\mathcal{F}$ can be written in the form
$$\mathcal{F}\left(x, y(x), y'(x), \ldots, y^{(k)}\right) = s(x) + \sum_{i=0}^{k} a_i(x) y^{(i)}(x) \tag{2.3}$$
with some functions $s, a_i : (\alpha, \beta) \to \mathbb{R}$, $i = 0, \ldots, k$. Note that, when $s(x) \equiv 0$ the linear equation is homogeneous.

A couple of remarks on the above formal definitions:

- Note that the functions s defined in the last two items coincide if $\mathcal{F}$ is linear (which also explains the same notation).
- If the differential equation is autonomous then s cannot depend on the independent variable x but can be a constant, so the equation still can be inhomogeneous.
- There are even more general differential equations, in particular for functions of several variables. For instance, the equation
$$\frac{\partial^2}{\partial t^2} u(x,t) - \frac{\partial^2}{\partial x^2} u(x,t) = 0$$
is a simple equation for the propagation of waves. Such equations that involve partial derivatives with respect to multiple (more than one) independent

variable are called *partial differential equations (PDE)*. In this module, we focus on differential equations for functions of one independent variable only. These are called *ordinary differential equations (ODE)*, even though in this module we may sometimes refer to them simply as differential equations for simplicity.

Let us briefly classify some of the differential equations.

Example 2.1.

1. Consider the equation

$$\dot{p}(t) = kp(t)\left(1 - \frac{p(t)}{p_m}\right),$$

which is used to describe the growth of a population $p(t)$ in time (more on this later).

We can write it in the form Equation 2.1 with

$$\mathcal{F}(t, p(t), \dot{p}(t)) = \dot{p}(t) - kp(t)\left(1 - \frac{p(t)}{p_m}\right).$$

1. Its order is one as only the first derivative occurs.
2. The independent variable is t, and the dependent variable p is the only function that depends on it. Therefore, the equation is autonomous.
3. It is homogeneous as setting $p(t)$ and $\dot{p}(t)$ to zero yields zero (i.e. there is no additive constant).
4. On the right-hand-side the quadratic term $p(t)^2$ occurs, so the equation is non-linear.

2. Now consider the equation for a mass-spring-damper system (again, more on this later):

$$m\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) = -c\frac{\mathrm{d}}{\mathrm{d}t}x(t) - kx(t).$$

Here m is the mass, c is the damping coefficient, and k is the spring constant; which are parameters.

Writing the equation in the form Equation 2.1 yields

$$\mathcal{F}(t, x(t), \dot{x}(t), \ddot{x}(t)) = m\ddot{x}(t) + c\dot{x}(t) + kx(t).$$

1. We see that the second derivative is the highest, so the equation is of order two.
2. The independent variable is t and does not explicitly appear in the equation. Therefore, the equation is autonomous.
3. We can write

$$\mathcal{F}(t, x(t), \dot{x}(t), \ddot{x}(t)) = s(t) + \sum_{i=0}^{2} a_i(t) \frac{d^i}{\mathrm{d}t^i} x(t)$$

with $a_2 = m$, $a_1 = c$, $a_0 = k$, and $s(t) = 0$. The equation therefore is linear.

4. As $s(t) = \mathcal{F}(t, 0, 0, 0) = 0$ (which we also knew from the previous point) the equation is homogeneous.

> **!** Important
>
> **Definition 2.2.** A function $y : (\alpha, \beta) \to \mathbb{R}$ is called a solution to the differential equation Equation 2.1 if it satisfies the equation.
>
> - The solution is called an *explicit solution*, if the dependent variable is given in terms of the independent variable as a combination of algebraic expressions or elementary functions.
> - An *implicit form* of the solution is an equation that relates the dependent and independent variables and involves no derivatives.

Example 2.2. The function $y(x) = 12e^{-2x}$ is an explicit solution to the differential equation $\frac{\mathrm{d}}{\mathrm{d}x} y(x) = -2y(x)$.
The expression

$$\log(y) - y + 4\log(x) - 2x + 4 = 0 \tag{2.4}$$

relates y and x but cannot be algebraically resolved for y as a function of x. However, assuming that $y = y(x) \neq 0$ is a differentiable function of x, then differentiating the expression Equation 2.4 with respect to x yields that

$$\left(\frac{1}{y(x)} - 1\right) \frac{\mathrm{d}}{\mathrm{d}x} y(x) = 2 - 4\frac{1}{x} \quad \Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}x} y(x) = \frac{y(x)}{1 - y(x)} \left(2 - 4\frac{1}{x}\right),$$

and Equation 2.4 is an implicit solution to this differential equation.

Remark 2.1. This distinction of a variable to formulate a differential equation from a solution might look a bit nit-picky but is similar to the careful distinction between a function, say g, and the expression $g(y)$, which is the value of g in a point y. In practice (and in the following), however, this distinction often is not made. In a slight abuse of notation, $y(x)$ sometimes is just a dependent variable used to formulate a differential equation such as Equation 2.1, and sometimes it denotes a solution, depending on the context.

In the remainder of this chapter we study differential equations for scalar functions that involve first derivatives. The differential equations are of the form

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = f\left(x(t), t\right), \quad t \in (\alpha, \beta), \tag{2.5}$$

where $f : \mathbb{R} \times (\alpha, \beta) \to \mathbb{R}$ with an interval $(\alpha, \beta) \subset \mathbb{R}$ is a given function.

2.2 Trivial differential equations

One can regard solving a differential equation of the form Equation 2.5 as the opposite to differentiating a given function. In differentiation calculus, we are given the function and want to find the derivative, or slope of the graph of the function. In differential equations, we know these slopes and want to find (graphs of) functions that fit them. In the literature, solving differential equations is therefore often called *integrating differential equations*.

This connection is easiest seen when considering the case that the given function f in Equation 2.5 is independent of $x(t)$ so that the differential equation reads

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = f(t), \quad t \in (\alpha, \beta). \tag{2.6}$$

This case is called *trivial*.

Theorem 2.1 (Fundamental Theorem of Calculus)**.** *Suppose $g : [a, b] \to \mathbb{R}$ is continuous and let*

$$G(x) := \int_a^x g(\tilde{x})\mathrm{d}\tilde{x}, \quad x \in [a, b].$$

Then G is an anti-derivative *of g, i.e., it satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}x}G(x) = g(x), \quad x \in (a, b).$$

Moreover, if $\tilde{G}$ is any other anti-derivative *of g then*

$$\int_a^b g(\tilde{x})\mathrm{d}\tilde{x} = \tilde{G}(b) - \tilde{G}(a),$$

and the two anti-derivatives of g differ by a constant only, so for some $c \in \mathbb{R}$

$$\tilde{G}(x) = G(x) + c.$$

> **Note**
>
> You will not need to reproduce the above theorem word-for-word *in this module*, but it will help to understand the underlying concepts and it will crop up again and again across your degree.

We see that solving Equation 2.6 amounts to finding any anti-derivative of f. Denoting it by F, the theorem says that then $F + c$ for any $c \in \mathbb{R}$ is an anti-derivative and, thus, a solution.

> **Important**
>
> **Definition 2.3** (General solution, particular solution)**.** The *general solution* to the trivial differential equation Equation 2.6 is given by
>
> $$x(t) = F(t) + c, \quad t \in (\alpha, \beta)$$
>
> where F is an anti-derivative of f and $c \in \mathbb{R}$ is any number.
> For a specific number c, $x(t) = F(t) + c$ is a *particular solution.*

So the general solution in fact is a whole family of solutions parametrised by c whilst the particular solution is a unique function. To single out a solution (or to determine a number c), an additional condition is required. As quite often the independent variable is time, this additional condition is called an *initial condition.*

> **Important**
>
> **Definition 2.4** (Initial value problem)**.** The *initial value problem* consists of finding a function $x : (\alpha, \beta) \to \mathbb{R}$ such that Equation 2.5 and the *initial condition*
>
> $$x(t_0) = x_0 \tag{2.7}$$
>
> for some $t_0 \in [\alpha, \beta]$ and some given number $x_0 \in \mathbb{R}$ are satisfied.

For the differential equation Equation 2.6 with such an initial condition Equation 2.7 the anti-derivative

$$x(t) = x_0 + \int_{t_0}^{t} f(\tilde{t})\mathrm{d}\tilde{t} \tag{2.8}$$

is the solution: indeed, as x_0 does not depend on t, it satisfies the differential equation because

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{t_0}^{t} f(\tilde{t})\mathrm{d}\tilde{t}\right) = f(t)$$

by the *Fundamental Theorem of Calculus*, and it also satisfies the initial condition because

$$x(t_0) = x_0 + \underbrace{\int_{t_0}^{t_0} f(\tilde{t})\mathrm{d}\tilde{t}}_{=0} = x_0.$$

Note that there is not an unknown constant c any more in the solution, this is a particular solution.

Example 2.3 (Spacecraft landing)**.** Consider a spacecraft landing on a planet (or moon). We want to write a simple equation for the motion of the spacecraft under the influence of gravity and thrust. For simplicity, we will neglect the motion of the planet, so we only consider the spacecraft's height $h(t)$ and its vertical velocity $v(t) = \dot{h}(t)$, both depending on time t. We assume that the spacecraft is propelled by rockets, their thrust is another force acting on it. From Newton's second law (force equals mass times acceleration) we can write

$$M\dot{v}(t) = -gM + u(t) \quad \Leftrightarrow \quad \dot{v}(t) = -g + \tfrac{1}{M}u(t)$$

where $M > 0$ is the mass (assumed constant here), $g > 0$ is the gravitation constant, and $u(t)$ is a given function describing the thrust by the spacecraft's rockets. We here assume for simplicity that $u(t) = \bar{u} \in \mathbb{R}$ is constant.

Assume now that, initially at time $t = t_0 = 0$, the velocity is known and given by $v_0 < 0$. We want to find out when the spacecraft comes to rest. In mathematical terms, we want to find a time $t^* > 0$ such that the solution to the initial value problem satisfies $v(t^*) = 0$.

Using the above result Equation 2.8, the solution is given by

$$v(t) = v_0 + \int_{t_0}^{t} \left(-g + \tfrac{1}{M}u(\tilde{t})\right) \mathrm{d}\tilde{t} = v_0 + \int_0^t \left(-g + \tfrac{1}{M}\bar{u}\right) \mathrm{d}\tilde{t} = v_0 + \left(-g + \tfrac{1}{M}\bar{u}\right) t.$$

The spacecraft comes to rest if $0 = v(t^*) = v_0 + (-g + \frac{1}{M}\bar{u})t^*$, which is the case if and only if

$$t^* = \frac{v_0}{g - \frac{1}{M}\bar{u}}.$$

We want that t^* is positive. As $v_0 < 0$ this requires that $g - \frac{1}{M}\bar{u} < 0$, or the thrust $\bar{u} > Mg$ has to be big enough. Poor spacecraft otherwise...

We may ask at which height the spacecraft comes to rest (assuming it does so). To formulate this questions in mathematical terms, let us assume that the initial height at time $t = 0$ is $h(0) = h_0 > 0$, and we are interested in the value of $h(t^*)$ where h satisfies the differential equation $\dot{h}(t) = v(t)$. As we know v already this is again an initial value problem with a trivial differential equation. Its solution is given by

$$h(t) = h_0 + \int_{t_0}^{t} v(\tilde{t})\mathrm{d}\tilde{t} = h_0 + \int_0^t \left(v_0 + \left(-g + \tfrac{1}{M}\bar{u}\right)\tilde{t}\right) \mathrm{d}\tilde{t} = h_0 + v_0 t + \frac{1}{2}\left(-g + \tfrac{1}{M}\bar{u}\right) t^2.$$

At time $t^* = \frac{v_0}{g - \frac{1}{M}\bar{u}}$ the height then is

$$h(t^*) = h_0 + \frac{v_0^2}{g - \frac{1}{M}\bar{u}} + \frac{1}{2}\frac{(-g + \frac{1}{M}\bar{u})v_0^2}{(g - \frac{1}{M}\bar{u})^2} = h_0 + \frac{1}{2}\frac{v_0^2}{g - \frac{1}{M}\bar{u}}.$$

We may also want that the spacecraft touches down at this time, i.e., $h(t^*) = 0$. We see that this is the case if and only if

$$2h_0 = \frac{v_0^2}{\frac{1}{M}\bar{u} - g} \quad \Leftrightarrow \quad \bar{u} = M\left(g + \frac{v_0^2}{2h_0}\right).$$

This is the thrust required for a smooth landing.

2.3 Well-posedness

A natural question is as to whether any differential equation of the form Equation 2.5 has a unique solution. We have seen already in the case of a trivial differential equation that we need to impose an additional condition to single out a particular solution. However, this is not sufficient. Consider the following examples:

Example 2.4 (Non-existence)**.** Consider the initial value problem

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = -\frac{x(t)^2}{t^2}, \quad x(0) = x_0 \in \mathbb{R}.$$

The right-hand-side function $f(x,t) = -x^2/t^2$ is not defined at $t = 0$. Intuitively, imposing a condition at time $t = 0$ seems problematic. Indeed, if we rewrite the differential equation in the form $x^2 + t^2 \frac{\mathrm{d}}{\mathrm{d}t}x = 0$ then evaluating at $t = 0$ yields that $x(0)^2 = 0$. The condition $x(0) = x_0 \in \mathbb{R}$ therefore cannot be satisfied if $x_0 \neq 0$, and the initial value problem has no solution then.

Example 2.5 (Non-uniqueness)**.** Consider the initial value problem

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = \sqrt{x(t)}, \quad x(0) = 0.$$

The function $x(t) = 0$ for all t obviously is a solution. However, it is not the only one. For any $c > 0$ the function

$$x_c(t) = \begin{cases} 0 & \text{if } t \leq c, \\ \frac{(t-c)^2}{4} & \text{otherwise,} \end{cases}$$

also is a solution. Indeed, if $t > c$ then

$$x_c'(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{(t-c)^2}{4}\right) = \frac{t-c}{2} = \sqrt{\frac{(t-c)^2}{4}} = \sqrt{x_c(t)},$$

and if $t \leq c$ then

$$x_c'(t) = 0 = \sqrt{x_c(t)},$$

where we note that in the point $t = c$ the limits of the derivatives from both sides exist and coincide so that the derivative exists, too.

Whilst these results are discouraging, one actually can ensure that there is a unique solution to Equation 2.5 with Equation 2.7 if the function f is sufficiently smooth, or "nice" enough. We will not specify here what this means as we are lacking the tools to prove these results as this stage. This problem is picked up and addressed in detail in later modules.

Remark 2.2. For the remainder of this chapter,

$$f \text{ is assumed to be "nice" unless stated otherwise.}$$

As a final remark, a problem is *well-posed* if it has a unique solution that "nicely" depends on the *data*. More precisely

$$\text{well-posedness} \;=\; \text{existence} \;+\; \text{uniqueness} \;+\; \text{stability}.$$

With *data* we mean parameters in f such as the spring constant in the mass-spring-damper problem earlier, but also initial values such as x_0 in Equation 2.7, and *stability* means that solutions for different data are not too far away from each other if the data points are close. Investigating stability requires the ability to measure distances between functions (the solutions), which requires concepts that will be introduced in later modules in great depth.

2.4 Solution techniques for linear equations

Let us now start to explicitly solve some non-trivial cases where the function f in Equation 2.5 depends on x. The simplest case is if f is linear in x,

$$f(x,t) = r(t)x + s(t),$$

with some functions $r, s : (\alpha, \beta) \to \mathbb{R}$. This is consistent with the definition of linearity in Equation 2.3 with $a_1 = r(t)$ and $a_0 = s(t)$, noting that the independent variable is denoted by t and the dependent variable by x. As per Remark 2.2, we assume that r and s are "nice" functions, therefore f is also "nice".

2.4.1 Homogeneous equations

We first consider the case $s \equiv 0$, which yields a homogeneous differential equation. The initial value problem then reads

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = r(t)x(t), \quad t \in (\alpha, \beta), \qquad x(t_0) = x_0. \tag{2.9}$$

If $r(t) = \bar{r}$ is constant then this means we are searching for a function such that its derivative is proportional to itself. We know of such functions: the exponential functions satisfy this property. Therefore we use an informed guess (also called an *ansatz*) to determine the solution.[1] Using the ansatz $x(t) = e^{\lambda t}$ we obtain

$$0 = \frac{\mathrm{d}}{\mathrm{d}t}x(t) - \bar{r}x(t) = \lambda e^{\lambda t} - \bar{r}e^{\lambda t},$$

and we see that $x(t) = e^{\bar{r}t}$ is a solution. Actually, any multiple of the ansatz,

$$x_c(t) = ce^{\bar{r}t}, \quad c \in \mathbb{R},$$

is also a solution, so we have found a general solution again. The initial condition singles out a particular solution by fixing c. The function

$$x(t) = x_0 e^{\bar{r}(t-t_0)} \tag{2.10}$$

satisfies $x(t_0) = x_0$ and thus is the solution to the initial value problem.

Consider now the case that r depends on the independent variable t. Let us try a similar ansatz,

$$x(t) = ce^{R(t)}$$

with $c \in \mathbb{R}$ and with a suitable function $R : (\alpha, \beta) \to \mathbb{R}$. Substituting this approach in the differential equation we see that R has to be such that

$$\dot{R}(t)ce^{R(t)} = \frac{\mathrm{d}}{\mathrm{d}t}x(t) = r(t)x(t) = r(t)ce^{R(t)}.$$

We see that R then has to satisfy $\dot{R} = r$, so R has to be an anti-derivative of r. If we choose

[1] This might seem a bit handwavy, but it is actually a valid (and common) mathematical technique.

$$R(t) = \int_{t_0}^{t} r(\tilde{t})\mathrm{d}\tilde{t}$$

then $R(t_0) = 0$, and the initial condition implies that

$$x_0 = x(t_0) = ce^{R(t_0)} = c.$$

! Important

Therefore, the solution of Equation 2.9 is given by

$$x(t) = x_0 e^{R(t)} \quad \text{with } R(t) = \int_{t_0}^{t} r(\tilde{t})\mathrm{d}\tilde{t}. \tag{2.11}$$

Example 2.6. Let us use the above ideas to solve the initial value problem

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) + 2tx(t) = 0, \quad x(10) = 3.$$

So here $t_0 = 10$ and $x_0 = 3$. Note that $r(t) = -2t$. According to Equation 2.11 we need to compute

$$R(t) = \int_{t_0}^{t} r(\tilde{t})\mathrm{d}\tilde{t} = \int_{10}^{t} -2\tilde{t}\mathrm{d}\tilde{t} = -\tilde{t}^2\Big|_{10}^{t} = -t^2 + 100.$$

Then the solution is given by

$$x(t) = x_0 e^{R(t)} = 3e^{100-t^2}.$$

Example 2.7 (Mixing problem)**.** Consider a tank containing 1000 litres of a mixture of water and chlorine. To reduce the amount of chlorine, fresh water is pumped into the tank at a rate of 6 l/s. The mixture is well-stirred and pumped out at a rate of 8 l/s. The initial concentration of chlorine is 0.02 g/l. We want to find i) the amount of chlorine in the tank as a function of time and ii) the interval of validity of the equation.

Let us start with the second question. We denote the volume of the fluid by $v(t)$. The equation governing the volume should state the variation of the volume in time is how much water comes in minus how much mixture comes out:

$$\frac{\mathrm{d}}{\mathrm{d}t}v(t) = 6 - 8 = -2.$$

This is a trivial differential equation, and with the initial condition $v(0) = 1000$ we obtain $v(t) = 1000 - 2t$. So when t reaches 500 then the tank is empty (and there is no chlorine left) and our equation is no longer valid.[2] Hence, the interval of validity is $t \in [0, 500]$.

We now denote the amount of chlorine by $c(t)$ in the interesting time interval $t \in [0, 500]$. The concentration is $q(t)/v(t)$, where $q(t)$ is the total amount of chlorine (not mixture!) in the tank at time t. The flow of chlorine out of the tank is given by the rate $\alpha = 8$ l/s times this concentration, and this is the only way how the amount of chlorine changes. Hence, we obtain the differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}c(t) = -\alpha\frac{c(t)}{v(t)} = -8\frac{c(t)}{1000 - 2t} = -\frac{4}{500 - t}c(t).$$

It comes with an initial value: $Q(0) = 0.02 \times 1000 = 20$. The problem thus is of the form Equation 2.9. With $r(t) = -4/(500 - t)$ we compute

$$R(t) = \int_0^t r(\tilde{t})\mathrm{d}\tilde{t} = 4\log(500 - t) - 4\log(500).$$

We obtain the solution

$$c(t) = c(0)e^{R(t)} = 20e^{-4\log(500-t)+4\log(500)} = 20\left(\frac{500 - t}{500}\right)^4.$$

2.4.2 Inhomogeneous equations and integrating factor

Consider the case where $s \neq 0$. The differential equation then reads

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = r(t)x(t) + s(t), \quad t \in (\alpha, \beta). \tag{2.12}$$

The exponential function from the previous section will prove quite useful in this case.

> **!** Important
>
> **Definition 2.5** (Integrating factor)**.** The function
>
> $$I(t) = e^{-R(t)}. \tag{2.13}$$
>
> where $R(t) = \int r(\tilde{t})\mathrm{d}\tilde{t}$ is an anti-derivative of r, is called an *integrating factor*. If an initial condition of the form $x(t_0) = x_0$ (as in Equation 2.7) is imposed then a

[2] Note that the equation and the solution would still hold from a mathematical point of view (i.e. there is nothing preventing $v(t)$ from being negative). However, the equation describes a real-world scenario, so a negative volume would be meaningless. This is a crucial reasoning, that will pop up repeatedly.

specific anti-derivative is chosen, namely

$$R(t) = \int_{t_0}^{t} r(\tilde{t})\mathrm{d}\tilde{t}.$$

Note that

$$\frac{\mathrm{d}}{\mathrm{d}t}I(t) = \frac{\mathrm{d}}{\mathrm{d}t}e^{-R(t)} = \left(-\frac{\mathrm{d}}{\mathrm{d}t}R(t)\right)e^{-R(t)} = -r(t)I(t). \tag{2.14}$$

Let us multiply the Equation 2.12 with $I(t)$ so that

$$I(t)\frac{\mathrm{d}}{\mathrm{d}t}x(t) - I(t)r(t)x(t) = I(t)s(t).$$

Then, using Equation 2.14, we can rewrite the left-hand side as

$$I(t)\frac{\mathrm{d}}{\mathrm{d}t}x(t) - I(t)r(t)x(t) = I(t)\frac{\mathrm{d}}{\mathrm{d}t}x(t) + x(t)\frac{\mathrm{d}}{\mathrm{d}t}I(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(I(t)x(t)\right)$$

and we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(I(t)x(t)\right) = I(t)s(t). \tag{2.15}$$

This is a trivial differential equation for $I(t)x(t)$, and such a transformation is exactly behind the idea of the integrating factor[3].

The initial condition for Equation 2.15 reads $I(t_0)x(t_0) = I(t_0)x_0$. Recalling the solution formula for such problems from Equation 2.8 (or just integrating with respect to t), we obtain that

$$I(t)x(t) = I(t_0)x_0 + \int_{t_0}^{t} I(\tilde{t})s(\tilde{t})\mathrm{d}\tilde{t}.$$

Substituting the integrating factor Equation 2.13 we end up with

$$x(t) = \frac{I(t_0)}{I(t)}x_0 + \int_{t_0}^{t}\frac{I(\tilde{t})}{I(t)}s(\tilde{t})\mathrm{d}\tilde{t} = e^{R(t)}x_0 + \int_{t_0}^{t} e^{R(t)-R(\tilde{t})}s(\tilde{t})\mathrm{d}\tilde{t}.$$

[3] More generally, so-called *exact equations* are obtained after transformation, but which are beyond the scope of this module. In you are interested, see Chapter 10 in Robinson's book for more detail.

! Important

Lemma 2.1. *The solution to the initial value problem Equation 2.12 and Equation 2.7 is given by*

$$x(t) = e^{R(t)}x_0 + \int_{t_0}^{t} e^{R(t)-R(\tilde{t})}s(\tilde{t})\mathrm{d}\tilde{t} \quad \text{with } R(t) = \int_{t_0}^{t} r(\tilde{t})\mathrm{d}\tilde{t}. \tag{2.16}$$

Algorithm 2.1. *An algorithm (or "recipe") to solve Equation 2.12, Equation 2.7 with an integrating factor is as follows.*

1. *Find an anti-derivative of r,*

$$R(t) = \int_{t_0}^{t} r(\tilde{t})\mathrm{d}\tilde{t}.$$

2. *Work out the integral*

$$x_s(t) = \int_{t_0}^{t} e^{R(t)-R(\tilde{t})}s(\tilde{t})\mathrm{d}\tilde{t}.$$

3. *The solution is then given by Equation 2.16,*

$$x(t) = e^{R(t)}x_0 + x_s(t).$$

Remark 2.3. This recipe is aligned with the solution formula we derived for Equation 2.16. There are other recipes to produce the solution with an integrating factor, some of which you may be familiar with from high school. In general, in this module alternative methods will be accepted in the assessed components, as long as they are i) mathematically correct and ii) clearly explained. You are still strongly encouraged to use the methods explained in this module.

The significant advantage of solution formulas such as Equation 2.16 (as opposed to some computational approaches) is of analytical nature. The formula tells us, how variable and parameter enter the solution and influence it. We can therefore analyse, how small perturbations such a due to measurement uncertainty can affect the solution. In other words, we can much more easily investigate the properties of the problem.

We will not much further investigate this topic here. However, note that small errors in r influence R and then, due to the e^R terms, are exponentially amplified. In turn, errors in s or x_0 influence the solution linearly only.

Example 2.8. You have just poured yourself a cup of coffee and want to drink it, but it is too hot. You need to let it cool down for a while and, while you wait, you decide to do some maths (as one does). You want to write an equation that describes the evolution of the temperature of the coffee over time, i.e. $T(t)$.[4]

Let us consider on how we can describe the cooling process. We assume that the ambient temperature is constant and denote it by $\bar{T}$. To describe the cooling process, we use the so-called Newton's law of cooling, which states that the rate at which the temperature of the coffee changes is proportional to the difference between the coffee and the ambient temperatures. In mathematical terms, we can write

$$\frac{\mathrm{d}}{\mathrm{d}t}T(t) = -k\left(T(t) - \bar{T}\right),$$

where $k > 0$ is a constant that determines how fast the coffee cools down (it would depend on the material of the mug, the amount of coffee, etc.). The negative sign indicates that if the coffee is hotter than the ambient temperature, it will cool down, and if it is cooler, it will warm up. Let us assume that when you poured the coffee at $t = 0$ the temperature was $T(0) = T_0$.

We need to solve the initial value problem, which is a linear inhomogeneous differential equation of the form Equation 2.12 with $r(t) = -k$ and $s(t) = k\bar{T}$ constant. We can use the recipe from alg. 2.1 to solve it. First, we compute

$$R(t) = \int_{t_0}^{t} r(\tilde{t})\mathrm{d}\tilde{t} = \int_{t_0}^{t} (-k)\mathrm{d}\tilde{t} = -k(t - t_0).$$

Then

$$\begin{aligned} x_s(t) &= \int_{t_0}^{t} e^{R(t)-R(\tilde{t})} s(\tilde{t})\mathrm{d}\tilde{t} \\ &= \int_{t_0}^{t} e^{-k(t-t_0)+k(\tilde{t}-t_0)} k\bar{T}\mathrm{d}\tilde{t} \\ &= \bar{T}\int_{t_0}^{t} k e^{k(\tilde{t}-t)}\mathrm{d}\tilde{t} \\ &= \bar{T}\left(e^{k(\tilde{t}-t)}\right)\Big|_{\tilde{t}=t_0}^{\tilde{t}=t} \\ &= \bar{T}\left(1 - e^{k(t_0-t)}\right). \end{aligned}$$

Finally, with $T_0 = T_0$,

[4] In a grimmer application, a similar approach is used in forensic science to determine the time of death based on the temperature of the body.

$$T(t) = e^{R(t)}x_0 + x_s(t) = e^{-kt}T(t_0) + \bar{T}\left(1 - e^{-kt}\right) = \left(T_0 - \bar{T}\right)e^{-kt} + \bar{T}.$$

By quickly inspecting the solution we see that, as one could imagine, the coffee will start at T_0 and will approach the ambient temperature $\bar{T}$ as $t \to \infty$.

Let us now calculate how long I need to wait until the coffee has cooled down to a temperature T^* so that I can drink it. We need to find a time t^* such that $T(t^*) = T^*$. Using the solution formula we have

$$T^* = \left(T_0 - \bar{T}\right)e^{-kt^*} + \bar{T},$$

so

$$t^* = \frac{1}{k}\log\left(\frac{T_0 - \bar{T}}{T^* - \bar{T}}\right).$$

Given that $T^* < T_0$ and $T^* > \bar{T}$, we see that $t^* > 0$ as expected. Now let us put some numbers in. Assume that the ambient temperature is $\bar{T} = 22°C$, the initial temperature of the coffee is $T_0 = 90°C$, and that I want to drink it at $T^* = 50°C$. If we assume that the cooling constant is $k = 5 \cdot 10^{-3}$ s^{-1}, then we have

$$t^* \approx 177 \text{ s} \approx 3 \text{ min}.$$

2.5 Further solution techniques

2.5.1 Separable equations

Let us now investigate the case where the function f in Equation 2.5 is a product such that the dependent and independent variables are separated:

$$f(x,t) = \xi(x)\tau(t)$$

with some functions $\xi : \mathbb{R} \to \mathbb{R}$ and $\tau : (\alpha, \beta) \to \mathbb{R}$. We will assume these to be "nice" so that the initial value problem

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = \xi(x(t))\tau(t), \quad t \in (\alpha, \beta), \qquad x(t_0) = x_0, \tag{2.17}$$

has a unique solution.

> **!** Important
>
> **Lemma 2.2.** *If x_0 is a zero of ξ then the constant function $x(t) = x_0$, $t \in (\alpha, \beta)$, is the solution to Equation 2.17}.*
> *If $\xi(x_0) \neq 0$ then the solution x to Equation 2.17 is such that $\xi(x(t)) \neq 0$ for all $t \in (\alpha, \beta)$.*

Proof. If x_0 is a zero of ξ then the constant function $x(t) = x_0$ satisfies both $\frac{\mathrm{d}}{\mathrm{d}t}x(t) = 0$ and $\xi(x(t))\tau(t) = \xi(x_0)\tau(t) = 0$ for all t, and thus the initial value problem Equation 2.17.

Consider now the case $\xi(x_0) \neq 0$ and let x denote the corresponding solution to Equation 2.17. Assume that there is a time $t_1 \in (\alpha, \beta)$ such that $\xi(x(t_1)) = 0$. Let y now denote the solution to Equation 2.17 but with the initial condition $y(t_1) = x(t_1)$ at the point t_1. Recall that the initial point in Equation 2.17 is arbitrary, and that the solution is unique. As x satisfies the differential equation and the initial condition we conclude that $y = x$. However, as $\xi(y(t_1)) = \xi(x(t_1)) = 0$, our previous result yields that the unique solution is also given by the constant function $y(t) = x(t_1)$, $t \in (\alpha, \beta)$. But then $\xi(y(t_0)) = 0 \neq \xi(x(t_0))$, which contradicts $y = x$. Therefore, $\xi(x(t)) \neq 0$ at all times $t \in (\alpha, \beta)$. □

Let us now study the second case in more detail. If $\xi(x(t)) \neq 0$ for all t we can divide the differential equation by $\xi(x(t))$ and obtain

$$\frac{1}{\xi(x(t))}\frac{\mathrm{d}}{\mathrm{d}t}x(t) = \tau(t).$$

Suppose now that $H(x)$ is an anti-derivative of $1/\xi(x)$, so that $H'(x) = \frac{1}{\xi(x)}$. By the chain rule

$$\frac{\mathrm{d}}{\mathrm{d}t}H(x(t)) = H'(x(t))\frac{\mathrm{d}}{\mathrm{d}t}x(t) = \frac{1}{\xi(x(t))}\frac{\mathrm{d}}{\mathrm{d}t}x(t).$$

Substituting this we get that

$$\frac{\mathrm{d}}{\mathrm{d}t}H(x(t)) = \tau(t).$$

We see that we have transformed the differential equation into a trivial differential equation for $H(x)$. Integrating with respect to t and using the initial condition yields that

$$H(x(t)) - H(x_0) = \int_{t_0}^{t} \tau(\tilde{t})\mathrm{d}\tilde{t}. \tag{2.18}$$

It remains to resolve this equation for $x(t)$.

> **!** Important
>
> **Algorithm 2.2.** *A popular* recipe *to solve Equation 2.17 consists of "multiplying the differential equation with* $\mathrm{d}t$*" so that*
>
> $$\mathrm{d}x = \xi(x)\tau(t)\mathrm{d}t,$$
>
> *then to divide by* ξ*,*
>
> $$\frac{1}{\xi(x)}\mathrm{d}x = \tau(t)\mathrm{d}t,$$
>
> *and finally to integrate,*
>
> $$\int_{x_0}^{x(t)} \frac{1}{\xi(\tilde{x})}\mathrm{d}\tilde{x} = \int_{t_0}^{t} \tau(\tilde{t})\mathrm{d}\tilde{t},$$
>
> *which is Equation 2.18 as* $H' = 1/\xi$*.*

Example 2.9. Let us use separation of variables to solve the initial value problem

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = 3t^2 e^{-x(t)}, \quad x(0) = 1.$$

Here $\xi(x) = e^x$ and $\tau(t) = 3t^2$. Using alg. 2.2 first obtain

$$e^x \mathrm{d}x = 3t^2 \mathrm{d}t$$

and then, with $t_0 = 0$ and $x_0 = 1$,

$$\int_1^{x(t)} e^{\tilde{x}}\mathrm{d}\tilde{x} = \int_0^t 3\tilde{t}^2\mathrm{d}\tilde{t}.$$

Therefore

$$e^{x(t)} - e^1 = t^3 \quad \Rightarrow \quad x(t) = \log\left(t^3 + e\right).$$

2.5.2 Substitution

Some differential equations that don't appear to satisfy any of the solution techniques or general methods previously described can still be evaluated using a transformation of variables. The below cases are not exhaustive, but serve to illustrate how substitution can be used to transform a differential equation into a form that we know how to solve.

2.5.2.1 Case 1: linear combinatons of y and x

! Important

Let us consider first order differential equations of the form

$$\frac{\mathrm{d}}{\mathrm{d}x}y(x) = f(ax+by) \tag{2.19}$$

for some constants $a, b \in \mathbb{R}$ and initial condition $y(x_0) = y_0$. Then an obvious choice of substitution would be $z = ax + by$, with $z(x_0) = ax_0 + by_0$. From this we see

$$\frac{\mathrm{d}}{\mathrm{d}x}z(x) = a + b\frac{\mathrm{d}}{\mathrm{d}x}y(x) = a + bf(ax+by)\,,$$

giving the differential equation

$$\frac{\mathrm{d}}{\mathrm{d}x}z(x) = a + bf(z)\,.$$

As the right hand side depends solely on z, we can then use separation of variables to get

$$\int_{x_0}^{x} \mathrm{d}\tilde{x} = \int_{z_0}^{z(x)} \frac{1}{a+bf(\tilde{z})}\mathrm{d}\tilde{z}\,,$$

then, by evaluating this integral (if possible) and substituting $z = ax + by$ back in to rearrange for y in terms of x, we therefore arrive at a solution to (Equation 2.19).

Example 2.10. Let us use substitution to solve the initial value problem

$$\frac{\mathrm{d}}{\mathrm{d}x}y(x) = 2(x+y)^2 \quad y(0) = 0.$$

So we set $z = x + y$, which gives us $z_0 = z(x_0) = x_0 + y_0 = 0$ and

$$\frac{\mathrm{d}}{\mathrm{d}x}z(x) = 1 + \frac{\mathrm{d}}{\mathrm{d}x}y(x) = 1 + 2z^2\,.$$

Separating variables and integrating (using the fact that $\int \frac{1}{1+z^2} = \tan^{-1}(z)$) gives us

$$\begin{aligned}
\int_0^x \mathrm{d}\tilde{x} = \int_0^{z(x)} \frac{1}{1+2\tilde{z}^2}\mathrm{d}\tilde{z} &= \int_0^{z(x)} \frac{1}{1+(\sqrt{2}\tilde{z})^2}\mathrm{d}\tilde{z} \\
x &= \frac{1}{\sqrt{2}}\tan^{-1}(\sqrt{2}\tilde{z})\Big|_0^{z(x)} \\
x &= \frac{1}{\sqrt{2}}\tan^{-1}(\sqrt{2}z)\,.
\end{aligned}$$

Rearranging for z we get

$$z(x) = \frac{1}{\sqrt{2}} \tan(\sqrt{2}x)\,,$$

and then we can substitute in $z(x) = x + y(x)$ to get the solution

$$y(x) = \frac{1}{\sqrt{2}} \tan(\sqrt{2}x) - x\,.$$

2.5.2.2 Case 2: quotients of y and x

! Important

Let us consider first order differential equations of the form

$$\frac{\mathrm{d}}{\mathrm{d}x} y(x) = f(y/x) \tag{2.20}$$

for some initial condition $y(x_0) = y_0$. Then an obvious choice of substitution would be $z = y/x$, with $z(x_0) = y_0/x_0$. Our equation is invariant under replacing x with ax and replacing y with ay, for any $a \in \mathbb{R}$. From this we see (from the product rule) that

$$\frac{\mathrm{d}}{\mathrm{d}x} y(x) = \frac{\mathrm{d}}{\mathrm{d}x} z(x)x + z(x)\,,$$

which rearranges to give

$$\frac{\mathrm{d}}{\mathrm{d}x} z(x) = \frac{1}{x}\left(\frac{\mathrm{d}}{\mathrm{d}x} y(x) - z(x)\right) = \frac{1}{x}(f(z) - z)\,.$$

Using separation of variables, we get

$$\int_{z_0}^{z(x)} \frac{1}{(f(\tilde{z}) - \tilde{z})} \mathrm{d}\tilde{z} = \int_{x_0}^{x} \frac{1}{\tilde{x}} \mathrm{d}\tilde{x} = \ln x - \ln x_0\,.$$

Example 2.11. For simplicity of notation, we will use y for $y(x)$ and z for $z(x)$ in this example.[5] Let us use substitution to solve the initial value problem

$$xy\frac{\mathrm{d}}{\mathrm{d}x} y - y^2 = (x + y)^2 e^{-y/x}$$

and $y(1) = 1$. This rearranges to give

[5] This is a trick I will also use in the blackboard sometimes to simplify the notation.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y}{x} + \left(\frac{x}{y} + 2 + \frac{y}{x}\right)e^{-y/x} = \frac{y}{x} + \frac{x}{y}\left(1 + \frac{y}{x}\right)^2 e^{-y/x}$$

So we let $z = y/x$, giving $z_0 = z(1) = 1$ and

$$\frac{dz}{\mathrm{d}x} = \frac{1}{x}(f(z) - z) = \frac{1}{x}\left(z + \frac{(1+z)^2}{z}e^{-z} - z\right) = \frac{1}{x}\left(\frac{(1+z)^2}{z}e^{-z}\right).$$

Separating variables and integrating gives

$$\int_1^z \frac{\tilde{z}e^{\tilde{z}}}{(1+\tilde{z})^2}\mathrm{d}\tilde{z} = \int_1^x \frac{1}{\tilde{x}}\mathrm{d}\tilde{x}\,.$$

Integrating the left hand side requires another change of variables, $u = 1 + z$, giving $u_0 = u(1) = 2$ and

$$e^{-1}\int_2^u \frac{(\tilde{u}-1)e^{\tilde{u}}}{\tilde{u}^2}\mathrm{d}\tilde{u} = e^{-1}\int_2^u \left(\frac{1}{\tilde{u}} - \frac{1}{\tilde{u}^2}\right)e^{\tilde{u}}\mathrm{d}\tilde{u} = e^{-1}\left(\frac{e^{\tilde{u}}}{\tilde{u}}\right)\Big|_2^u,$$

which is integrable for the implicit solution

$$\ln x = \left(\frac{e^{\frac{y}{x}}}{1+\frac{y}{x}} - \frac{e}{2}\right).$$

2.5.2.3 Case 3: the Bernoulli equation

The Bernoulli equation is of the form

$$\frac{\mathrm{d}}{\mathrm{d}x}y(x) + P(x)y(x) = Q(x)y(x)^n\,, \quad n \neq 1\,.$$

Changing variables to $z = y^{1-n}$ gives

$$\frac{\mathrm{d}}{\mathrm{d}x}z(x) = (1-n)y(x)^{-n}\frac{\mathrm{d}}{\mathrm{d}x}y(x)\,,$$

and the differential equation

$$\frac{\mathrm{d}}{\mathrm{d}x}z(x) + (1-n)P(x)z(x) = (1-n)Q(x)\,,$$

which is first order linear and therefore can be solved using the integrating factor method described in Section Section 2.4.2.

2.6 Autonomous equations

Let us now consider the special case that the given function f in Equation 2.5 is independent of t so that the differential equation reads

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = f(x(t)), \quad t \in (\alpha, \beta). \tag{2.21}$$

Recall that such equations where the independent variables doesn't appear anywhere other than as a parameter of the dependent variable are called *autonomous*. These equations can be very challenging to integrate, with no method producing an explicit solution.

However, there are tools that give us some insight into the qualitative behaviour of solutions.

2.6.1 Stationary points and stability

> **!** Important
>
> **Definition 2.6** (Stationary point)**.** A stationary point (or equilibrium point) of an autonomous differential equation (Equation 2.21) is a point x^* such that $f(x^*) = 0$.

This means that a stationary point x^* is a constant solution to Equation 2.21. Indeed, if $f(x^*) = 0$ then $x(t) = x^*$ for all t is a solution. This means that if at any point our solution equals x^*, $x(t) = x^*$ (or equivalently, the initial condition $x(t_0) = x^*$), then our solution will never move away from this point.

To find stationary points of a differential equation, we have to set $f(x) = 0$ and calculate all the solutions of that equation.

Example 2.12. Consider the equation

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = x^2 - 1.$$

Then $f(x) = x^2 - 1 = (x + 1)(x - 1)$ and setting $f(x) = 0$ gives the stationary points $x_1^* = -1$ and $x_2^* = 1$.

An interesting question now is what happens when the solution is near to (but not exactly at) a stationary point. In other words, we want to investigate whether the solution will move closer to or further away from the stationary point. This is called the *stability* of the stationary point. In order for the following investigation to make sense we need to assume that $t \in [t_0, \infty)$, so we can look at the long-term behaviour of solutions.

Important

Definition 2.7. A stationary point x^* of an autonomous differential equation of the form Equation 2.21 is called *stable* if nearby solutions remain close as the independent variable grows.

Definition 2.8. A stationary point x^* of an autonomous differential equation of the form Equation 2.21 is called *unstable* if nearby solutions move away from x^* as the independent variable grows.

Lemma 2.3. *If x^* is a stationary point of an autonomous differential equation of the form Equation 2.21, i.e. we have that $f(x^*) = 0$, then if*

$$f'(x^*) > 0: \quad x^* \text{ is unstable,}$$
$$f'(x^*) < 0: \quad x^* \text{ is stable.}$$

It's worth noting that not all stationary points can be classified as either *stable* or *unstable*: in particular we have not included the case $f'(x^*) = 0$. As we will see below, such stationary points can be stable, unstable or a third option, called a *saddle point*.

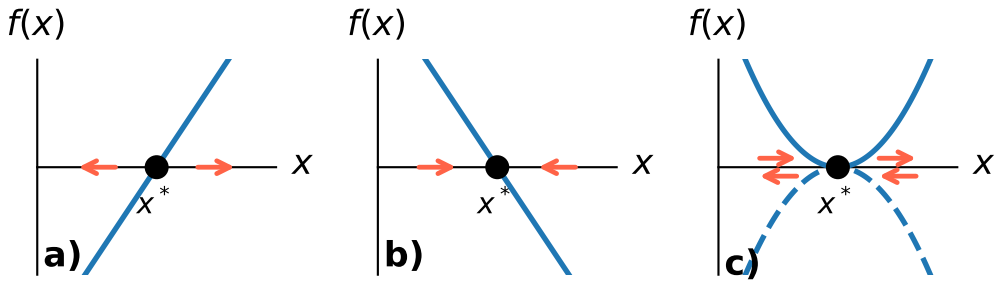


Figure 2.1: Different scenarios for the stability of a stationary point. a) Case 1 (unstable), b) Case 2 (stable), c) Case 3 (saddle point) for both f > 0 (solid) and f < 0 (dashed).

This can be demonstrated by considering all possible scenarios (as shown in Figure 2.1):

Case 1: Consider $f(x) < 0$ if $x < x^*$ and that $f(x) > 0$ if $x > x^*$, where x are points near the x^*, say, $x \in [x^* - \varepsilon, x^* + \varepsilon]$ for some small $\varepsilon > 0$.

1. Suppose now that x is a solution to Equation 2.21 that starts at a point $x(t_0) < x^*$. Then $\dot{x}(t_0) = f(x(t_0)) < 0$, so the solution will decay and move further away from x^*. Using the same argument, this will continue for $t > t_0$ as long as $x(t) \in [x^* - \varepsilon, x^*)$, and ultimately $x(t)$ will leave the small interval.
2. Similarly if $x(t_0) > x^*$. Then $\dot{x}(t_0) = f(x(t_0)) > 0$, so the solution will increase and move away from x^* and finally end up outside of $[x^* - \varepsilon, x^* + \varepsilon]$.

Therefore nearby solutions move away from x^* as the independent variables grows and x^* is *unstable*.

Case 2: Consider now the opposite case that $f(x) > 0$ if $x < x^*$ and that $f(x) < 0$ if $x > x^*$, $x \in [x^* - \varepsilon, x^* + \varepsilon]$ for some small $\varepsilon > 0$. Following the same arguments as for Case 1:

1. Solutions to Equation 2.21 that start at $x(t_0) < x^*$ increase as t increases.
2. Solutions that start at $x(t_0) > x^*$ will decay towards x^* as t increases.

Therefore nearby solutions will move towards x^* and the stationary point x^* is *stable*.

Case 3: Consider now the case that $f(x) \geq 0$ for all $x \in [x^* - \varepsilon, x^* + \varepsilon]$ for some small $\varepsilon > 0$, i.e. the stationary point is a turning point of f that only touches 0 at x^* (Note: this is equivalent to considering the alternative scenario $f(x) \leq 0$ for all $x \in [x^* - \varepsilon, x^* + \varepsilon]$).

1. Solutions to Equation 2.21 that start at $x(t_0) < x^*$ increase towards x^* as t increases.
2. Solutions that start at $x(t_0) > x^*$ will increase away from x^* as t increases and will eventually leave the small interval.

> **!** Important
>
> We call stationary points with this property (stable from one side and unstable from the other) *saddle points*.

If we can differentiate the function f in Equation 2.21 (with respect to x) then the sign of $f'(x^*)$ can be used as a criterion to decide on the stability of the point. If $f'(x^*) > 0$ then f is increasing locally around x^*, (the scenario outlined in Case 1) and x^* is *unstable*. Similarly, if $f'(x^*) < 0$ then f is decreasing locally around x^* (Case 2), and we see that x^* then is *stable*.

We also see that if x^* is a *saddle point* then we will have $f'(x^*) = 0$ (Case 3), however $f'(x^*) = 0$ is **not** a sufficient criteria for x^* to be a saddle point. For example, consider the scenario where x^* is an inflection point, rather than a turning point, of f, with f locally increasing around fx^* then we would still have $f'(x^*) = 0$ but x^* would be *unstable* as in Case 1. Stationary points where $f'(x^*) = 0$ must therefore be carefully considered.

The connection with the derivative will later on be studied again in greater detail in the context of linearisation around stationary points of systems of differential equations (see Section 6.6).

2.6.2 Phase lines

> **!** Important
>
> A *phase line* is a popular way to illustrate the dynamics of an autonomous differential equation such as Equation 2.21. This is a (usually horizontal) line for the points $x \in \mathbb{R}$ with a couple of arrows to the left or right that indicate the direction of the evolution and are proportional to the value of $f(x)$. Figure 2.2 gives an impression of a phase line for the following example.

The phase line is another topic that we will discuss later on in the context of systems of differential equations (then called *phase diagram* or *phase portrait*, see Section 6.5).

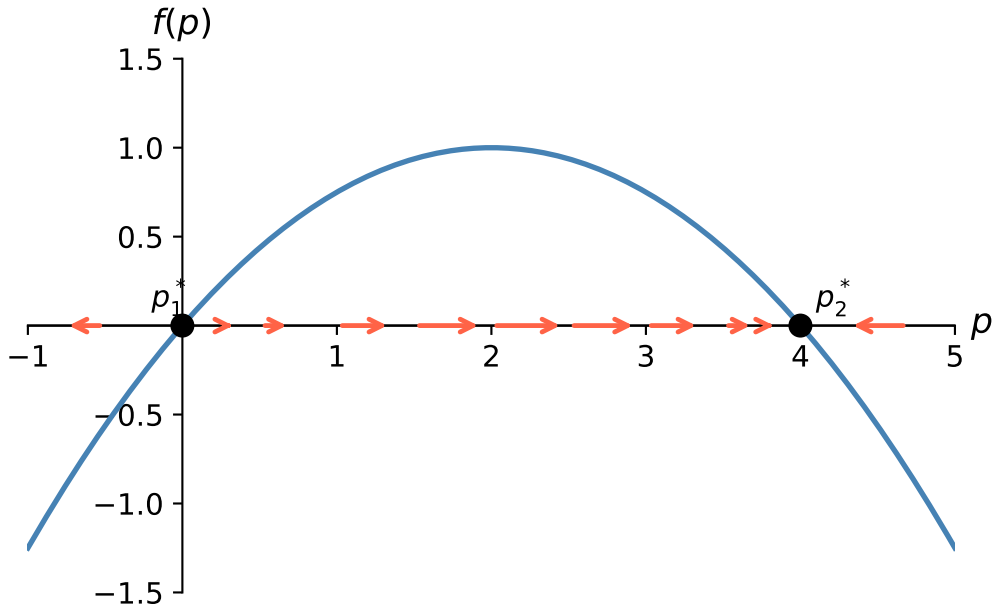


Figure 2.2: Phase line for the population growth equation in Example 2.13. The parameters are $k = 1$ and $p_\mathrm{m} = 4$, and the orange arrows indicate the sign and magnitude of $f(p)$.

Example 2.13 (Population growth)**.** Consider the following equation (sometimes called the logistic equation) for the growth of a population of a certain species:

$$\dot{p}(t) = kp(t)\left(1 - \frac{p(t)}{p_\mathrm{m}}\right).$$

Here $p(t)$ is the population of the species at time t, $k > 0$ is the growth rate of the population and $p_\mathrm{m} > 0$ is the maximum population size (or *carrying capacity*), after

which the population cannot grow further. This equation is a special case of the general autonomous differential equation Equation 2.21, with $f(p) = kp(1 - p/p_\mathrm{m})$. We are typically interested in the dynamics with initial populations $p_0 \in (0, p_\mathrm{m})$ at any time $t_0 \in \mathbb{R}$ (for simplicity, assume that $t_0 = 0$). That is, populations cannot be negative (you cannot have a negative number of individuals) and the population cannot exceed the maximum population size p_m.

Here, the right-hand side of the autonomous equation is

$$f(p) = kp(1 - p/p_\mathrm{m}),$$

which has two zeros at $p_1^* = 0$ and $p_2^* = p_\mathrm{m}$. Note that f is differentiable with $f'(p) = k(1 - 2p/p_\mathrm{m})$. Using the criterion involving the derivative we can also conclude that

$$f(p_1^*) = 0,\ f'(p_1^*) = k > 0 \quad \Rightarrow \quad p_1^* \text{ is unstable,}$$

and

$$f(p_2^*) = 0,\ f'(p_2^*) = -k < 0 \quad \Rightarrow \quad p_2^* \text{ is stable.}$$

As an alternative approach to the derivative criterion, we can draw the phase line for this equation (see Figure Figure 2.2). We see that $f(p)$ intersects the horizontal axis at $p_1^* = 0$ and $p_2^* = p_\mathrm{m}$ (as expected). In the interval $(0, p_\mathrm{m})$, $f(p) > 0$, so the arrows point to the right, and in the intervals $(-\infty, 0)$ and (p_m, ∞), $f(p) < 0$, so the arrows point to the left (even though these are not physically meaningful for the population growth equation). This means that solutions starting at a population $p_0 \in (0, p_\mathrm{m})$ will increase towards $p_2^* = p_\mathrm{m}$ as t increases. In other words, solutions starting near $p_1^* = 0$ will move away from it, while solutions starting near $p_2^* = p_\mathrm{m}$ will move towards it. This is consistent with our previous conclusion that p_1^* is unstable and p_2^* is stable.

In this case one can actually explicitly work out the solution (using techniques outlined in Section Section 2.5.1). It is given by

$$p(t) = \frac{p_0 p_\mathrm{m}}{(p_\mathrm{m} - p_0)e^{-kt} + p_0}.$$

The function indeed increases and converges to $p_2^* = p_\mathrm{m}$ as $t \to \infty$, so this is the stable point that attracts nearby solutions.

2.6.3 Bifurcations

As we have seen in the previous section, we are able to evaluate the behaviour of solutions to autonomous differential equations close to stationary points. We shall now consider how these stationary points can depend on our choice of parameters. Indeed, sometimes a small change in a parameter can cause a change in the number fixed points and/or their stability.

> ! Important
>
> **Definition 2.9.** A bifurcation (or bifurcation point) in a non-linear differential equation is when a small change in a parameter results in a qualitative change in long-term solutions to the equation.

Examples of bifurcations include the creation and destruction of fixed points, or a change in the stability of existing fixed points.

It's worth noting that changes in parameters are likely to result in changes in the *value* of fixed points, but this does not count as a bifurcation if the number of fixed points and their stability stays the same and any change in value is "smooth". For example, in the logistic population growth equation above, changing the maximum population size p_m would change the values of the fixed point p_2^*, but not its stability or the total number of fixed points – this is not a bifurcation.

2.6.3.1 Saddle-node bifurcation

Consider the non-linear autonomous differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = r + x(t)^2\,.$$

Stationary points of this equation occur at all points x^* such that $f(x^*) = 0$, giving

$$r + (x^*)^2 = 0, \qquad \Longrightarrow \qquad (x^*)^2 = -r, \qquad \Longrightarrow \qquad x^* = \pm\sqrt{-r}.$$

This yields three scenarios:

1. $r > 0$: $x(t)$ is a real-valued function, so there are no stationary points x^* (they are complex).
2. $r = 0$: There is one stationary point at $x^* = 0$. Since $f(x) = r + x^2$, differentiating gives $f'(x) = 2x$ and therefore $f'(x^*) = 0$, so $x^* = 0$ is a saddle point.

3. $r < 0$: In this case $-r > 0$ and therefore $x^* = \sqrt{-r}$ gives two fixed points, $x_1^* = \sqrt{-r} > 0$ and $x_2^* = -\sqrt{-r} < 0$. To evaluate their stability, we see that $f'(x_1^*) = 2\sqrt{r} > 0$, therefore x_1^* is unstable, and $f'(x_2^*) = -2\sqrt{r} < 0$, therefore x_2^* is stable.

We can also visualise this by drawing phase lines for the three scenarios (see Figure 2.3).

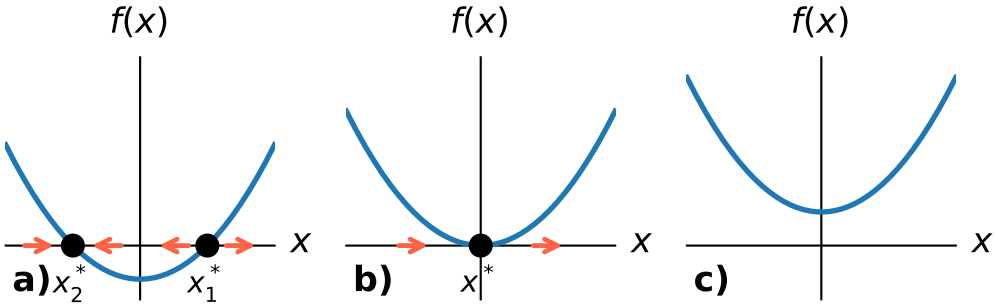


Figure 2.3: Phase line diagrams demonstrating the saddle-node bifurcation: a) $r < 0$; b) $r = 0$; and c) $r > 0$.

! Important

A bifurcation diagram is given by plotting our stationary points, x^*, as a function of the parameter that we are varying (in the above example, this would be r). We draw these using solid lines for stable stationary points and dashed lines for unstable stationary points (see Figure 2.4 for a bifurcation diagram of the saddle-node bifurcation).

! Important

Definition 2.10. A saddle-node bifurcation is any bifurcation where two stationary points collide, briefly becoming a single stationary point before annihilating each other nd leaving no stationary points.

2.6.3.2 Transcritical bifurcation

Consider the non-linear autonomous differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = rx(t) - x(t)^2.$$

Stationary points of this equation occur at all points x^* such that $f(x^*) = 0$, giving

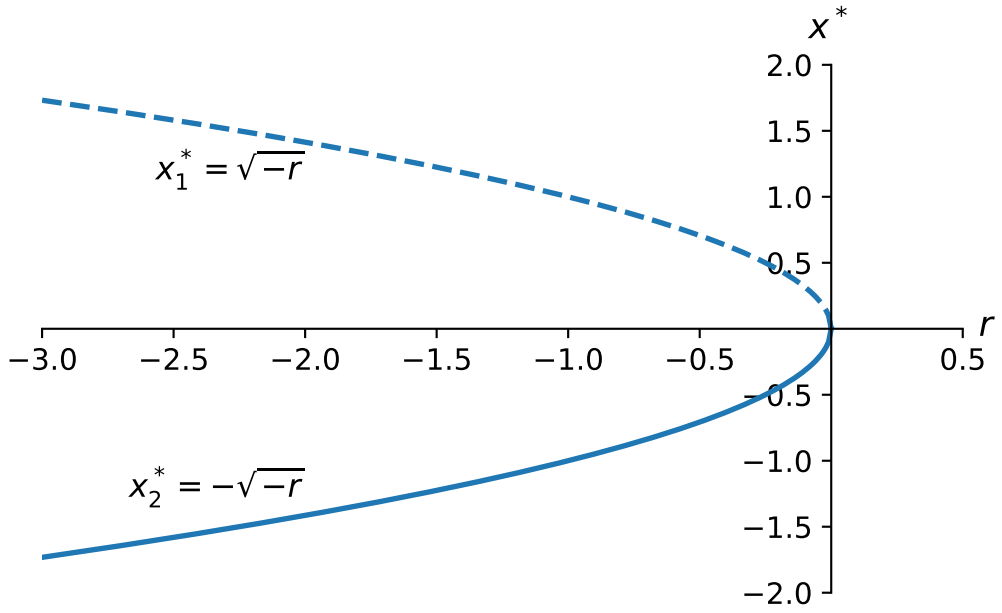


Figure 2.4: Bifurcation diagram for the saddle-node bifurcation. Solid line represents the stable stationary point, dashed line represents the unstable stationary point.

$$rx^* - (x^*)^2 = x^*(r - x^*) = 0\,.$$

Therefore, there are two stationary points if $r \neq 0$: $x_1^* = 0$ and $x_2^* = r$. When $r = 0$ we have that $x_1^* = x_2^* = 0$, giving only one stationary point.

Stability of a stationary point, x^*, is found by examining the sign of $f'(x^*) = r - 2x$. Evaluating at the stationary points we obtain $f'(x_1^*) = r$ and $f'(x_2^*) = -r$. Then, we need to consider:

1. $r > 0$: $f'(x_1^*) > 0$ so it is unstable, and $f'(x_2^*) < 0$ so it is stable.
2. $r < 0$: $f'(x_1^*) < 0$ so it is stable, and $f'(x_2^*) > 0$ so it is unstable.
3. $r = 0$: we cannot use the derivative, so we need to consider the sign of f around the stationary point. We see that $f(x) = -x^2 < 0$ for $x \in \mathbb{R} \setminus \{0\}$, therefore x^* is a saddle point (see Figure 2.5).

This means that there is a change of stability of the fixed points when r changes sign. See Figure 2.6 for the relevant bifurcation diagram.

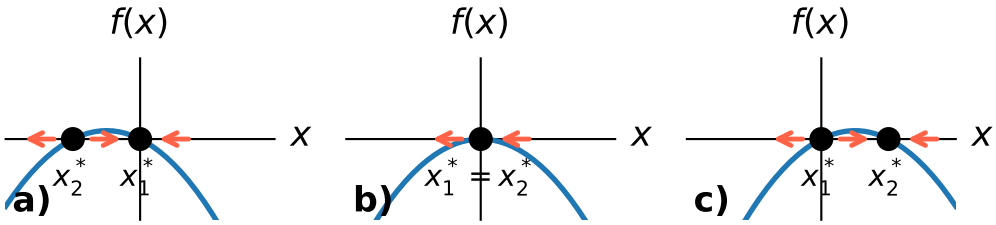


Figure 2.5: Phase line diagrams demonstrating the transcritical bifurcation: a) $r < 0$; b) $r = 0$; and c) $r > 0$.

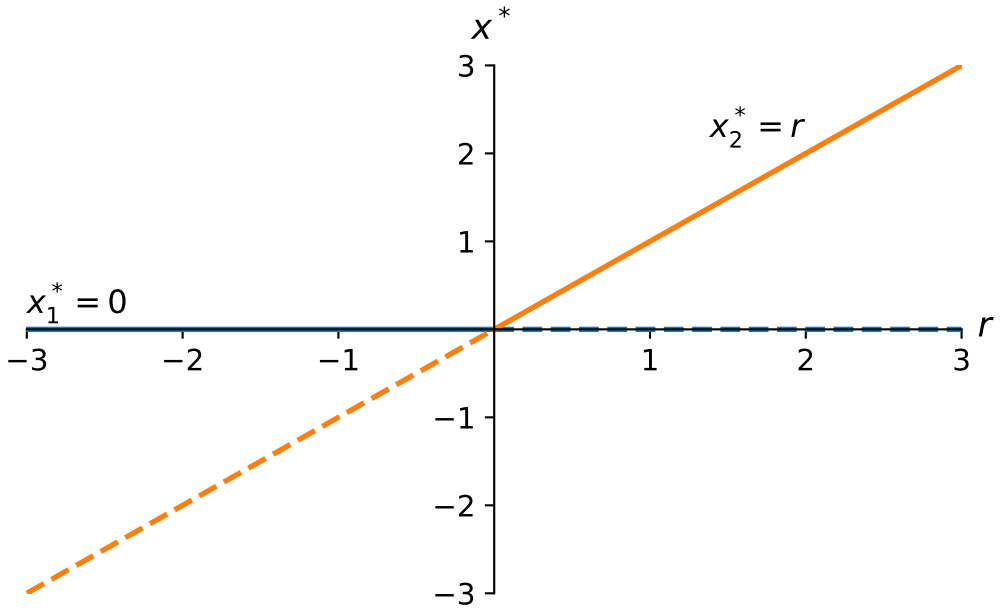


Figure 2.6: Bifurcation diagram for the transcritical bifurcation. Solid line represents the stable stationary point, dashed line represents the unstable stationary point.

> **!** Important
>
> **Definition 2.11.** A transcritical bifurcation is a bifurcation where there is an exchange of stabilities between two fixed points.

2.6.3.3 Pitchfork bifurcation

Consider the non-linear autonomous differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = rx(t) - x(t)^3$$

Stationary points of this equation occur at all points x^* such that $f(x^*) = 0$, giving

$$rx^* - (x^*)^3 = x^*(\sqrt{r} + x^*)(\sqrt{r} - x^*) = 0\,.$$

We also have that $f'(x) = r - 3x^2$ and now we can examine all three possible cases:

1. $r < 0$: As $x(t)$ is real-valued, there is only one stationary point at $x^* = 0$ and $f'(x^*) = f'(0) = r < 0$, therefore the fixed point is stable.
2. $r = 0$: There is only one stationary point at $x^* = 0$ and $f'(x^*) = 0$. This means we cannot use the sign of f at the stationary point to determine its stability, however we see that $f(x) = -x^3$ is < 0 if $x > 0 = x^*$ and > 0 if $x < 0 = x^*$, therefore x^* is stable. We can also see this by plotting $f(x)$ against x for $r = 0$ and drawing the phase line (see Figure 2.8).
3. $r > 0$: There are three stationary points at $x^* = 0$, $x_1^* = \sqrt{r}$ and $x_2^* = -\sqrt{r}$. We have that $f'(0) = r > 0$, therefore $x^* = 0$ is unstable, and $f'(\sqrt{r}) = f'(-\sqrt{r}) = r - 3r = -2r < 0$, therefore x_1^* and x_2^* are stable.

See Figure 2.7 for the bifurcation diagram.

> **!** Important
>
> **Definition 2.12.** A pitchfork bifurcation is a bifurcation where a differential equation transitions between having one stationary point and having three stationary points.
>
> 1. *Super-critical pitchfork bifurcation:* A pair of stable stationary points are created at the bifurcation point and the existing stationary point switches from stable to unstable.
> 2. *Sub-critical pitchfork bifurcation:* A pair of unstable fixed points are created at the bifurcation point and the existing stationary point switches from unstable to stable.

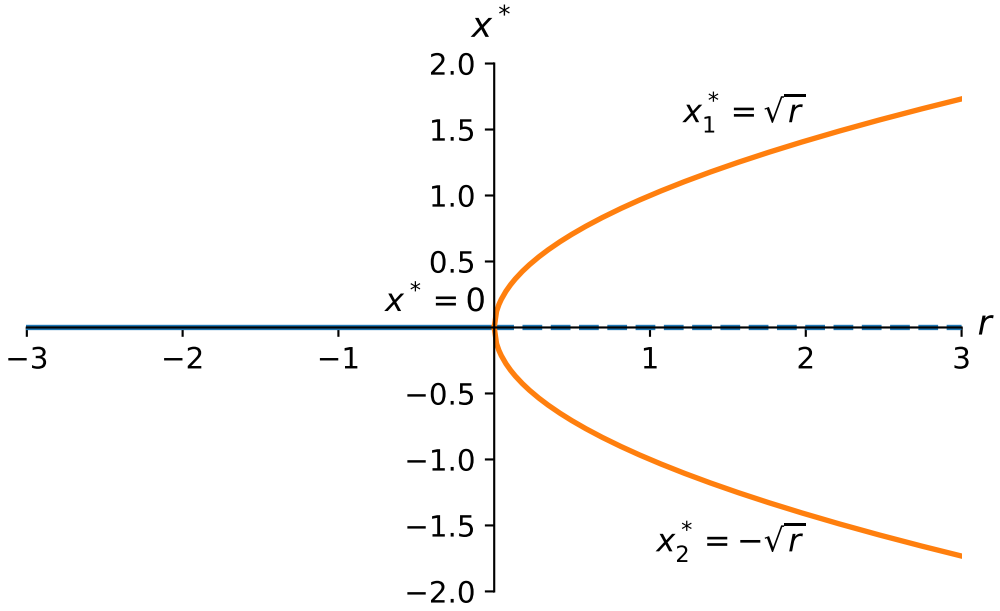


Figure 2.7: Bifurcation diagram for the pitchfork bifurcation.

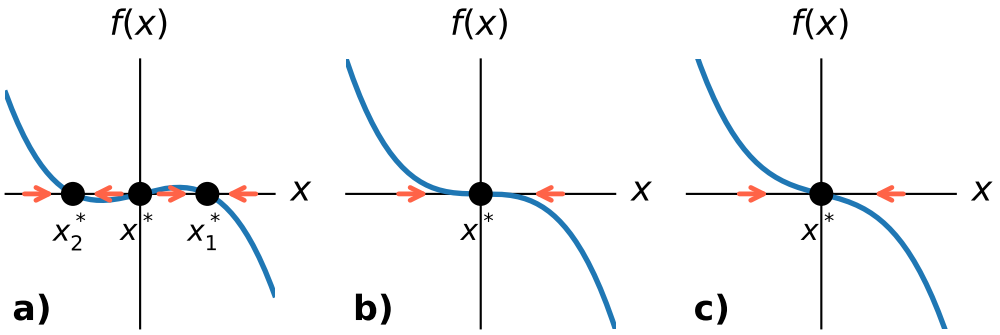


Figure 2.8: Phase line diagrams demonstrating the pitchfork bifurcation: a) $r < 0$; b) $r = 0$; and c) $r > 0$.

Example 2.14 (Fishery)**.** Recall the logistic equation for the growth of a given population (Example 2.13), let us say fishes. As we have seen, the equation describes the growth of a population that is limited by a maximum population size. However, let us now consider a scenario where the population is subject to fishing. In this case, we can modify the equation to account for that:

$$\frac{\mathrm{d}}{\mathrm{d}t}p(t) = rp(t)\left(1 - \frac{p(t)}{p_\mathrm{m}}\right) \underbrace{-cp(t)}_{\text{fishing term}} .$$

Here, the fishing term (with $c > 0$) is proportional to the population size, meaning that the more fish there are, the more can be caught. We want to consider the stationary points and their stability based on the value of c (i.e. how much fishing is taking place).

First, we need to compute the stationary points:

$$f(p^*) = rp^*\left(1 - \frac{p^*}{p_\mathrm{m}}\right) - cp^* = 0 \quad \Longrightarrow \quad p_1^* = 0, \quad \text{and} \quad p_2^* = p_\mathrm{m}\left(1 - \frac{c}{r}\right).$$

Note that when $c = 0$, we fall back to the original logistic equation that we studied in Example 2.13. The stationary point p_1^* always exists, but p_2^* only exists if $c < r$. If $c > r$, then $p_2^* < 0$ and is not physically meaningful so we can ignore it. Now let us consider the derivative of f:

$$f'(p) = r - c - \frac{2r}{p_\mathrm{m}}p.$$

Therefore, we have that

$$f'(p_1^*) = r - c, \quad \text{and} \quad f'(p_2^*) = -(r - c).$$

Note that there is a change in the number of stationary points at $c = r$ and also the sign of the derivatives changes at that point. Therefore, we need to consider three cases:

1. $0 < c < r$: In this case we have two stationary points. From the derivative we see that $f'(p_1^*) = r - c > 0$, therefore p_1^* is unstable, and $f'(p_2^*) = -(r - c) < 0$, therefore p_2^* is stable.
2. $c > r$: In this case we have only one stationary point, p_1^*, and $f'(p_1^*) = r - c < 0$, therefore p_1^* is stable.
3. $c = r$: In this case we have only one stationary point as $p_1^* = p_2^* = 0$. We have that $f'(p_1^*) = r - c = 0$, so we cannot use the derivative to determine stability. However, we can see that in this case $f(p) = -rp^2/p_\mathrm{m} < 0$ for all $p > 0$ (we do not need to consider p < 0), therefore p_1^* is stable.

With this information, we can draw the bifurcation diagram for this system (see Figure 2.9).

The physical interpretation of this is that if the fishing rate is small ($c < r$), the population will always end up stabilising at a positive value p_2^*. However, if the fishing rate is too high ($c \geq r$), the population will eventually die out (i.e. reach $p_1^* = 0$) no matter what the initial condition is.

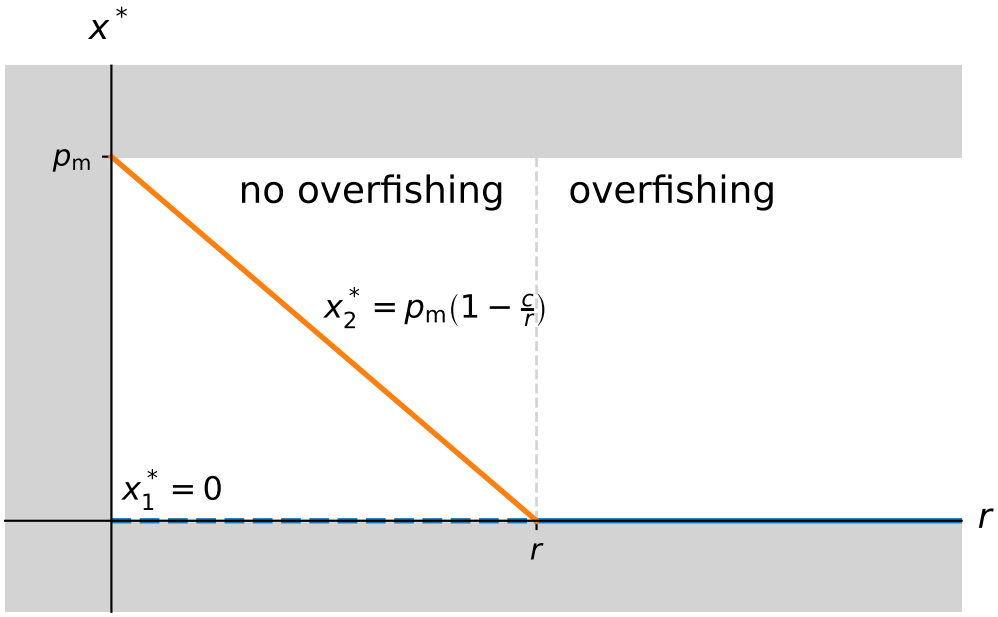


Figure 2.9: Bifurcation diagram for the fishery equation. Solid line represents the stable stationary point, dashed line represents the unstable stationary point. Note that we constraint our bifurcation diagram to the physically meaningful region $p^* \in [0, p_\mathrm{m}]$ and $c \geq 0$ (non-physical regions are shaded).

2.7 Linear second order equations

We now want to look into higher order linear differential equations, however we confine mostly to second order equations. Second order equations already showcase the essential ideas and difficulties, and the methods can be generalised to higher order equations, even though with a higher complexity. Moreover, second order equations are of importance in many applications as they can describe oscillatory behaviour.

Therefore we consider differential equations of the form

$$a(t)\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) + b(t)\frac{\mathrm{d}}{\mathrm{d}t}x(t) + c(t)x(t) = s(t), \tag{2.22}$$

with given functions $a(t) \neq 0$, $b(t)$, $c(t)$, and $s(t)$, $t \in (\alpha, \beta)$.

First, a brief observation. A special case is $a(t) = 1$, $b(t) = c(t) = s(t) = 0$:

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} x(t) = 0.$$

Integrating twice we obtain that

$$x(t) = C_1 t + C_0$$

where $C_1, C_0 \in \mathbb{R}$ are integration constants. This already shows that, in order to identify a specific solution we need to impose two conditions. This motivates a new definition of initial value problem.

! Important

Definition 2.13 (Initial value problem)**.** The *initial value problem for linear second order equations* consists of finding a function $x : (\alpha, \beta) \to \mathbb{R}$ such that the differential equation Equation 2.22 and the *initial conditions*

$$x(t_0) = x_0, \quad \frac{\mathrm{d}}{\mathrm{d}t} x(t_0) = v_0 \tag{2.23}$$

for some $t_0 \in [\alpha, \beta]$ and some given numbers $x_0, v_0 \in \mathbb{R}$ are satisfied.

Remark 2.4. Throughout this chapter and unless stated otherwise we assume that the data a, b, c, and s are "nice" enough so that the initial value problem in Definition 2.13 has a unique solution.

There are abstract results that address this problem. However, these results usually do not produce explicit solutions. Our strategy for this purpose is to exploit the linearity as follows:

1. We first study the homogeneous equation ($s = 0$ in Equation 2.22). Thanks to linearity this will require solving two problems. The outcome will be a general solution called the *complementary function* with two parameters. These enable us to account for two initial conditions Equation 2.23.
2. The inhomogeneous case then requires finding a specific solution called the *particular integral* of Equation 2.22. Thanks to linearity again, we can then use the results for the homogeneous case to account for the initial condition.

2.7.1 Homogeneous equation and complementary function

We consider now the case of a homogeneous equation with $s = 0$ in Equation 2.22,

$$a(t)\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) + b(t)\frac{\mathrm{d}}{\mathrm{d}t}x(t) + c(t)x(t) = 0, \tag{2.24}$$

and aim for describing the set of solutions.

Lemma 2.4. *Assume that x_1 and x_2 are two solutions of Equation 2.24. Then so is*

$$x(t) = l_1 x_1(t) + l_2 x_2(t), \quad t \in [\alpha, \beta],$$

for all $l_1, l_2 \in \mathbb{R}$.

This lemma can be generalised to higher order in a straightforward way.

Proof. Substituting x in Equation 2.24 we see that indeed

$$\begin{aligned}
&a(t)\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) + b(t)\frac{\mathrm{d}}{\mathrm{d}t}x(t) + c(t)x(t) \\
&\quad = a(t)\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(l_1 x_1(t) + l_2 x_2(t)\right) + b(t)\frac{\mathrm{d}}{\mathrm{d}t}\left(l_1 x_1(t) + l_2 x_2(t)\right) + c(t)\left(l_1 x_1(t) + l_2 x_2(t)\right) \\
&\quad = l_1 a(t)\frac{\mathrm{d}^2 x_1}{\mathrm{d}t^2} + l_2 a(t)\frac{\mathrm{d}^2 x_2}{\mathrm{d}t^2} + l_1 b(t)\frac{\mathrm{d}x_1}{\mathrm{d}t} + l_2 b(t)\frac{\mathrm{d}x_2}{\mathrm{d}t} + l_1 c(t)x_1(t) + l_2 c(t)x_2(t) \\
&\quad = l_1\left(a(t)\frac{\mathrm{d}^2 x_1}{\mathrm{dx}_1 t^2} + b(t)\frac{\mathrm{d}x_1}{\mathrm{d}t} + c(t)x_1(t)\right) + l_2\left(a(t)\frac{\mathrm{d}^2 x_2}{\mathrm{d}t^2} + b(t)\frac{\mathrm{d}x_2}{\mathrm{d}t} + c(t)x_2(t)\right) \\
&\quad = l_1 \times 0 + l_2 \times 0 = 0.
\end{aligned}$$

□

Definition 2.14 (Linearly independent functions)**.** Two functions x_1, x_2 on an interval $[\alpha, \beta]$ are *linearly independent*[6] if the only solution to

$$l_1 x_1(t) + l_2 x_2(t) = 0 \text{ for any } l_1, l_2 \in \mathbb{R}$$

is $l_1 = l_2 = 0$.

[6] You may be familiar with the concept of linear independence from linear algebra. Here, the vector space is that of functions, which can be pointwise added and multiplied with a scalar.

Note that linear independence implies that $x_1 \neq 0$ and $x_2 \neq 0$ (as whole functions, at some points t their value is permitted to be zero).

The concept generalises to an arbitrary number of functions n, but in the particular case here ($n = 2$) we can give a more intuitive definition of linear independence: if the functions are linearly dependent then they are multiples of each other. For instance, if $l_1 \neq 0$ (otherwise necessarily $l_2 \neq 0$ and we divide by l_2) then

$$x_1(t) = -\frac{l_2}{l_1} x_2(t).$$

! Important

Definition 2.15 (Complementary function)**.** Given any two linearly independent solutions x_1, x_2 of Equation 2.24, the *general solution* (or set of solutions)

$$l_1 x_1(t) + l_2 x_2(t) \text{ for any } l_1, l_2 \in \mathbb{R}$$

is called the *complementary function.*

Corollary 2.1. *Assume that x_1 and x_2 are two linearly independent solutions of Equation 2.24. Then the solution to the initial value problem in Equation 2.24 with initial conditions in Equation 2.23 is given by*

$$x(t) = l_1 x_1(t) + l_2 x_2(t), \quad t \in [\alpha, \beta],$$

where l_1, l_2 are such that

$$\begin{aligned} x_0 &= x(t_0) = l_1 x_1(t_0) + l_2 x_2(t_0), \\ v_0 &= v(t_0) = l_1 \frac{\mathrm{d}}{\mathrm{d}t} x_1(t_0) + l_2 \frac{\mathrm{d}}{\mathrm{d}t} x_2(t_0). \end{aligned} \tag{2.25}$$

2.7.2 Inhomogeneous equation and particular integral

Let us now assume that $s \neq 0$ in Equation 2.22, i.e. our differential equation is inhomogeneous. A convenient objective then is to find a solution x_p that satisfies

$$x_p(t_0) = 0, \quad \frac{\mathrm{d}}{\mathrm{d}t} x_p(t_0) = 0. \tag{2.26}$$

This is called the particular integral.

Lemma 2.5. *Assume x_1 and x_2 are two linearly independent solutions of Equation 2.24 (homogeneous) and x_p is a solution to Equation 2.22 (inhomogeneous) fulfilling Equation 2.26, then the solution to the initial value problem Equation 2.22 with initial conditions Equation 2.23 is given by*

$$x(t) = x_\mathrm{p}(t) + l_1 x_1(t) + l_2 x_2(t), \quad t \in [\alpha, \beta],$$

where l_1 and l_2 are such that

$$\begin{aligned} x_0 &= l_1 x_1(t_0) + l_2 x_2(t_0), \\ v_0 &= l_1 \frac{\mathrm{d}}{\mathrm{d}t} x_1(t_0) + l_2 \frac{\mathrm{d}}{\mathrm{d}t} x_2(t_0). \end{aligned}$$

Proof. Substituting our solution x from Equation 2.27 into our inhomogeneous differential equation Equation 2.22 (and using dot notation to represent derivatives of x with respect to time and dropping the argument t in the dependent variables for brevity) we get:

$$l_1 \left(a(t)\ddot{x}_1 + b(t)\dot{x}_1 + c(t)x_1\right) + l_2 \left(a(t)\ddot{x}_2 + b(t)\dot{x}_2 + c(t)x_2\right) + a(t)\ddot{x}_\mathrm{p} + b(t)\dot{x}_\mathrm{p} + c(t)x_\mathrm{p} = s(t).$$

As x_1 and x_2 are solutions to the homogeneous equation

$$a(t)\ddot{x} + b(t)\dot{x} + c(t)x = 0,$$

we have that these terms cancel to zero and we are left with

$$a(t)\ddot{x}_\mathrm{p} + b(t)\dot{x}_\mathrm{p} + c(t)x_\mathrm{p} = s(t),$$

which is true by definition of x_p as a solution to the inhomogeneous case where $s \neq 0$. Since $x_\mathrm{p}(t_0) = \dot{x}_\mathrm{p}(t_0) = 0$, we have that

$$\begin{aligned} x_0 &= l_1 x_1(t_0) + l_2 x_2(t_0) + x_\mathrm{p}(t_0) = l_1 x_1(t_0) + l_2 x_2(t_0) + 0, \\ v_0 &= l_1 \dot{x}_1(t_0) + l_2 \dot{x}_2(t_0) + \dot{x}_\mathrm{p}(t_0) = l_1 \dot{x}_1(t_0) + l_2 \dot{x}_2(t_0) + 0. \end{aligned}$$

□

Remark 2.5. In practice, finding x_p can be challenging. But note that the initial conditions Equation 2.26 do not have to be satisfied. In fact, if $x_p(t_0) \neq 0$ and $\dot{x}_p(t_0) \neq 0$ then the equation $x(t) = l_1 x_1(t) + l_2 x_2(t) + x_\mathrm{p}(t)$ is still a solution, and we simply need to include $x_\mathrm{p}(t_0)$ and $\dot{x}_\mathrm{p}(t_0)$ when using our initial conditions to calculate l_1 and l_2, giving us the following result.

Important

Lemma 2.6. *Assume x_1 and x_2 are two linearly independent solutions of Equation 2.24 (homogeneous) and x_p is a solution to Equation 2.22 (inohomogeneous), then the solution to the initial value problem Equation 2.22 with initial conditions Equation 2.23 is given by*

$$x(t) = x_\mathrm{p}(t) + l_1 x_1(t) + l_2 x_2(t), \quad t \in [\alpha, \beta], \tag{2.27}$$

where l_1 and l_2 are such that

$$\begin{aligned} x_0 &= x_\mathrm{p}(t_0) + l_1 x_1(t_0) + l_2 x_2(t_0), \\ v_0 &= \frac{\mathrm{d}}{\mathrm{d}t} x_\mathrm{p}(t_0) + l_1 \frac{\mathrm{d}}{\mathrm{d}t} x_1(t_0) + l_2 \frac{\mathrm{d}}{\mathrm{d}t} x_2(t_0). \end{aligned} \tag{2.28}$$

2.8 Linear and second order with constant coefficients

To illustrate how the procedure outlined in the previous section works let us investigate the case of an equation Equation 2.22 with constant coefficients,

$$a \frac{\mathrm{d}^2}{\mathrm{d}t^2} x(t) + b \frac{\mathrm{d}}{\mathrm{d}t} x(t) + c x(t) = s(t), \tag{2.29}$$

where we assume that $a \neq 0$ (otherwise the order of the differential equation would be less than two).

We first need to find the solutions to the homogeneous equation, so assume that $s = 0$ and study

$$a \frac{\mathrm{d}^2}{\mathrm{d}t^2} x(t) + b \frac{\mathrm{d}}{\mathrm{d}t} x(t) + c x(t) = 0, \tag{2.30}$$

We have seen that a function of the form $e^{\lambda t}$ solved the linear first order equations with constant coefficients, so let us try this form. Substituting it in the differential equation Equation 2.30 yields that

$$0 = a\lambda^2 e^{\lambda t} + b\lambda e^{\lambda t} + c e^{\lambda t} = \left(a\lambda^2 + b\lambda + c\right) e^{\lambda t}.$$

! Important

Definition 2.16 (Auxiliary equation)**.** The equation

$$a\lambda^2 + b\lambda + c = 0$$

is called the *auxiliary equation* or *characteristic equation* of the differential equation Equation 2.30.

So if λ is a solution to the auxiliary equation then $e^{\lambda t}$ solves the differential equation. The solutions to this quadratic equation are given by

$$\lambda_{1,2} = -\frac{b}{2a} \pm \frac{1}{2a}\sqrt{b^2 - 4ac}.$$

We have to distinguish several cases, depending on the sign of $b^2 - 4ac$.

2.8.1 Real roots

! Important

If $b^2 - 4ac > 0$ then $\lambda_1 \neq \lambda_2$ are two real roots of the auxiliary equation. The functions

$$x_1(t) = e^{\lambda_1 t}, \quad x_2(t) = e^{\lambda_2 t}$$

then are two linearly independent solutions of Equation 2.30. Let us now find the coefficients l_1 and l_2. For simplicity, let us assume that

$$t_0 = 0.$$

Otherwise replace t with $t - t_0$ in the following computations.
We proceed, starting with

$$x(t) = l_1 x_1(t) + l_2 x_2(t) = l_1 e^{\lambda_1 t} + l_2 e^{\lambda_2 t},$$

the initial conditions Equation 2.28 yield that

$$\begin{aligned} x_0 &= x(0) = l_1 x_1(0) + l_2 x_2(0) = l_1 + l_2, \\ v_0 &= \dot{x}(0) = l_1 \dot{x}_1(0) + l_2 \dot{x}_2(0) = l_1 \lambda_1 + l_2 \lambda_2. \end{aligned}$$

Solving this system (e.g. by substituting $l_2 = x_0 - l_1$ in the second equation and resolving for l_1) yields that

$$l_1 = \frac{x_0 \lambda_2 - v_0}{\lambda_2 - \lambda_1}, \quad l_2 = \frac{x_0 \lambda_1 - v_0}{\lambda_1 - \lambda_2}$$

and thus the solution

$$x(t) = \frac{x_0\lambda_2 - v_0}{\lambda_2 - \lambda_1} e^{\lambda_1 t} + \frac{x_0\lambda_1 - v_0}{\lambda_1 - \lambda_2} e^{\lambda_2 t}.$$

Example 2.15. Consider the initial value problem

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) + \frac{\mathrm{d}}{\mathrm{d}t}x(t) - 6x(t) = 0, \qquad x(0) = 5, \quad \frac{\mathrm{d}}{\mathrm{d}t}x(0) = 0.$$

We know from Corollary Corollary 2.1 that the solution then is given by

$$x(t) = l_1 x_1(t) + l_2 x_2(t).$$

The auxiliary equation reads

$$\lambda^2 + \lambda - 6 = 0$$

and has the solutions

$$\lambda_{1,2} = -\frac{1}{2} \pm \sqrt{\frac{1}{4} - 6} = \frac{1}{2}\left(1 \pm 5\right), \quad \Rightarrow \quad \lambda_1 = 2,\ \lambda_2 = -3,$$

so

$$x(t) = l_1 e^{\lambda_1 t} + l_2 e^{\lambda_2 t} = 3e^{2t} + 2e^{-3t}.$$

2.8.2 Complex roots

Important

If $b^2 - 4ac < 0$ then

$$\lambda_{1,2} = p \pm iq \quad \text{where } p = -\frac{b}{2a},\ q = \frac{\sqrt{4ac - b^2}}{2a}.$$

The functions

$$\tilde{x}_1(t) = e^{(p+iq)t}, \quad \tilde{x}_2(t) = e^{(p-iq)t}$$

are linearly independent but complex-valued solutions to Equation 2.30. We are interested in two linearly independent real-valued solutions, which we would obtain if we followed the same approach as in the previous case, only that this would require

many complex-values calculations. Instead, we will rearrange the expression so we can work in the real domain only. First, note that the arguments in Lemma 2.4 still go through if $l_1, l_2 \in \mathbb{C}$. Second, by Euler's formula

$$e^{\pm iqt} = \cos(qt) \pm i \sin(qt).$$

Therefore, we can define the real-valued functions

$$x_1(t) = \frac{1}{2}\left(\tilde{x}_1(t) + \tilde{x}_2(t)\right) = e^{pt}\cos(qt),$$
$$x_2(t) = \frac{1}{2i}\left(\tilde{x}_1(t) - \tilde{x}_2(t)\right) = e^{pt}\sin(qt).$$

As they are linear combinations of solutions to Equation 2.30, they will also be solutions. They are also linearly independent, as can be seen by the fact that they are not multiples of each other. Therefore, we will use instead x_1 and x_2 to build our general solution.

Let us again assume that $t_0 = 0$ but solve an example directly, following the lines around Equation 2.25.

Example 2.16. Consider the initial value problem

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) - 2\frac{\mathrm{d}}{\mathrm{d}t}x(t) + 26x(t) = 0, \qquad x(0) = 3, \quad \frac{\mathrm{d}}{\mathrm{d}t}x(0) = -12.$$

Then $a = 1$, $b = -2$, $c = 26$, and then

$$p = -\frac{b}{2a} = 1, \quad q = \frac{\sqrt{4ca - b^2}}{2a} = \frac{\sqrt{104 - 4}}{2} = 5,$$

so we obtain that

$$x_1(t) = e^t\cos(5t), \quad x_2(t) = e^t\sin(5t).$$

The general solution thus is

$$x(t) = l_1x_1(t) + l_2x_2(t) = l_1e^t\cos(5t) + l_2e^t\sin(5t), \quad l_1, l_2 \in \mathbb{R}.$$

Note that

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = l_1\left(e^t\cos(5t) - 5e^t\sin(5t)\right) + l_2\left(e^t\sin(5t) + 5e^t\cos(5t)\right).$$

To satisfy the initial conditions we have to solve

$$\begin{aligned} 3 &= l_1 x_1(0) + l_2 x_2(0) = l_1, \\ -12 &= l_1 \dot{x}_1(0) + l_2 \dot{x}_2(0) = l_1 + 5l_2. \end{aligned}$$

Substituting the result for l_1 into the second equation yields $l_2 = -3$. Hence,

$$x(t) = 3e^t \left(\cos(5t) - \sin(5t)\right).$$

2.8.3 Repeated real root

> **!** Important
>
> In the case $b^2 - 4ac = 0$ the auxiliary equation has only one root
>
> $$\lambda := \lambda_{1,2} = -\frac{b}{2a}.$$
>
> Then $x_1(t) = e^{\lambda t}$ is only one solution. A convenient second, linearly independent solution is given by[7]
>
> $$x_2(t) = te^{\lambda t}.$$

To see that this indeed is a solution, note that

$$\frac{b}{a} = -2\lambda$$

and

$$b^2 - 4ac = 0 \quad \Rightarrow \quad c = \frac{b^2}{4a} \quad \Rightarrow \quad \frac{c}{a} = \frac{1}{4}\frac{b^2}{a^2} = \lambda^2.$$

Now

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} x_2(t) &= e^{\lambda t} + \lambda t e^{\lambda t}, \\ \frac{\mathrm{d}^2}{\mathrm{d}t^2} x_2(t) &= 2\lambda e^{\lambda t} + \lambda^2 t e^{\lambda t}. \end{aligned}$$

[7] This second solutions drops out of the blue here. It is fairly natural, though, but deriving it requires some tools from linear algebra, namely the *Jordan Canonical Form* and the *Matrix Exponential*. Details are left for future modules.

Altogether,

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}t^2}x_2(t) + \frac{b}{a}\frac{\mathrm{d}}{\mathrm{d}t}x_2(t) + \frac{c}{a}x_2(t) \\ &= 2\lambda e^{\lambda t} + \lambda^2 t e^{\lambda t} - 2\lambda\left(e^{\lambda t} + \lambda t e^{\lambda t}\right) + \lambda^2 t e^{\lambda t} \\ &= e^{\lambda t}\left(2\lambda + \lambda^2 t - 2\lambda - 2\lambda^2 t + \lambda^2 t\right) = 0,\end{aligned}$$

so $x_2(t)$ indeed solves Equation 2.30.

Example 2.17. Consider the initial value problem

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) + 6\frac{\mathrm{d}}{\mathrm{d}t}x(t) + 9x(t) = 0, \qquad x(-1) = 2, \quad \frac{\mathrm{d}}{\mathrm{d}t}x(-1) = -5.$$

Then $a = 1$, $b = 6$, $c = 9$, and indeed $b^2 - 4ac = 0$. Noting that $\lambda = -b/2a = -3$, $t_0 = -1$, $x_0 = 2$, and $v_0 = -5$ and hence that the solution is given by

$$x(t) = l_1 x_1(t) + l_2 x_2(t) = l_1 e^{-3t} + l_2 t e^{-3t} = (3 + t)\, e^{-3(t+1)}.$$

The last step of this statement involves substituting for $x(-1) = 2$ and $\dot{x}(-1) = -5$ and solving the resulting equations for l_1 and l_2.

2.8.4 Inhomogeneous equation

> **!** Important
>
> Let us now assume that $s \neq 0$ and investigate the inhomogeneous equation Equation 2.29. As discussed we may try to find a *particular integral* x_p. However, there is no general technique to do so, other than by the method of "inspired guesswork".

Let us consider two examples.

Example 2.18. Let us try to find the solutions to

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) + \frac{\mathrm{d}}{\mathrm{d}t}x(t) - 2x(t) = t^2.$$

With initial condition $x(0) = \dot{x}(0) = 1$. The solutions to the homogeneous equation can be found using the methods in the previous sections and are given in terms of

$$x_1(t) = e^{-2t}, \quad x_2(t) = e^t.$$

As $s(t)$ is a polynomial we may try to find a particular integral in the form of a polynomial, too. Choosing the same degree let us try the ansatz

$$x_p(t) = p_2 t^2 + p_1 t + p_0, \quad \text{with some } p_i \in \mathbb{R}.$$

Substituting this and its derivatives into, the differential equation yields that

$$t^2 = \frac{\mathrm{d}^2}{\mathrm{d}t^2} x_p(t) + \frac{\mathrm{d}}{\mathrm{d}t} x_p(t) - 2x_p(t) = 2p_2 + (2p_2 t + p_1) - 2(p_2 t^2 + p_1 t + p_0).$$

Equating coefficients yields the equations

$$-2p_2 = 1, \quad 2p_2 - 2p_1 = 0, \quad 2p_2 + p_1 - 2p_0 = 0.$$

We see that $p_2 = -1/2$, then $p_1 = p_2 = -1/2$ and finally $p_0 = p_2 + p_1/2 = -3/4$ so that

$$x_p(t) = -\frac{1}{2}t^2 - \frac{1}{2}t - \frac{3}{4}.$$

With $x(0) = \dot{x}(0) = 1$ we obtain the solution

$$x(t) = -\frac{1}{2}t^2 - \frac{1}{2}t - \frac{3}{4} + l_1 e^{-2t} + l_2 e^t = -\frac{1}{2}t^2 - \frac{1}{2}t - \frac{3}{4} + \frac{1}{12}e^{-2t} + \frac{5}{3}e^t.$$

Example 2.19. Let us investigate the same equation but with $s(t) = e^{-t}$,

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} x(t) + \frac{\mathrm{d}}{\mathrm{d}t} x(t) - 2x(t) = e^{-t}.$$

In this case we may try with an exponential function for the particular integral, too. Substituting

$$x_p(t) = ke^{-t}, \quad k \in \mathbb{R},$$

into the differential equation yields

$$e^{-t} = ke^{-t} - ke^{-t} - 2ke^{-t} = -2ke^{-t},$$

and we see that $k = -1/2$ yields that particular integral

$$x_p(t) = -\frac{1}{2}e^{-t}.$$

Again, the initial conditions Equation 2.26 are not satisfied but Remark Remark 2.5 tells us how to incorporate any initial conditions.
Note that $x_1(t) = e^t$ and $x_2(t) = e^{-2t}$ are solutions to the homogeneous equation. So if, say,

$$s(t) = e^{-2t}$$

then the ansatz $x_p(t) = ke^{-2t}$ will not work. Remembering what we did in the case of repeated roots of the auxiliary equation we may try with $x_p(t) = p(t)e^{-2t}$ where $p(t)$ is a polynomial. In this specific case, $x_p = -\frac{1}{3}te^{-2t}$ turns out to work.

2.8.5 Application: mass-spring-damper system

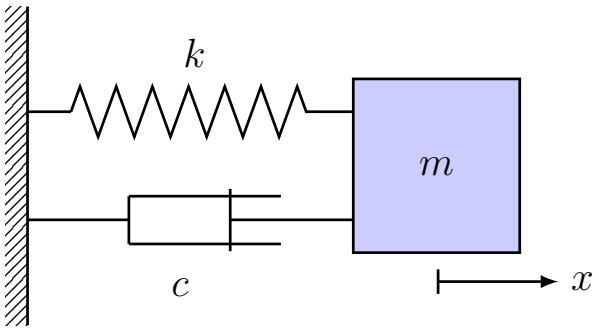


Figure 2.10: Sketch of a mass-spring-damper system.

Consider the system composed of a mass m attached to a spring with spring constant k and a damper[8] with damping coefficient c, as shown in Figure 2.10. The mass is free to move in one dimension, and we denote the position along that direction $x(t)$. The force exerted by the spring is proportional to the displacement of the mass from its equilibrium position (which wlog[9] we define it to be $x = 0$), and is given by Hooke's law as $F_k = -kx(t)$. The force exerted by the damper is proportional to the velocity of the mass, and is given by $F_d = -c\dot{x}(t)$. The equation of motion for the mass can be derived using Newton's second law, which states that the sum of the forces acting on the mass is equal to the mass times its acceleration:

[8] A damper is a device that provides resistance to motion, typically through a viscous fluid or mechanical friction.
[9] Without loss of generality.

$$m\ddot{x}(t) = -kx(t) - c\dot{x}(t). \tag{2.31}$$

Initially ($t = 0$), we assume that the mass is displaced from its equilibrium position and released from rest, which can be expressed as the initial conditions

$$x(0) = x_0, \quad \dot{x}(0) = 0. \tag{2.32}$$

Equation 2.31 is of the form Equation 2.30 with $a = m$, $b = c$, and $c = k$. The auxiliary equation then reads

$$m\lambda^2 + c\lambda + k = 0$$

and has the solutions

$$\lambda_{1,2} = -\frac{c}{2m} \pm \frac{1}{2m}\sqrt{c^2 - 4mk}.$$

Depending on the parameters we have the following solution regimes:

2.8.5.1 Underdamped case

If $c^2 - 4mk < 0$ then Section 2.8.2 gives us the linearly independent solutions

$$x_1(t) = e^{pt}\cos(qt), \quad x_2(t) = e^{pt}\sin(qt),$$

of the differential equation, where $p = -c/(2m)$ and $q = \frac{1}{2m}\sqrt{4mk - c^2}$. Starting with $x(t) = l_1x_1(t) + l_2x_2(t)$, $l_1, l_2 \in \mathbb{R}$, and accounting for the initial conditions Equation 2.32 we obtain the solution

$$x(t) = x_0e^{pt}\left(\cos(qt) - \frac{p}{q}\sin(qt)\right).$$

We obtain an oscillatory behaviour in time with exponential decay of the oscillation amplitude, to which we refer to as underdamped (i.e. there is damping but the system still oscillates), and the behaviour is illustrated in Figure 2.11. Note that, in the case when $c = 0$ we have $p = 0$ so the oscillation amplitude does not decay in time. This is the undamped case, illustrated in Figure 2.12.

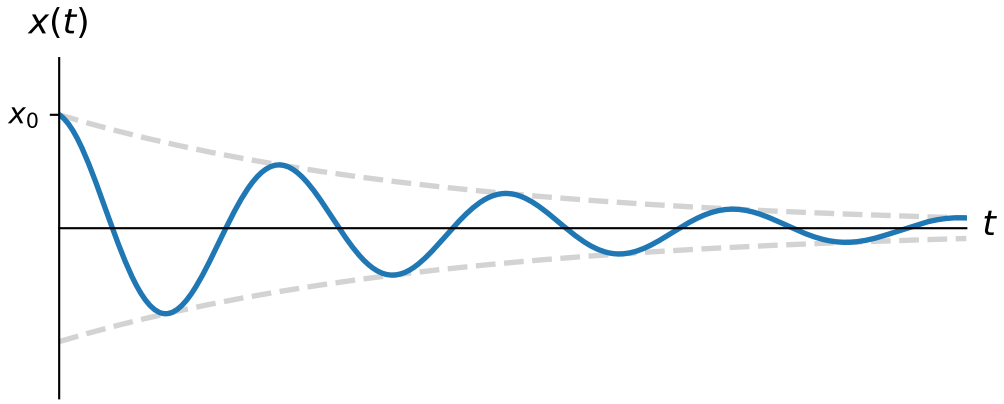


Figure 2.11: **Underdamped** case of a mass-spring-damper system. Note how the system oscillates but the amplitude (shown by the dashed lines) decays exponentially in time

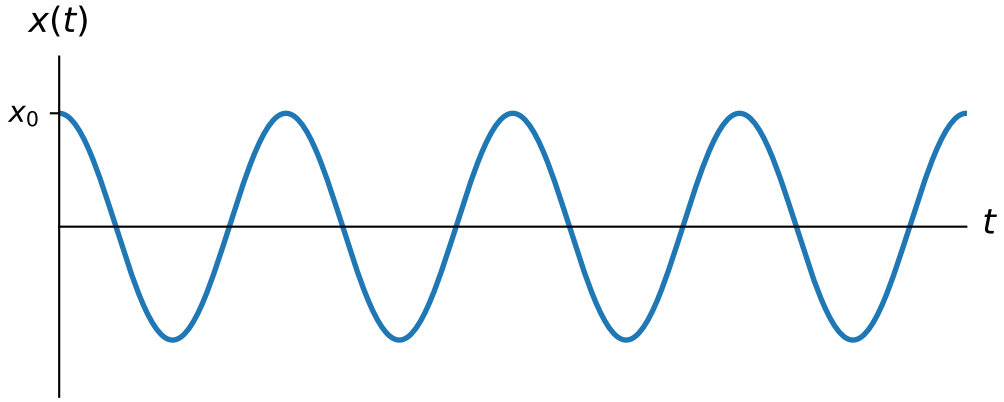


Figure 2.12: **Undamped** case of a mass-spring-damper system. Note how the system oscillates with constant amplitude

2.8.5.2 Overdamped case

If $c^2 - 4mk > 0$ then Section 2.8.1 yields the linearly independent solutions

$$x_1(t) = e^{\lambda_1 t}, \quad x_2(t) = e^{\lambda_2 t},$$

so

$$x(t) = l_1 e^{\lambda_1 t} + l_2 e^{\lambda_2 t}.$$

Accounting for the initial conditions the solution of initial value problem Equation 2.31, Equation 2.32 is given by

$$x(t) = x_0 \left(\frac{\lambda_2}{\lambda_2 - \lambda_1} e^{\lambda_1 t} + \frac{\lambda_1}{\lambda_1 - \lambda_2} e^{\lambda_2 t} \right).$$

Note that $\lambda_1, \lambda_2 < 0$ because $\frac{1}{2m}\sqrt{c^2 - 4mk} < \frac{1}{2m}\sqrt{c^2} = c/(2m)$, so $x(t)$ decays exponentially fast in time, with no oscillations. This is known as the overdamped case, and its behaviour can is shown in Figure 2.13.

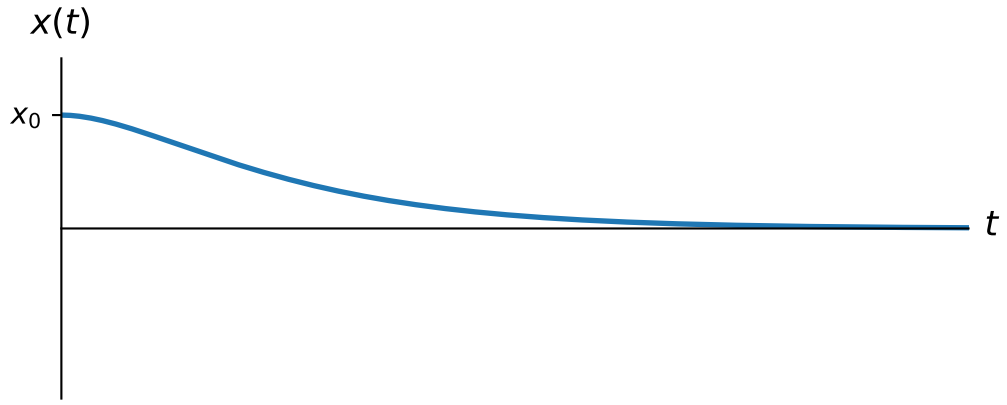


Figure 2.13: **Overdamped** case of a mass-spring-damper system. Note how the system now does not oscillate, and just decays to the

2.8.5.3 Critically damped case

If $c^2 - 4mk = 0$ then we have a repeated eigenvalue $\lambda = -c/(2m)$. Following the arguments in Section 2.8.3 one can work out the solution to the initial value problem Equation 2.31, Equation 2.32:

$$x(t) = x_0 e^{\lambda t}(1 - \lambda t).$$

From Figure 2.14, this behaviour looks similar to the overdamped case. However, the decay is faster than in the overdamped case as the eigenvalue is larger in magnitude. This is known as the critically damped case.

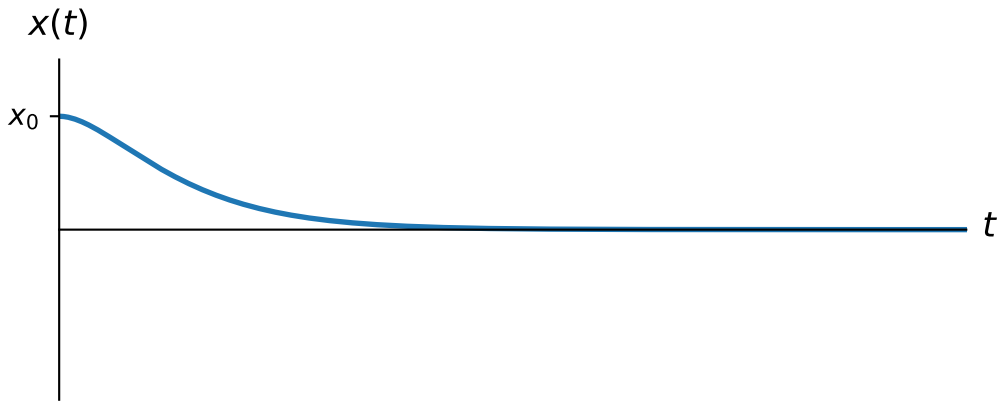


Figure 2.14: **Critically damped** case of a mass-spring-damper system. Note how the system returns to equilibrium quickly as without oscillating.

In many practical applications (e.g. vehicle suspensions or seismic dampers), the critically damped case is the ideal as it allows the system to return to equilibrium as quickly as possible without oscillating.

3 Difference Equations

Aims of the chapter

By the end of this chapter, you should be able to:

- Explain what a difference equation is.
- Solve first order linear difference equations.
- Explain Euler's method to approximate solutions to differential equations.
- Solve second order linear difference equations with constant coefficients.
- Analyse the behaviour of first order autonomous difference equations.
- Draw a cobweb diagram to visualise the behaviour of first order autonomous difference equations.

3.1 Introduction to difference equations

In the previous chapter, we learnt how to determine solutions for (certain types of) ordinary differential equations. Those solutions told us the value of the dependent variable x for any value of the independent variable $t \in [\alpha, \beta]$. However, sometimes we will be interested instead in obtaining an equation relating one time step (e.g. day/hour/year) to the next one.

Example 3.1 (Compound interest)**.** Imagine you want to invest a sum of money p in a bank account that pays a given interest r per year. Let us denote x_n as the amount of money in the account at a given year n. At the beginning (year zero) we have

$$x_0 = p.$$

After one year, the amount of money in the account will be the initial amount plus the interest earned, which is r times the initial amount:

$$x_1 = p(1 + r).$$

After another year, the interest paid will be computed based on the money in the account at the end of the previous year (not the initial amount), so we have

$$x_2 = x_1(1 + r).$$

We could compute the amount of money in the account at any year n by repeating this process, but we can also write a general formula for the amount of money in the account at year n based on the amount of money in the account at year $n - 1$:

$$x_n = x_{n-1}(1 + r).$$

This is an example of a difference equation and, in this chapter, we will learn how to solve these types of equations and analyse their behaviour.

! Important

Definition 3.1 (Difference equation)**.** A difference equation relates the value of a variable at one time step to its value at the previous k time steps (for a given k):

$$x_n = f(x_{n-1}, x_{n-2}, \ldots, x_{n-k}, n) \tag{3.1}$$

where f is some function. In order to be able to solve it, we need to specify the values of the variable at the first k time steps (often known, unsurprisingly, as the initial conditions).

Remark 3.1. The notation for the variables used is completely arbitrary. For example, we could rewrite the definition above as

$$y_m = g(y_{m-1}, y_{m-2}, \ldots, y_{m-k}, m)$$

where g is some function. The only difference is that we have changed the name of the variable from x to y and the index from n to m. Note that we can also shift the index by any constant. For example, the three equation below are equivalent:

$$\begin{aligned} x_n &= x_{n-1}(1 + r), \\ x_{n+1} &= x_n(1 + r), \\ x_{n+73} &= x_{n+72}(1 + r). \end{aligned}$$

As we will repeatedly see throughout this chapter, there are very strong similarities between difference and differential equations, including its properties.[1]

[1] Be careful as there are some key difference, as we will see later. For example, on the stability criterion for autonomous equations.

! Important

Definition 3.2. For a difference equation of the form Equation 3.1, we define:

1. The *order* of a difference equation is the difference between the highest and the lowest indices of x.[2] Note that this is independent of the notation we use: for the equations above, they are all first order. The order also specifies the number of initial conditions we need to specify in order to be able to solve the equation.
2. The difference equation is *autonomous* if the function f does not depend explicitly on the index n.
3. The difference equation is homogeneous if setting all instances of the dependent variable x to zero yields zero, i.e.

$$s(n) = f(0, 0, \dots, 0, n) = 0.$$

Otherwise, the difference equation is *inhomogeneous*.

4. The difference equation is linear if the function f is linear in all instances of the dependent variable x, i.e. if it can be written as

$$f(x_{n-1}, x_{n-2}, \dots, x_{n-k}, n) = s(n) + \sum_{i=1}^{k} a_i(n) x_{n-i},$$

for some given functions $s, a_i : \mathbb{N} \to \mathbb{R}$, $i = 1, \dots, k$.

Example 3.2. The difference equation

$$x_{n+2} - 5x_{n+1} + 3x_n = \cos \frac{n}{63}$$

can be classified as follows:

1. Of order 2, since the highest index is $n + 2$ and the lowest is n.
2. Non-autonomous, since the right-hand side depends explicitly on n.
3. Inhomogeneous, since setting $x_n = 0$ for all n yields $\cos \frac{n}{63} \neq 0$.
4. Linear, since it can be written as a linear combination of the instances of x plus an arbitrary function of n.

Example 3.3. The difference equation

[2] This happens to be the number we defined as k in Definition 3.1.

$$x_{n+7} - \cos x_{n+3} = 3\pi$$

can be classified as follows:

1. Of order 4, since the highest index is $n + 7$ and the lowest is $n + 3$.
2. Autonomous, as none of the terms depend explicitly on n (other than the dependent variable).
3. Inhomogeneous, since setting $x_n = 0$ for all n yields $3\pi \neq 0$.
4. Non-linear, since the term $\cos x_{n+3}$ is not a linear function of x_{n+3}.

3.2 First order linear difference equations

Let us now consider first order difference equations

$$x_{n+1} = f(x_n, n),$$

and, in particular, the linear[3] case

$$x_{n+1} = a(n)x_n + b(n),$$

with a given x_0, where $a, b : \mathbb{N} \to \mathbb{R}$ are given functions.

3.2.1 Autonomous case

Let us constrain first to the autonomous case, where $a(n) = a$ and $b(n) = b$ are constants, so that we have

$$x_{n+1} = ax_n + b. \tag{3.2}$$

[3]Nonlinear difference equations can have extremely complicated solutions, and we will only consider them briefly later, in section Section 3.5.

3.2.1.1 Homogeneous case ($b = 0$)

First we consider the case where $b = 0$, which gives

$$x_{n+1} = ax_n.$$

In order to figure out the solution, we will compute the first few terms of the sequence and try to find a pattern.

$$\begin{aligned} x_1 &= ax_0, \\ x_2 &= ax_1 = a^2x_0, \\ x_3 &= ax_2 = a^3x_0, \\ &\vdots \\ x_n &= a^nx_0. \end{aligned} \tag{3.3}$$

! Important

Proposition 3.1. *The solution to the linear autonomous homogeneous first order difference equation*

$$x_{n+1} = ax_n$$

with a given initial condition x_0 is given by

$$x_n = a^nx_0.$$

Remark 3.2. To formally prove that this is a solution we would need to check that indeed it satisfies Equation 3.2 (quite trivial in this case, but not so much in the upcoming examples).

From the solution, we can see that there is a special case. If $x_0 = 0$, we have that $x_n = 0$ for all $n \in \mathbb{N}$. In this case, if we start at zero we stay zero. This is analogous to a *stationary point* of a differential equation and we will see a more rigorous definition of this later on in Section 3.5. Moreover, we see that if $|a| < 1$, then for any x_0, the solution will tend to zero as $n \to \infty$, while if $|a| > 1$ the solution will tend to infinity. The case where $|a| = 1$ yields some interesting behaviour: if $a = 1$ then $x_n = x_0$ for all n, while if $a = -1$ then $x_n = (-1)^nx_0$ which oscillates between x_0 and $-x_0$. Again, more on this in Section 3.5.

3.2.1.2 Inhomogeneous case ($b \neq 0$)

We now consider the inhomogeneous case, such that

$$x_{n+1} = ax_n + b$$

where a and $b \neq 0$ are constants and for some initial value x_0.

As before, we compute a few terms and infer the solution:

$$\begin{aligned}
x_1 &= ax_0 + b, \\
x_2 &= a(ax_0 + b) + b = a^2x_0 + ab + b, \\
x_3 &= a(a^2x_0 + ab + b) + b = a^3x_0 + a^2b + ab + b, \\
&\vdots \\
x_n &= a^nx_0 + b\sum_{k=0}^{n-1} a^k = \begin{cases} a^nx_0 + b\frac{1-a^n}{1-a} & \text{if } a \neq 1, \\ x_0 + bn & \text{if } a = 1. \end{cases}
\end{aligned}$$

If $a \neq 1$ we can rearrange this to get

$$x_n = a^n\left(x_0 - \frac{b}{1-a}\right) + \frac{b}{1-a}.$$

3.2.2 Non-autonomous case

Finally, we want to consider the non-autonomous case:

$$x_{n+1} = a(n)x_n + b(n),$$

where $a(n)$ and $b(n)$ are functions of n and we have an initial value x_0.

Using the same iterative method as in the autonomous cases, it can be shown that solutions to this equation are of the form

$$x_n = \prod_{k=0}^{n-1} a(k)x_0 + \sum_{k=0}^{n-1} b(k) \prod_{j=k}^{n-1} \frac{a(j)}{a(k)},$$

where $\prod_{k=0}^{n} f(k)$ represents the product from $k = 0$ to $k = n$ of $f(k)$.

3.3 Euler's method

Difference equations are not only interesting in their own right, but they also have a very important application in approximating solutions to differential equations. Consider an ordinary differential equation of the form

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(t, x), \qquad x(0) = x_0. \tag{3.4}$$

As we discussed in the previous chapter, unless $f(t, x)$ is of a very specific form, we will not be able to find an explicit solution $x(t)$ to this equation. In that case, we can help us with a computer.

When a computer draws a graph of a function, it effectively draws a series of dots representing the function. If the dots are dense enough, they will look like a line, but they are still just a series of dots. So, if we want to draw a graph of the solution $x(t)$ to equation (Equation 3.4), we could find a series of dots that approximate the solution $x(t)$ at a discrete set of values of t, rather than the continuous solution. There are multiple ways of doing so, but the simplest one is called the Euler's method.[4]

The key idea is to take small time steps h an approximate the solution at $t + h$ based on the solution at t. Then, we can use a Taylor expansion to relate $x(t + h)$ to $x(t)$:

$$x(t + h) = x(t) + h\dot{x}(t) + \frac{h^2}{2}\ddot{x}(t) + \dots. \tag{3.5}$$

We can approximate this by ignoring all but the first two terms and, using Equation 3.4, we obtain

$$x(t + h) \approx x(t) + hf(t, x(t)). \tag{3.6}$$

Remark 3.3. This is equivalent to assuming that $\dot{x}(t)$ is a constant, which is rarely the case, and can lead to significant errors.[5]

[4] Named after the Swiss mathematician Leonhard Euler (1707-1783) who proposed this method two centuries before electronic computers were invented.

[5] There are improved methods that take into account the fact that $\dot{x}(t)$ is not constant. For example, the *improved Euler's method* assumes that the second derivative is constant, and provides a better approximation. The *Runge-Kutta Method* takes this further by assuming that the fourth derivative is constant, yielding even higher accuracy. However, these methods are very much more complicated to put in to practice, and we will not study them here: those interested should consider taking MA2K4 Numerical Methods and Computing.

Another way to think about this approximation is to consider the definition of the derivative:

$$\dot{x}(t) = \lim_{h \to 0} \frac{x(t+h) - x(t)}{h}.$$

The definition holds in the limit, but we could approximate it by taking a small (but non-zero) value of h. If you replace it in Equation 3.4, you will get the same approximation as above.

We have effectively obtained a difference equation, but let us redefine the notation to make it more apparent. Let us assume that we know the value of $x(t)$ at $t = 0$, so we define $x_0 = x(0)$ and $x_n = x(nh)$, for $n \in \mathbb{N}$. Then, we can advance our solution using Equation 3.6 to get

$$\begin{aligned} x_0 &= x(0) \\ x_1 &= x(h) = x(0+h) = x(0) + h\dot{x}(0) = x_0 + hf(0, x_0) \\ x_2 &= x(2h) = x(h+h) = x(h) + hf(x(h)) = x_1 + hf(h, x_1) \\ &\vdots \\ x_{k+1} &= x_k + hf(kh, x_k) \end{aligned} \tag{3.7}$$

At each step we have everything we need to compute the next value *without* having to know the explicit solution $x(t)$.

> **!** Important
>
> **Algorithm 3.1** (Euler's method)**.** *We can approximate the first order differential equation (Equation 3.4) using the formula*
>
> $$x_{n+1} = x_n + hf(nh, x_n)$$
>
> *where $h \ll 1$ is a small constant time step.*

An example of how the method works is shown in Figure 3.1.

3.4 Second order linear difference equations with constant coefficients

Now we turn our attention to second order linear difference equations with constant coefficients. That is equations of the form

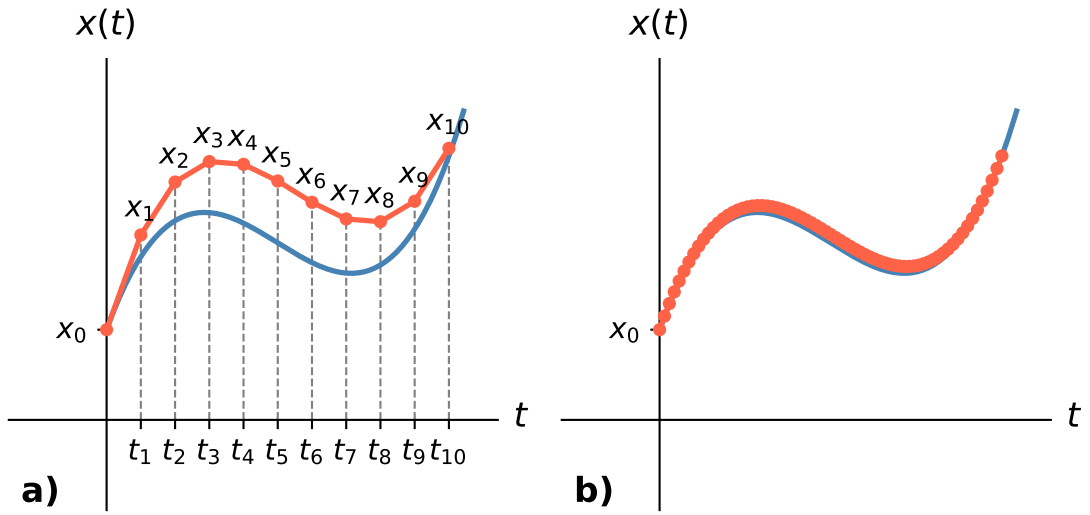


Figure 3.1: Sketch of the Euler's method. The blue curve is the exact solution of a given differential equation, while the red dots represent the approximation obtained using Euler's method. Panel a) shows the case with a larger step size, and panel b) shows the case with a smaller step size, showing the improved accuracy.

$$x_{n+2} + ax_{n+1} + bx_n = f_n$$

We already saw for first order equations that we could obtain solutions of the form of a^n, so we try the analogous approach here.[6] As we discussed earlier, for second order equations we need two initial conditions so, unless otherwise stated, we will assume that x_0 and x_1 are given.

3.4.1 Homogeneous case

In this case, we consider the case where $f_n = 0$, i.e. equations of the form

$$x_{n+2} + ax_{n+1} + bx_n = 0 \tag{3.8}$$

Appealing to previous solutions, we "guess" a solution $x_n = Ak^n$, giving us

$$Ak^{n+2} + Aak^{n+1} + Abk^n = 0$$

[6] Note that the a^n is analogous to e^{at} in the differential equations case. This is one of the differences between difference and differential equations, so make sure to not mix them up.

which is true if

$$Ak^n(k^2 + ak + b) = 0$$

So, either $k = 0$ and $x_n \equiv 0$, which isn't particularly interesting, or alternatively, k is a root of the *auxiliary equation*.

! Important

Definition 3.3 (Auxiliary equation)**.** The auxiliary equation of the second order linear difference equation with constant coefficients (Equation 3.8) is

$$k^2 + ak + b = 0.$$

Then, the suitable values of k are given by the quadratic formula,

$$k = \frac{-a \pm \sqrt{a^2 - 4b}}{2}.$$

As with the differential equations case, we need to consider three distinct cases.

3.4.1.1 Distinct real roots

! Important

In this case both $x_n = Ak_1^n$ and $x_n = Bk_2^n$ are solutions so by linearity the general solution is

$$x_n = Ak_1^n + Bk_2^n,$$

where A and B are constants determined by the initial conditions.

Example 3.4 (Fibonacci's rabbits)**.** Let's now consider a historical example. In 1202 the Italian mathematician Leonardo Fibonacci posed the following problem in his book *Liber Abaci*. He wanted to predict the growth of a population of rabbits based on the following assumptions:

1. We start with one baby rabbit.[7]
2. A rabbit can't give birth until it is 2 years old.
3. A single unisex adult rabbit gives birth to one baby per year.
4. A rabbit never dies.[8]

[7] I am sure you can see some fundamental biological flaw in this assumption, but let us bear with him for now.

[8] At least in the time frame of this problem.

Consider the Fibonacci sequence, given by the difference equation

$$x_{n+2} = x_{n+1} + x_n \quad : \quad x_0 = x_1 = 1 \tag{3.9}$$

Using these assumptions, we can construct Table 3.1, which shows the number of rabbits in each year, split into adults, one-year-olds and newborns.

Table 3.1: Table of Fibonacci's rabbits.

Year	0	1	2	3	4	5	6	...
Adult	0	0	1	1	2	3	5	...
One year	0	1	0	1	1	2	3	...
Newborns	1	0	1	1	2	3	5	...
TOTAL (x_n)	1	1	2	3	5	8	13	...

From the table, we can see that the number of rabbits in year n is equal to the number of rabbits in year $n-1$ plus the number of babies born in year n. But the number of babies in year n is just equal to the number of adults in year n, which is itself the number of rabbits in year $n-2$, since two years later all the rabbits will be adults. This gives us the formula:

$$\underbrace{x_n}_{\text{rabbits in year } n} = \underbrace{x_{n-1}}_{\text{rabbits in year } n-1} + \underbrace{x_{n-2}}_{\text{rabbits in year } n-2 = \text{ adults in year } n}$$

Let us now solve this difference equation. We first rewrite it as

$$x_{n+2} - x_{n+1} - x_n = 0.$$

This gives us the auxiliary equation $k^2 - k - 1 = 0$, with roots $k = \frac{1 \pm \sqrt{5}}{2}$ (where $k = \frac{1+\sqrt{5}}{2}$ is known as the "golden ratio"[9]). Plugging this into our general solution gives

$$x_n = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n.$$

We have the initial conditions $x_0 = x_1 = 1$ which we can substitute into the general solution to give us a system of equations for A and B:

[9]This quantity appears recurrently in art but also in nature.

$$x_0 = A + B = 1,$$
$$x_1 = A\left(\frac{1+\sqrt{5}}{2}\right) + B\left(\frac{1-\sqrt{5}}{2}\right) = 1.$$

Solving them yields

$$A = \frac{1+\sqrt{5}}{2\sqrt{5}}, \quad \text{and} \quad B = \frac{\sqrt{5}-1}{2\sqrt{5}},$$

so the solution to the Fibonacci sequence is

$$x_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^{n+1}.$$

Interestingly, despite of all the $\sqrt{5}$ in the solution, x_n is an integer for all $n \in \mathbb{N}$. This can be shown by induction, but we will not go into that here. What we can easily prove is that the ratio of two consecutive terms in the Fibonacci sequence tends to the golden ratio as $n \to \infty$:

$$\frac{x_n}{x_{n-1}} = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}}{\left(\frac{1+\sqrt{5}}{2}\right)^{n} - \left(\frac{1-\sqrt{5}}{2}\right)^{n}}.$$

Taking the limit as $n \to \infty$, and noting that $\left|\frac{1-\sqrt{5}}{2}\right| < 1$, gives

$$\lim_{n\to\infty} \frac{x_n}{x_{n-1}} = \lim_{n\to\infty} \frac{\left(\frac{1+\sqrt{5}}{2}\right)^{n+1}}{\left(\frac{1+\sqrt{5}}{2}\right)^{n}} = \frac{1+\sqrt{5}}{2}.$$

3.4.1.2 Complex roots

> **!** Important
>
> As it happened with differential equations, with complex roots we could just plough ahead and write the solution in terms of complex numbers, but given that the solution must be real it is more convenient to write the roots in the form $k_\pm = re^{\pm i\theta}$, where $r^2 = p^2 + q^2$ and $\theta = \arctan\frac{q}{p}$. Then, the general solution is
>
> $$x_n = r^n\left(A\cos n\theta + B\sin n\theta\right) \tag{3.10}$$

We will not go into detail about this, but it follows by considering

$$C\left(re^{i\theta}\right)^n + \overline{C}\left(re^{-i\theta}\right)^n$$

where $\overline{C}$ is the complex conjugate of C (with C chosen so that the solution is real), and by noting that $re^{i\theta} = r(\cos\theta + i\sin\theta)$ and that $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ (de Moivre's theorem).

Example 3.5. We want to solve

$$x_{n+2} - 2x_{n+1} + 2x_n = 0.$$

The auxiliary equation is

$$k^2 - 2k + 2 = 0, \qquad \Longrightarrow \qquad k = 1 \pm i = \sqrt{2}e^{\pm\frac{i\pi}{4}}$$

so, plugging this into Equation 3.10, we get

$$x_n = (\sqrt{2})^n\left(A\cos\left(\frac{n\pi}{4}\right) + B\sin\left(\frac{n\pi}{4}\right)\right).$$

3.4.1.3 Repeated real root

Once again, we look to differential equations for inspiration. We need a second solution, so we try $x_n = nk^n$ and, as we will show below, this is indeed a solution.

> **!** Important
>
> Therefore, the general solution is
>
> $$x_n = Ak^n + Bnk^n. \tag{3.11}$$

We know that Ak^n is a solution, so we just need to show that Bnk^n is also a solution. In order to do so, we just substitute into the difference equation. We obtain

$$B(n+2)k^{n+2} + aB(n+1)k^{n+1} + bBnk^n = 0$$

which, rearranged, gives

$$Bnk^n\left(k^2 + ak + b\right) + Bk^n\left(2k^2 + ak\right) = 0.$$

The factor in brackets in the first term is zero as it satisfies the auxiliary equation. To show that the factor in brackets in the second term is zero, note that when we have a repeated root the in the auxiliary equation it must be of the form $k = -\frac{a}{2}$. Therefore, the identity is satisfied and Bnk^n is indeed a solution and Equation 3.11 is the general solution.

Example 3.6. Find a general solution of

$$x_n - 6x_{n-1} + 9x_{n-2} = 0.$$

The auxiliary equation is just

$$k^2 - 6k + 9 = 0, \qquad \Longrightarrow \qquad (k-3)^2 = 0, \qquad \Longrightarrow \qquad k = 3.$$

So, from Equation 3.11, the general solution is just

$$x_n = (A + Bn)3^n.$$

3.4.2 Inhomogeneous Case

> **!** Important
>
> In this subsection we will consider difference equations of the form
>
> $$x_{n+2} + ax_{n+1} + bx_n = f_n. \tag{3.12}$$
>
> Since Equation 3.12 is linear, we start by finding the solution to the homogeneous case, the *complementary function*, e.g. $x_n = Ak_1^n + Bk_2^n$.
> We then find a *particular solution*[10] that gives f_n on the right-hand side. Then, by linearity, the general solution is simply the complementary function and the particular solution added together.

As it happened with differential equations, there is no magic recipe to find a particular solution. In this subsection we will study two particular forms of f_n that are common in applications, and we will see how to find a particular solution for them.

[10] Obviously *particular integral* would make no sense in this context since there are no differentials or integrals.

3.4.2.1 f_n is a polynomial in n

> **!** Important
>
> When f_n is a polynomial in n, we try as a particular solution a polynomial of the same degree. For example, if $f_n = n^2$, we try $x_n = An^2 + Bn + C$.

Note that, if f_n is a polynomial of degree zero (i.e. a constant), the method is consistent with the results found in Section 3.2.1.2. In that case, the particular solution is also a constant.

Example 3.7. Solve

$$x_n - x_{n-1} - 6x_{n-2} = -36n. \tag{3.13}$$

We consider first the homogeneous case $x_n - x_{n-1} - 6x_{n-2} = 0$ to find the complementary function. The auxiliary equation yields

$$k^2 - k - 6 = 0, \qquad \Longrightarrow \qquad (k-3)(k+2) = 0, \qquad \Longrightarrow \qquad k = 3, -2,$$

which gives the complementary function

$$x_n = A \cdot 3^n + B(-2)^n. \tag{3.14}$$

For the particular solution, since f_n is a polynomial of degree 1, we try the most general first-degree polynomial possible, i.e. $x_n = Cn + D$, and substitute this into Equation 3.13:

$$Cn + D - C(n-1) - D - 6C(n-2) - 6D = -36n$$

giving $C = 6, D = 13$, so the particular solution is $x_n = 6n + 13$.

Combining this with Equation 3.14 gives us the general solution

$$x_n = A3^n + B(-2)^n + 6n + 13.$$

3.4.2.2 f_n is an exponential, c^n

This is the "equivalent" case to that of e^{ct} in differential equations.[11] Once again, we have to be careful when c^n is also part of the complementary function but, once again, the ideas from differential equations are carried over.

[11] And, again, do not mix both cases up!

! Important

So, for the particular solution when $f_n = c^n$, try:

$$\begin{aligned} x_n &= Dc^n && \text{if } c \text{ is not a root of the auxiliary equation,} \\ x_n &= Dnc^n && \text{if } c \text{ is a root,} \\ x_n &= Dn^2c^n && \text{if } c \text{ is a repeated root.} \end{aligned}$$

We will now illustrate it with two examples.

Example 3.8. Find the general solution of

$$x_{n+2} + x_{n+1} - 6x_n = 12(-2)^n.$$

First we find the complementary function. Our auxiliary equation is $k^2 + k - 6 = 0$, giving roots $k = 2, -3$ and so we have our complementary function

$$x_n = A \cdot 2^n + B(-3)^n.$$

We have that $(-2)^n$ is not a solution of the homogeneous case, so we try

$$x_n = D(-2)^n$$

as a particular solution. Substituting this in gives

$$D(-2)^{n+2} + D(-2)^{n+1} - 6D(-2)^n = 12(-2)^n,$$

which yields $D = 3$. Bringing everything together gives us our general solution:

$$x_n = A \cdot 2^n + B(-3)^n - 3(-2)^n.$$

Example 3.9. Solve

$$x_{n+2} + x_{n+1} - 6x_n = 40 \cdot 2^n.$$

As in Example 3.8, the complementary function is $x_n = A \cdot 2^n + B(-3)^n$. However, this time 2^n *is* a solution of the homogeneous case, so we try

$$x_n = Dn2^n$$

as a particular solution. Substituting this into the equation gives

$$D(n+2)2^{n+2} + D(n+1)2^{n+1} - 6Dn2^n = 40 \cdot 2^n$$

so $D = 4$, giving us our general solution

$$x_n = A \cdot 2^n + B(-3)^n + 4n \cdot 2^n.$$

3.5 First order (autonomous) nonlinear difference equations

Like with differential equations, solving nonlinear difference equations (even first order ones) can be extremely difficult. In these situations we need to find alternative methods to understand how the solution behaves. Again, we will turn our attention to first order autonomous equations which, for the case of difference equations, are of the form

$$x_{n+1} = f(x_n) \tag{3.15}$$

given some x_0. Remember that this is *autonomous* because f is a function of x_n, but not of n explicitly, i.e. it does not involve terms like n^2 or $e^n x_n$.

Calculating our sequence for Equation 3.15 we get

$$\begin{aligned} x_1 &= f(x_0), \\ x_2 &= f(x_1) = f(f(x_0)) = f^2(x_0), \\ &\vdots \\ x_n &= f^n(x_0). \end{aligned}$$

> **!** Important
>
> Here for shorthand purposes we have used the notation $f^2(x) = f(f(x))$, or more generally,
>
> $$\underbrace{f(f(\dots f(f(}_{n} x)) \dots)) = f^n(x)$$
>
> Note that $f^n(x) \neq (f(x))^n$, and also that $f^n(x) \neq f^{(n)}(x) = \frac{d^n}{dx^n} f(x)$.

If $f(x_n)$ is *nonlinear*, then (most of the time) we *cannot* write an explicit solution of the form

$$x_n = g(n)$$

for some function g. Therefore, like in Section 2.6, we will have to resort to qualitative methods to understand the behaviour of the solution. In the particular case of difference equations, as we will see, the solution can show much more interesting behaviours.

We already mentioned *stationary points* of difference equations in section Section 3.2, but we now give a more rigorous definition.

> ! Important
>
> **Definition 3.4** (Stationary point)**.** A *stationary point* of the difference equation
>
> $$x_{n+1} = f(x_n)$$
>
> is a point x_* such that $f(x_*) = x_*$. This means that if $x_n = x_*$, then $x_{n+1} = f(x_n) = x_*$ for all n.

Remark 3.4. Note that the definition of stationary point is different from the one for differential equations (Definition 3.4). In that case we were looking for points x^* satisfying $f(x^*) = 0$, while here we are looking for points x_* satisfying $f(x_*) = x_*$. This is a common mistake in exams, so make sure you understand the difference.

In Section 3.2 we also saw that the solution to the linear difference equation could be stable or unstable depending on the value of the coefficient: stable if $|a| < 1$ and unstable if $|a| > 1$. The question is: does this generalise to nonlinear equations? Fortunately for us, it does.

> ! Important
>
> **Lemma 3.1.** *If x_* is a stationary point of the autonomous difference equation*
>
> $$x_{n+1} = f(x_n),$$
>
> *then the stationary point is*
>
> 1. Stable *if* $|f'(x_*)| < 1$.
> 2. Unstable *if* $|f'(x_*)| > 1$.

Proof. Consider that we start at a point near the stationary point, i.e. $x_0 = x_* + h$ with $|h| \ll 1$. Then, taking a Taylor expansion of f around x_* we have

$$\begin{aligned} x_1 &= f(x_0) \\ &= f(x_* + h) \\ &= f(x_*) + hf'(x_*) + \frac{h^2}{2} f''(x_*) + \dots . \end{aligned}$$

For sufficiently small h, and using the fact that $f(x_*) = x_*$, we approximate x_1 by

$$x_1 \approx x_* + hf'(x_*).$$

Let us carefully consider how the value of x_n has changed after one iteration. If $|f'(x_*)| < 1$, then we can rearrange the expression above into $|x_1 - x_*| < |h| = |x_0 - x_*|$, i.e. we have moved closer to x_*. More formally, one could show that if $|f'(x_*)| < 1$, then $f^n(x_* + h) \to x_*$ as $n \to \infty$. This means that if $ |f'(x_)/ < 1 $ the solution tends towards the stationary point, so x_* is* stable. *Conversely if $|f'(x_*)| > 1$ then x_* is* unstable*.

Remark 3.5. Note that the sequence $x_{n+1} = f(x_n)$ need *not* be monotonic, hence why we had to work with absolute values. The sequence could converge to (or diverge from) the stationary point by oscillating around it. This is another fundamental difference between difference and differential equations, and it is one of the reasons why difference equations can show more complicated behaviour than differential equations (as we will see next).

□

3.6 The logistic equation & chaos

Let us now consider a discrete version of the logistic equation,[12] often called the logistic map:

$$x_{n+1} = \lambda x_n(1 - x_n),$$

where $\lambda > 0$ represents the growth rate. We now look for its stationary points, i.e. x_* such that

$$x_* = \lambda x_*(1 - x_*), \qquad \implies \qquad \lambda x_*^2 + (1 - \lambda)x_* = 0.$$

This yields two stationary points:

[12] This version is fundamentally different to that in Example 2.13, that is, it is not what you would get if you simply discretised the differential equation, say with the Euler method.

$$x_* = 0, \quad \text{and} \quad x_* = \frac{\lambda - 1}{\lambda}.$$

Note that, as x_n represents a population, we are only interested in the case $x_n \geq 0$, so the second stationary point is only physically relevant when $\lambda \geq 1$.

Since $f(x) = \lambda x(1 - x)$, the stability of the stationary point will be determined by

$$f'(x) = \lambda(1 - 2x).$$

Consider first $x_* = 0$. Then $f'(0) = \lambda$, so $x_* = 0$ is an unstable stationary point if $\lambda > 1$, while if $\lambda < 1$ then our growth rate for small x_n will kill off the population and $x_* = 0$ is stable. On the other hand, the stationary point $x_* = (\lambda - 1)/\lambda$ yields $f'(\frac{\lambda-1}{\lambda}) = 2 - \lambda$. So x_* is stable if $|2 - \lambda| < 1$, that is if $1 < \lambda < 3$.

But what happens if $\lambda > 3$?[13] In that case we get $f'(\frac{\lambda-1}{\lambda}) \leq -1$ and both stationary points are unstable. This is a breaking difference with the differential equation case. To illustrate this, consider Figure 3.2, which sketches a phase line for the logistic map when $\lambda > 3$.

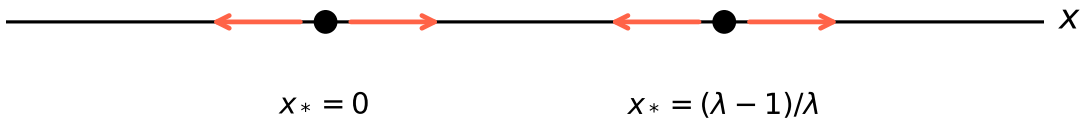


Figure 3.2: Sketch of a phase line for the logistic map when $\lambda > 3$. Both stationary points are unstable, so a nearby solution will be pushed away from them. Note that this is for purely illustrative purposes in the comparison with the differential equation case, in the next section we will see a better way of studying the logistic map.

Both stationary points are unstable, therefore a nearby solution will be pushed away from them. In the case of a differential equation, because we must move continuously, this would mean that there should be a stable stationary point in between where these solutions could converge too.[14] However, in the case of a difference equation, we can jump over stationary points, which enables us to only have two unstable stationary points. But it is not that these solutions will diverge to infinity. In fact, things are about to become a lot more complicated.

[13] Remember this stationary point is only applicable if $\lambda \geq 1$.
[14] Or that we made a mistake in our calculations.

Remark 3.6. Our analysis above did not determine what happens when $\lambda = 1$ or $\lambda = 3$. In these cases, the derivative criterion does not help us and one needs to study the equation in more detail.

If $\lambda = 1$, then both stationary points overlap at $x_* = 0$. If we start with $x_0 > 0$, then $x_1 = x_0(1 - x_0) < x_0$, and so the sequence will decrease monotonically to 0. As x_n must be non-negative, we do not need to study the other case, but, if you are curious, you would find that starting at a negative point the solution would be pushed towards $-\infty$.

If $\lambda = 3$, the stationary point $x_* = 0$ is unstable, and we ony need to carefully study the other stationary point $x_* = 2/3$. In that case, we have $x_1 = 3x_n(1 - x_n)$ and, if you compute a few iterations (you really want to use a computer here), you will see that the solution converges to the stationary point but extremely slowly. In fact, it is often that the case that at the threshold of stability (effectively at a bifurcation point) the solution will do extremely weird things, so it is not always trivial to assess what is going on.

3.6.1 Cobweb diagrams

We saw that phase lines were useful when studying differential equations, but these no longer make sense for difference equations because x_n is a sequence of numbers rather than a continuous function. However, graphical representations are a great way to get a feeling for how solutions of an equation behave.

For difference equations we can use *cobweb diagrams*, which are constructed by plotting the graphs of $x_{n+1} = x_n$ and $x_{n+1} = f(x_n)$ on the same set of axes (with x_n on the horizontal axis and x_{n+1} on the vertical axis). The two graphs cross where $x_{n+1} = f(x_n) = x_n$, i.e. at the stationary points.

> ! Important
>
> **Algorithm 3.2.** *The procedure for drawing a cobweb diagram is as follows:*
>
> 0. *Draw the graphs of $x_{n+1} = f(x_n)$ and $x_{n+1} = x_n$ on the same axes.*
> 1. *Start at x_0 on the x_n axis.*
> 2. *Go up to the graph of f and across to the x_{n+1} axis to get x_1.*
> 3. *Draw a line from x_1 to graph of $x_n = x_{n+1}$, then down to x_n axis.*
> 4. *Repeat from item 2.*
>
> *This method is illustrated in Figure 3.3.*

Cobweb diagrams can be drawn by hand but, as we shall see, accuracy is vital so they are best created by computer software such as MATLAB or Python (as I used here).

To see the behaviour of the logistic equation in detail, we will consider a number of examples.

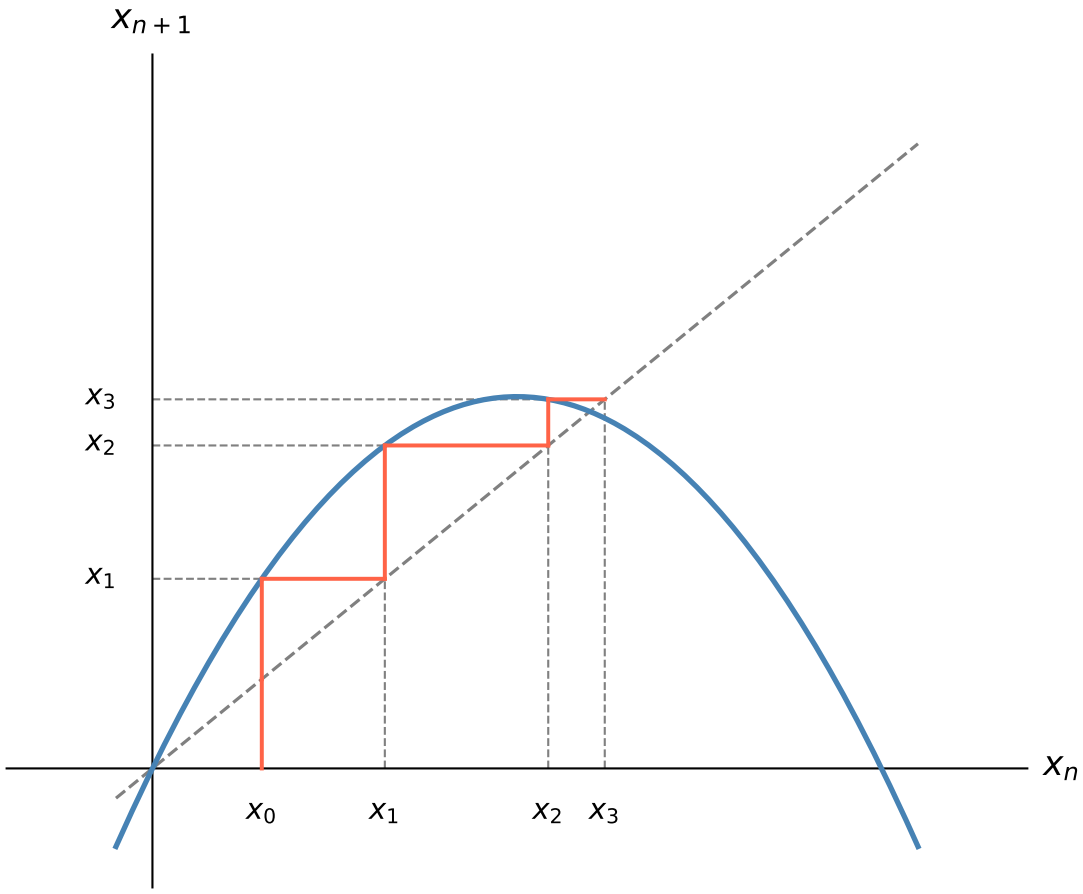


Figure 3.3: Sketch of the method to produce a cobweb diagram. The blue line is $f(x_n)$ while the grey dashed line is $x_{n+1} = x_n$. The red lines show the method to find x_1, x_2, and x_3 from an initial value of x_0.

3.6.1.1 Stable example: $\lambda = 2.5$

We will now produce cobweb diagrams for various values of λ. The first case we will consider is $\lambda = 2.5$, shown in Figure 3.4, which is a stable case. In this case, we simply spiral in towards the stationary point $x_* = \frac{\lambda-1}{\lambda}$.

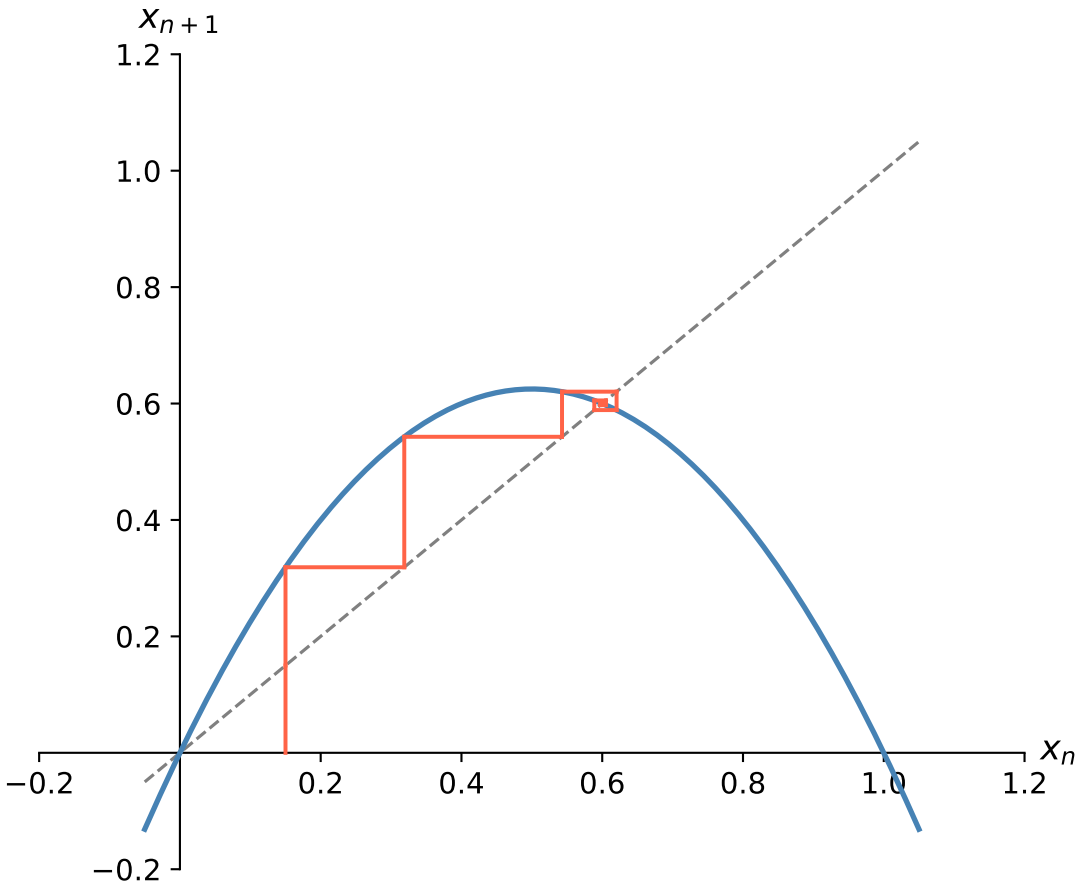


Figure 3.4: The logistic equation with $\lambda = 2.5$, with initial value $x_0 = 0.15$, iterated 50 times. As we can see, the solution simply spirals into the stationary point $x_* = \frac{\lambda-1}{\lambda} = 0.6$.

3.6.1.2 Period two orbit: $\lambda = 3.2$

Now the stationary point $x_* = \frac{\lambda-1}{\lambda}$ is *unstable*, with the cobweb diagram plotted in Figure 3.5.

As we can see, when $\lambda = 3.2$, the solution is attracted to a "period two orbit", i.e. two alternating points; each repeated every 2 iterations. Even though we concluded that the

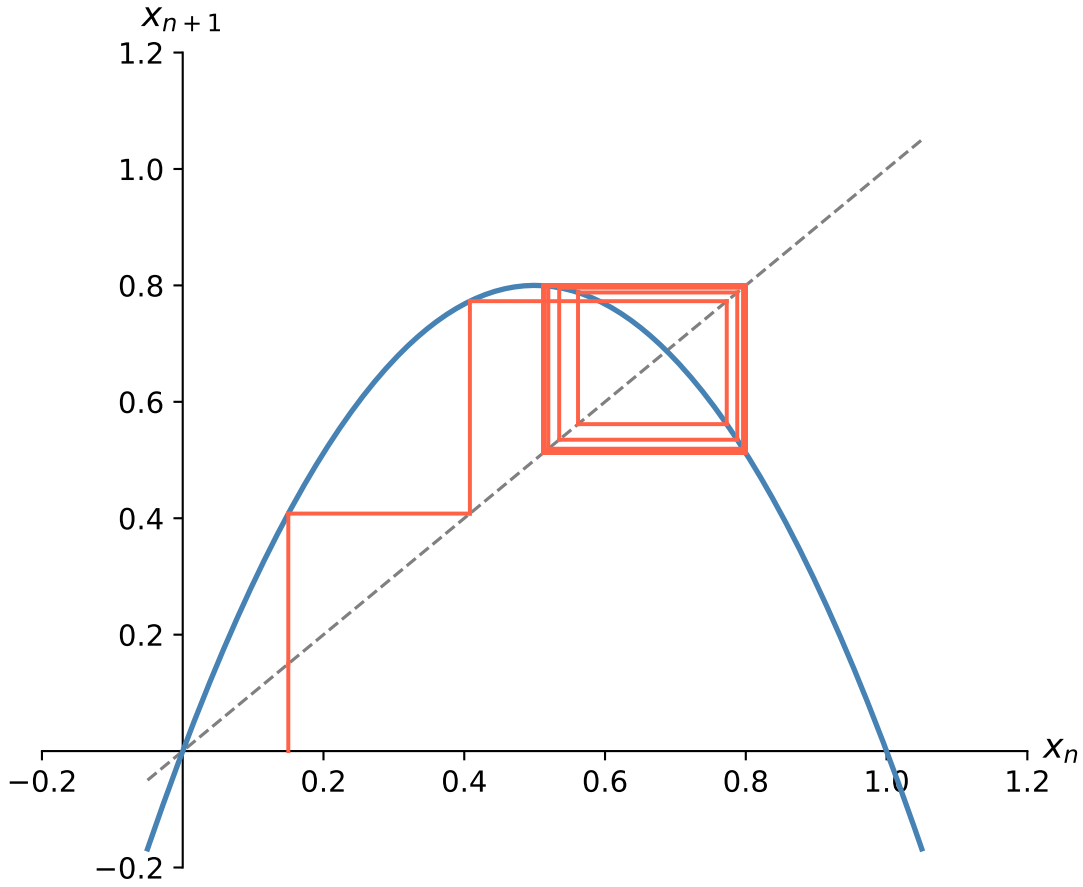


Figure 3.5: The logistic equation with $\lambda = 3.2$, with initial value $x_0 = 0.15$, iterated 50 times. Here the solution settles down to oscillate with period 2.

stationary point is unstable, in some sense, it feels like it is stable as we get attracted towards the orbit and do not go away from it. In fact we can show that each point x_* of the orbit is a stationary point of $f^2(x) = f(f(x))$, i.e. $f^2(x_*) = x_*$.

For $f(x) = \lambda x_n(1 - x_n)$, we have

$$f^2(x_n) = \lambda\left[\lambda x_n(1 - x_n)\right]\left(1 - \left[\lambda x_n(1 - x_n)\right]\right).$$

To find values of x_* we now need to solve a quartic equation. The formula for this exists, but is messy! However, we already know two of the factors: we know that $x_* = 0$ and $x_* = \frac{\lambda-1}{\lambda}$ will solve the quartic, since if x_* solves $f(x_*) = x_*$, then

$$f^2(x_*) = f(f(x_*)) = f(x_*) = x_*.$$

So we divide the quartic by $x_n(x_n - \frac{\lambda-1}{\lambda})$, giving

$$\lambda x_n^2 - (1 + \lambda)x_n + \left(1 + \frac{1}{\lambda}\right) = 0,$$

which has roots

$$x_\pm = \frac{(1 + \lambda) \pm \sqrt{(\lambda + 1)(\lambda - 3)}}{2\lambda}.$$

These exist provided $\lambda > 3$, which is consistent with our previous findings, and for our particular value of $\lambda = 3.2$ they are

$$x_+ \approx 0.80, \qquad x_- \approx 0.51,$$

as observed in Figure 3.5.

Even though we will not go into detail here, $\lambda = 3$ is a bifurcation point where the stability of the stationary point $x_* = \frac{\lambda-1}{\lambda}$ changes and new solutions to $f^2(x) = x$ are created. These points are stable stationary points of $f^2(x_n)$ if $3 < \lambda < 1 + \sqrt{6} \approx 3.45$. If we call the two points x_+ and x_-, then we find that $f(x_+) = x_-$ and $f(x_-) = x_+$, and we have a period two orbit. However, beyond $\lambda \approx 3.45$, these stationary points become unstable.

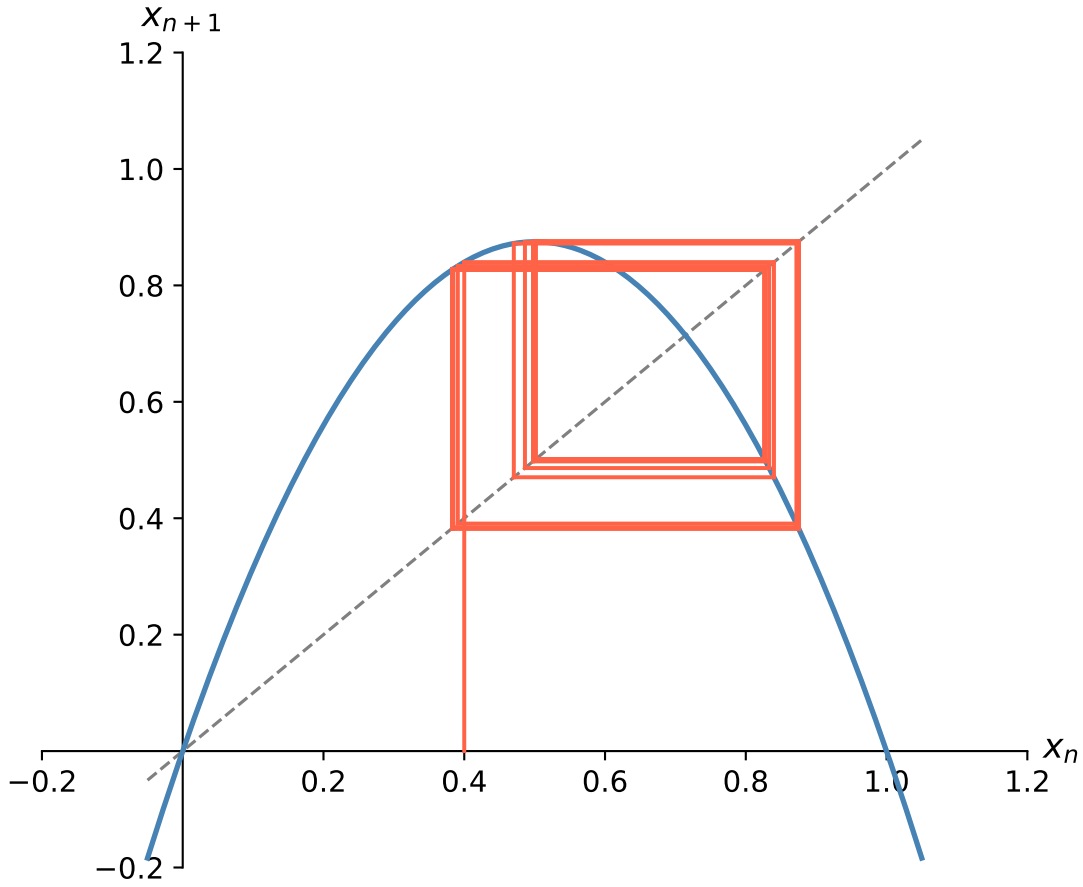


Figure 3.6: The logistic equation with $\lambda = 3.5$, with initial value $x_0 = 0.4$, iterated 50 times. Here the solution settles down to oscillate with period 4.

3.6.1.3 Period four orbit: $\lambda = 3.5$

The cobweb diagram appears in Figure Figure 3.6. In this case we no longer have a period two orbit, but rather a period four orbit, in which $f^4(x_*) = x_*$.

If we then increase λ further in small increments, we get periods of 8, 16, 32, etc. This is called a period-doubling cascade.[15] Beyond $\lambda \approx 3.57$ the solutions we observe are apparently random. In fact it is an example of chaos (see Figure 3.7).[16]

! Important

Definition 3.5 (Period k orbits)**.** Period k orbits of an autonomous first order difference equation

$$x_{n+1} = f(x_n)$$

are given by solutions of

$$f^k(x_*) = x_*$$

which are not either stationary points or belonging to a period j orbit where $j < k$.

[15] The Wikipedia article about the logistic map has a very cool animation on how the cobweb diagram evolves as we increase λ.

[16] We deliberately do not provide a formal definition of chaos as it is complicated. But from the figure you get the idea that the behavious is "messy" at the very least.

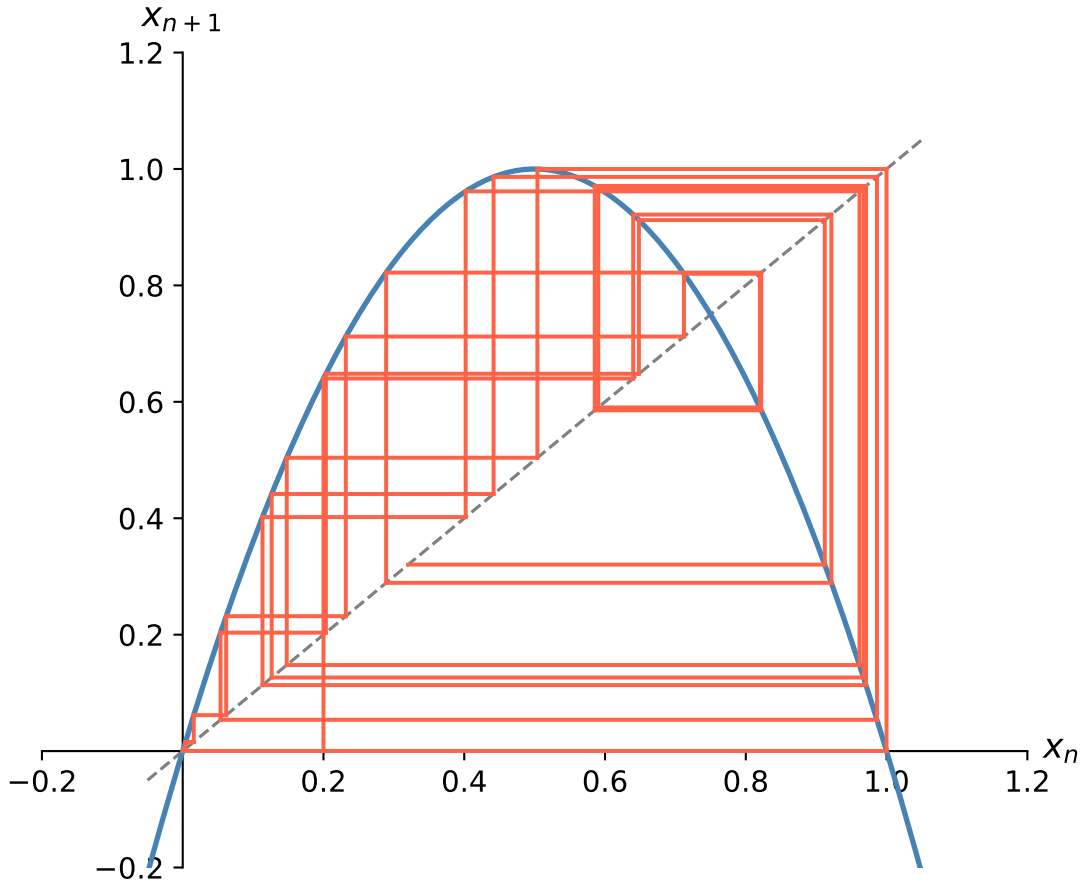


Figure 3.7: The logistic equation with $\lambda = 4$, with initial value $x_0 = 0.2$, iterated 30 times.

4 Mathematical Modelling

Aims of the chapter

By the end of this chapter, you should be able to:

- Explain what a mathematical model is.
- Describe the modelling cycle.
- Classify mathematical problems as forward, inverse, or control problems.

In the previous two chapters we have introduced differential equations and discussed methods for their solution. In some of the examples, however, we have seen that the equations were motivated from a real-world phenomenon and that we used the solutions to better understand such phenomena. In some cases, we even had to formulate the equations ourselves, based on observations and assumptions about the phenomenon (think of the mixing problem, Example 2.7). This is the essence of *mathematical modelling* (or modelling for short), which we will discuss in this chapter.[1]

4.1 What is a model?

Let us start with a definition.

Important

Definition 4.1 (Mathematical model)**.** A mathematical model is an **abstract representation** of a concrete **system** using **mathematical concepts** and language. It is a representation of the system that captures its essential characteristics and behaviour, allowing for analysis, prediction, and understanding of the system's dynamics.

Note that I have bolded the three key concepts in the definition. It is an abstract representation because it is not the system itself, but an idealised version of it. It is always related to a system that it tries to represent. And we use mathematical concepts as a tool to describe and analyse this system.

[1] In fact, we often refer to the equations as *models* when they are used to describe a phenomenon. In the previous chapters I deliberately avoided this terminology to avoid confusion.

Generally, the purpose of mathematical modelling is gaining insight and expanding knowledge by describing behaviours and trends in real-world problems, and by making predictions about these. As we will see, in the modelling process, assumptions are made that involve simplification and reduction to essential features, characteristics, or quantities of interest. Suitable variables and parameters for these are identified and then related in mathematical equations. This process is guided by some principles and systematic procedures, and it hopefully leads to equations that can be solved and further analysed. In fact, the choices made to formulate the model are informed by available tools, methodology, and techniques for its study. Before we discuss the modelling process in more detail, let us look at an example (or rather revisit it).

Example 4.1 (Mass-spring-damper: derivation revisited)**.** Let us consider again the mass-spring-damper system we introduced in Section 2.8.5. We outlined the physical principles that lead to the differential equation, but looks in detail into the various assumptions:

1. The only forces acting on the mass are the spring force and the damping force (i.e. we neglect gravity, friction, and any other forces).
2. The spring can be assumed to be linear with a constant k, i.e. the spring force is proportional to the displacement of the mass from its equilibrium position.[2] This is usually a good approximation for small displacements, but for larger displacements the spring may behave nonlinearly.
3. The damper can be assumed to be linear as well with a constant c, i.e. the damping force is proportional to the velocity of the mass. Again, this is usually a good approximation for small velocities, but for larger velocities the damper may behave nonlinearly. We also assume that the damping does not depend on the position of the mass (i.e. we will not hit the wall of the damper[3]).

We denote the position of the mass as $x(t)$ and, for simplicity, we define the origin to be the position where the spring is "relaxed" (i.e. when it exerts no force on the mass). Then, as we saw in Section 2.8.5, we can use our assumptions to write the forces that both the spring and the damper exert on the mass:

$$F_k = -kx(t), \quad F_c = -c\frac{\mathrm{d}}{\mathrm{d}t}x(t).$$

The negative signs denote that the forces oppose the displacement and velocity, respectively. In order to build the model, we also need to relate the forces to the position of the mass (or, in this case, its acceleration). We use Newton's second law to relate the forces to the acceleration of the mass. This would not be considered an assumption, but rather a principle that is well established and experimentally verified. It states that the

[2] This is known as Hooke's law, named after Robert Hooke, and English polymath from the 17th century. Trivia fact: Hooke coined the term "cell" for the basic unit of biological tissue.

[3] Dampers usually contain a piston in a viscous fluid.

sum of the forces acting on an object is equal to the mass of the object multiplied by its acceleration and, given that we assumed the only forces acting on the mass are the spring and damping forces, we have

$$m\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) = F_k + F_c = -kx(t) - c\frac{\mathrm{d}}{\mathrm{d}t}x(t).$$

This example illustrates the modelling process. In this case we were quite lucky that people like Newton and Hooke did most of the heavy lifting for us, and we could use well established principles to formulate the model. However, in many other applications, we have to make more assumptions and simplifications, and we have to be careful that these are reasonable and do not lead to a model that is too far from reality as we will see in Section 4.3

4.2 Mathematical problems

Models are just a tool to solve a *mathematical problem*, often also formulated in the form of a *mathematical question*. Whilst a mathematical model lists variables and relations between these, the problem states which variable are assume known and which variable one wishes to identify or find. In short, the mathematical problem tells us what we want to achieve with the model. We then apply mathematical methods, tools, and techniques to that problem in order to solve or analyse it and, from it, make predictions and draw other useful conclusions from the system we want to understand.

! Important

There are three broad types of problems:

1. *Forward problems*, or *evaluation questions*. Given all needed information about a process or phenomenon, an we quantitatively predict its other properties and how it will function?

   Examples: What is the maximum attainable speed of this car? How quickly will this disease spread through the population of this city?

2. *Inverse problems*, or *detection questions*. If some information about a phenomenon is not directly available, can it be "reverse engineered" from observations?

   Examples: How can we use data from CT scans to estimate the location of a tumour? Can we determine the damping of an oscillating pendulum from the decay of its time series height data?

3. *Control problems*, or *design questions*: Can we create a solution that best meets a specified goal?

> Examples: What shape is best for a paper airplane to make it fly furthest? How should a pill be coated to release its drug at a constant rate over an entire day?

Example 4.2 (from April exam 25-26)**.** Let us now ground these concepts by looking at the mass-spring-damper system again. Consider the following real-world scenarios; are they a forward, inverse, or control problem?

1. **Predict the position of the mass at a given time.** This is a forward problem, as we need to determine the variable $x(t)$ at a given time t from the model (as it is not stated otherwise, we assume that we are given the initial conditions and parameters).
2. **Decide which damper we need to achieve critical (i.e. optimal) damping.** This is an control problem, as we aim to design a system that meets a specific goal, in this case achieving critical damping.
3. **Compute the time it will take for the oscillations to decay below a certain threshold.** This is a forward problem, as we need to determine the variable $x(t)$, similarly to the first scenario. Despite sounding like an inverse problem, as we are given a threshold and need to find the time, we are still solving for $x(t)$ and then determining when it crosses the threshold, so it is a forward problem.
4. **Calculate the spring constant from experimental data.** This is an inverse problem, as we are aiming to determine a parameter of the system (the spring constant) from measurements.
5. **Determine the forcing necessary to keep the system oscillating at a given frequency and amplitude.** This is a control problem, as we are aiming to influence the system to achieve a specific behaviour.

4.3 The modelling cycle

Now that we have introduced the concepts of a mathematical model and a mathematical problem, let us consider how one goes about creating them for a real-world phenomenon. To do so, we introduce the concept of a *modelling cycle*, shown in Figure 4.1, which is a systematic approach to mathematical modelling. It consists of several stages that are iteratively applied to refine the model and ensure its accuracy and relevance.

We start on with a real-world phenomenon or process that we want to study. In order to do so, we formulate a mathematical model that captures the essential characteristics of the phenomenon. This involves identifying suitable variables and parameters, and establishing relationships between them through equations, making a series of assumptions in the process. This model will be the core of the mathematical problem we want to study. Then, we study the mathematical problem using suitable mathematical tools, which will strongly depend on the nature of the model and the problem. The goal is to obtain

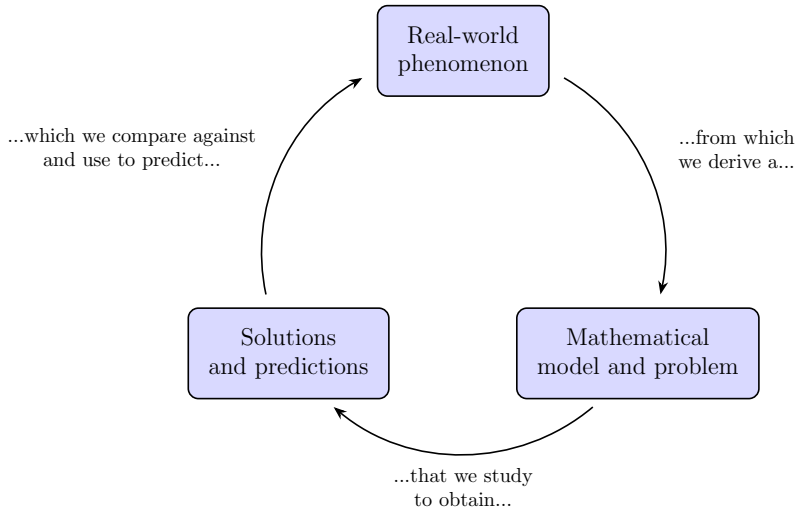


Figure 4.1: Illustration of the modelling cycle.

insight on the problem either by computing analytical solutions or, if it is not possible, using alternative methods, like approximating the solution numerically (e.g. Section 3.3) or studying its qualitative behaviour (e.g. Section 2.6). From the solution we can then draw conclusions and make predictions about the real-world phenomenon. Finally, we close the cycle by validating these conclusions and predictions against observations and measurements of the real-world problem. If the outcome is not satisfactory, we may need to refine the model by revisiting the assumptions and relationships made in the formulation stage, and then repeat the cycle until we are happy with the outcome.

The bulk of this module focuses on the study of the equations that arise from the modelling process, so let us now briefly discuss the other two steps of the modelling cycle, namely model formulation and model validation.

4.3.1 Model formulation

The first step is to formulate a mathematical model that captures the essential characteristics of the phenomenon we are interested in. In Example 4.1 we saw a real (yet quite simple) example, but often it is a lot more complicated than that. In this section we will consider two other examples, one arising from chemistry and one from electrical engineering, to illustrate the process of model formulation. We will also see a more complicated example about modelling a disease outbreak in the next section, when we discuss the validation of models. But, before that, let us expand on the general principles of model formulation.

The model formulation step involves transforming a real-world phenomenon into a mathematical representation. Inevitably, no matter how complicated our model ends

up being, this representation will be a simplified version of the real-world. The process involves identifying the suitable *variables* that quantify the aspects of the phenomenon that we are interested in (and thus deciding which ones are we happy to ignore).

Once we have defined the variables, we need to establish relationships between them, which is usually done by making a series of assumptions. These assumptions are based on our understanding of the phenomenon and may involve simplifications, idealisations, or approximations. The goal is to capture the essential characteristics of the phenomenon while keeping the model tractable. Sometimes these relationships are guided by well-established principles, such as Newton's laws in Example 4.1, but in many cases they are based on empirical observations and may be more ad-hoc, for example when in Example 2.14 we assumed that the fishing rate is proportional to the population. The resulting relationships are expressed in mathematical form: they are often in the form of equations (differential or difference for the models in this module) but they could be as simple as an explicit empirical formula, or as complicated as a coupled system of partial differential equations. In any case, the relationships are expressed in mathematical terms.

> **Note**
>
> The examples below are purely to illustrate the model formulation process. They draw on principles from other areas and which you are not expected to know. In do not expect you to know the details of the examples in the exam, or of the underpinning physics. If an example is used in an exam question, all the necessary information and guidance will be provided.

Example 4.3 (Law of mass action)**.** We want to model a chemical reaction. Modelling chemical reactions is very complicated, but we start with a very simple model here. Consider a reaction of the form

$$A + B \underset{k_2}{\overset{k_1}{\rightleftharpoons}} C. \tag{4.1}$$

This means that a chemical A reacts with a chemical B to produce a chemical C. In this case we assume that one mol of A reacts with one mol of B to produce one mol of C. The reaction is reversible (i.e. C can break down into A and B again) and the rate at which each reaction occurs[4] is determined by the constants k_1 (left to right) and k_2 (right to left).

We can invoke the *law of mass action* to translate the chemical reaction in Equation 4.1 into a set of differential equations, which are called *rate equations*. It basically states that the evolution of the concentration[5] $r(t)$ for a given reactant R involved in j_R reactions, is given by

[4] That is, how fast the reaction takes place.
[5] The amount of substance per unit volume.

$$\frac{\mathrm{d}}{\mathrm{d}t}r(t) = \sum_{k=1}^{j_R} X_k - \sum_{k=1}^{j_R} Y_k,$$

where, X_k and Y_k represent the creation and consumption rates of R by reaction k, respectively, and are proportional to the product of the concentrations of the reactants involved in the reaction, with a proportionality constant given by the reaction rate constant. For the reactants A, B, and C in the above example Equation 4.1, we would obtain

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}a(t) &= k_2 c(t) - k_1 a(t) b(t), \\
\frac{\mathrm{d}}{\mathrm{d}t}b(t) &= k_2 c(t) - k_1 a(t) b(t), \\
\frac{\mathrm{d}}{\mathrm{d}t}c(t) &= k_1 a(t) b(t) - k_2 c(t),
\end{aligned}$$

where $a(t)$, $b(t)$, and $c(t)$ are the concentrations of A, B, and C at time t, respectively. The first term on the right-hand side of each equation is the creation rate (hence the $+$ sign) and the second term is the consumption rate (hence the $-$ sign). Note that the creation rate of A and B is the same as the consumption rate of C, and vice versa, which is a consequence of mass conservation.

If several molecules of a reactant are involved in a reaction then we have to count each in the creation rate and the consumption rate. Let us consider a more complicated example and assume that n_A molecules of A and n_B molecules of B react to n_C molecules of C and n_D molecules of D,

$$n_A A + n_B B \xrightarrow{k} n_C C + n_D D.$$

The numbers n_A, n_B, n_C, n_D are called *stoichiometries* and they appear both as prefactors in the creation/consumption rates (i.e. how many mols of each species are created/consumed) and in the exponents of the concentrations (i.e. how many mols of each species are needed for the reaction to occur). The corresponding rate equations[6] read:

[6] Here we have multiple differential equations coupled together, which is known as a system of differential equations. We will see how to study them in Chapter 6.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}a(t) &= -n_A k a(t)^{n_A} b(t)^{n_B},\\
\frac{\mathrm{d}}{\mathrm{d}t}b(t) &= -n_B k a(t)^{n_A} b(t)^{n_B},\\
\frac{\mathrm{d}}{\mathrm{d}t}c(t) &= n_C k a(t)^{n_A} b(t)^{n_B},\\
\frac{\mathrm{d}}{\mathrm{d}t}d(t) &= n_D k a(t)^{n_A} b(t)^{n_B}.
\end{aligned}$$

Note that this reaction is irreversible, as C and D cannot combine to produce A and B again. The actual *rate* here is the same for all reactants and given by

$$\frac{1}{n_A}\frac{\mathrm{d}}{\mathrm{d}t}a(t) = \frac{1}{n_B}\frac{\mathrm{d}}{\mathrm{d}t}b(t) = -\frac{1}{n_C}\frac{\mathrm{d}}{\mathrm{d}t}c(t) = -\frac{1}{n_D}\frac{\mathrm{d}}{\mathrm{d}t}d(t) = k a(t)^{n_A} b(t)^{n_B}.$$

Note that then

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{1}{n_A}a(t) + \frac{1}{n_C}c(t)\Big) = 0, \tag{4.2}$$

and analogously for the combinations (A, D), (B, C) and (B, D). These are identities that express *mass conservation*, namely, that all atoms (or, respectively, their masses) that are present before the reaction are also present after the reaction, just bound in different molecules.[7]

Example 4.4 (Hansen-Samuelson multiplier-accelerator model)**.** We want to model the evolution of the gross domestic product (GDP) of a country over time. The GDP is the total value of goods and services produced in a country in a given period of time, and it is an important indicator of the economic health of a country. We will write a very (very!) simplified model, known as the Hansen-Samuelson model.[8] It is based on the following assumptions:

1. We are only interested in the GDP at discrete time intervals, say every year, and we denote the GDP at a given time t by Y_t.
2. The GDP is the sum of the consumption C_t (i.e. how much people spend), the investment I_t (i.e. how much businesses invest in new capital), and the government spending G_t (i.e. how much the government spends on public services and infrastructure). Mathematically, $Y_t = C_t + I_t + G_t$.

[7] Mass conservation is another of those well established physical principles, like Newton's laws, that we can use to formulate certain models.

[8] Written by US economist Paul Samuelson, who credited his fellow economist Alvin Hansen for the idea. Samuelson was awarded the Nobel prize in economics in 1970.

3. The consumption at a given year is proportional to the GDP of the previous year, $C_t = \alpha Y_{t-1}$, with $0 < \alpha < 1$. This represents the idea that people only spend a fraction of their income.
4. The investment at a given year is proportional to the change in consumption from the previous year, $I_t = \beta(C_t - C_{t-1})$, with $\beta > 0$. This represents the idea that businesses invest more when the economy is growing and therefore there is more demand, and less when it is shrinking.
5. The government spending is assumed to be constant, $G_t = G > 0$. That is, government's actions are not influenced by the state of the economy.[9]

Combining all the assumptions above (i.e. plugging 3-5 into 2) we obtain

$$Y_t = \alpha(1 + \beta)Y_{t-1} - \alpha\beta Y_{t-2} + G.$$

This is a second order linear difference equation with constant coefficients, much like the ones we studied in Chapter 3. We can study it using the same techniques, and we will see that the GDP will either converge to a steady state, oscillate around it, or grow to infinity, depending on the values of α and β.

In contrast with Example 4.3, this model is not based at all on any well established physical principles. Instead it is based on a series of assumptions that are made to capture the essential characteristics of the economy (and thus human nature). Parameters α and β are not something that we can go and measure in the lab or get from a book, and they must be estimated from data. Therefore, it is paramount to parameterise and validate the model against reality, which is the focus of the next section.

4.3.2 Model validation

Once we have formulated a model and studied the associated mathematical problem, we need to validate the model against reality. Crucially, this step will tell us if we can stop or if we need to around the modelling cycle again and refine our model.

When it comes to comparing the model with reality, there are multiple aspects to consider. Typically, we will want to compare the prediction of the model with observations and measurements of the real-world phenomenon. Ideally, we want *quantitative agreement*, that is comparing the values predicted by the model with the values measured in reality, and computing the error between them. However, this is not always possible, and for simple models for complex systems we may have to settle for *qualitative agreement*, that is comparing the general behaviour of the model with the general behaviour of the phenomenon. For example, if we are modelling a disease outbreak, we may not be able

[9] If you have been following the news in the past few years, you will be well aware that this assumption is not realistic at all.

to predict the exact number of infected individuals at a given time, but we may be able to predict the general trend of the outbreak (e.g. whether it will grow or decline).

In order to make the comparison, we need to have the parameter values for the specific system we want to study. For example, in Section 2.8.5 we found the equation governing the mass-spring-damper system but, for a specific configuration, unless I know the mass, spring constant, and damping coefficient, I will not be able to make quantitative predictions of the system. In this case, we could go to the lab and measure each quantity independently and plug them into our model. However, in more complex cases (e.g. Example 4.4 for the GDP model) we will not be able to measure the parameters directly, and we will have to estimate them from data. This is known as *parameter estimation* or *model calibration*, and it is basically an inverse sub-problem that we need to solve first in order to eventually solve our problem.

After the validation step, we can decide if we are happy with the insight and predictions we have obtained from the model, or if we missed any critical bits. In that case, we need to go back and refine our model: we need to revise the assumptions we made in the formulation step, and possibly add more variables and relationships to capture the essential characteristics of the phenomenon. We then repeat complete the cycle and reassess.

Finally, once we are happy with the model, we can use it to make predictions and draw conclusions about the real-world phenomenon. It is important to always keep in mind the real-world system and not trust blindly the outcome of the model. Often we will be able to validate the model against a certain scenario, but we will want to predict other scenarios. In that case, we need to be aware of the model limitations and that we are not extrapolating beyond the range of validity of the model. Because in this case we do not have data to validate the model against, we need to be extra careful and critical about the results obtained.[10]

Example 4.5 (Covid-19)**.** Luckily for us, the Covid-19 pandemic is now over, but it was a very relevant case on which we used modelling to tackle a very real (and serious) problem. We also have a lot of data available, which makes it a good example to illustrate the modelling cycle.

We will focus on the first wave of the pandemic in England, which took place in March 2020. The data we have is the number of confirmed Covid-19 infections per day, which was collected by the UK Health Security Agency and is available through their dashboard. In Figure 4.2 we show the data for the first 120 days of the pandemic, starting on 01/03/2020 (day 0 in the graph).

Suppose now we are 30 days into the outbreak and would like to make a prediction about the number of infected people, as shown in Figure 4.3. The simplest approach we

[10]There is also always the risk of making mistakes in the calculations, so that is another good reason to stop and think whether the results make any sense. A classical example of this is when you end up with a negative population or concentration, which is clearly not physically possible.

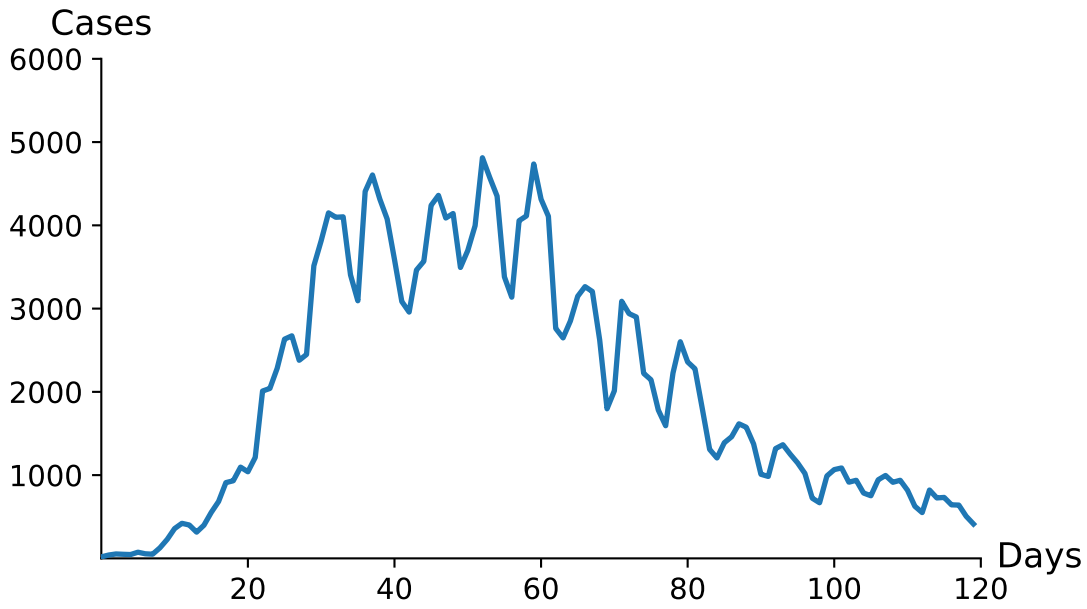


Figure 4.2: Confirmed COVID-19 cases between 01/03/2020 and 28/06/2020 in England.

could take would be to simply fit some curves to the data and use their extensions for a prediction. This means we only use data processing tools and completely ignore any knowledge we may have about disease spreading and any context of the data.

We observe that the numbers effectively increase, but there are many increasing functions that could describe the trend. Writing t for the time measured in days, for instance, a quadratic polynomial of the form $p(t) = mt^2$ with a suitable number $m > 0$ or an exponential function of the form $f(t) = se^{ct}$ with parameters $s, c > 0$ could effectively capture the growth. For some suitable parameters m, s, and c we indeed seem to be able to get good fits in that the curves nicely run through the data points for the first 30 days, as shown in Figure 4.4.

Let us now use these curves for a prediction: in Figure 4.5 we plot the data and the curves for the whole 120 days. It does not look very good. Non of the approaches is able to describe the fact that the infection numbers reach a maximum before decaying again. Of course, for the fits we had chosen functions that increase, however this was inspired by the trend in the data for the first 30 days. In order to recover the observed behaviour of the infections numbers for the later days we need to combine the data with some additional knowledge about what they mean and describe.

Mathematical models of infectious disease epidemiology track the infected individuals in a population using assumptions about transmission and recovery of the disease. A classical approach is the *SEIR model*, on which many later, refined models are based. The population, denoted by N, is assumed constant in time and split into *susceptible*

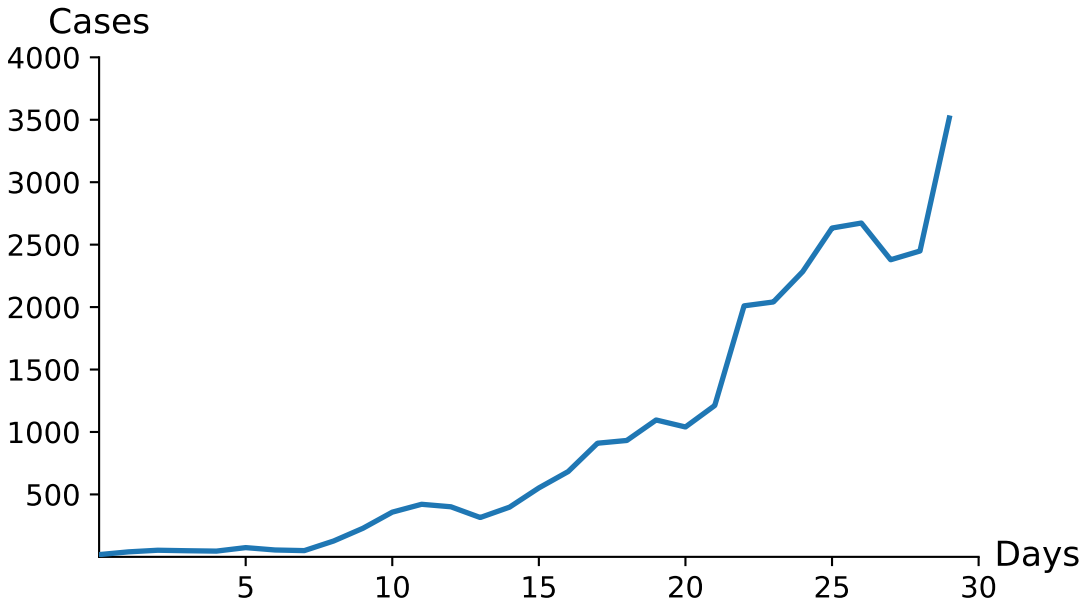


Figure 4.3: Confirmed COVID-19 cases between 01/03/2020 and 30/03/2020 in England.

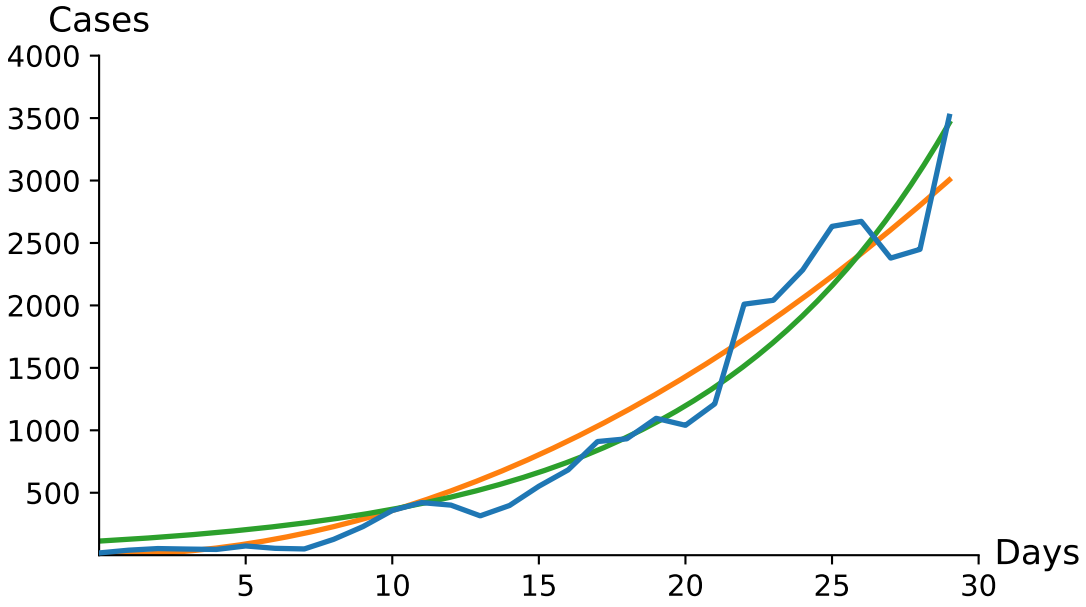


Figure 4.4: Fitted models for the first 30 days. Orange is the quadratic and green is the exponential.

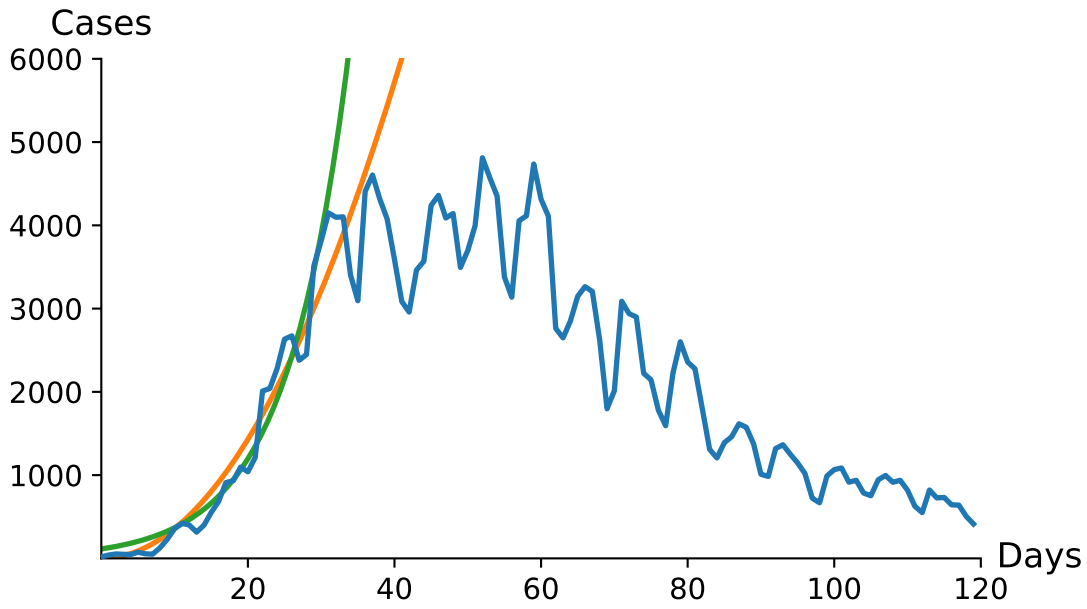


Figure 4.5: Fitted models for the first 30 days validated against data for the first 120 days. Orange is the quadratic and green is the exponential.

individuals $S(t)$, *exposed* (but not yet infectious) individuals $E(t)$, *infected* (and infectious) individuals $I(t)$, and *recovered*[11] individuals $R(t)$. As indicated, these population compartments depend on time t measured here in days, but they have to sum up to the overall population at any time,

$$S(t) + E(t) + I(t) + R(t) = N \quad \forall t.$$

Assumptions are then made about the changes of the compartments:

1. Susceptible individuals get in contact with other people, and the chance that another person is infectious is $I(t)/N$. We assume that they become exposed from such a contact at a rate that is proportional to a factor $\beta > 0$.
2. Exposed individuals become (ill and) infectious at a rate with a factor denoted by $\epsilon > 0$.
3. Individuals from the compartment $I(t)$ recover at a rate with a factor denoted by $\gamma > 0$.

Denoting the change with a time derivative we obtain a set of differential equations, one equation for each compartment:

[11] At least in the best case. Somewhat cynically, the notation $R(t)$ also allows for the interpretation as a *removed* individual...

$$\begin{aligned}
\frac{dS}{\mathrm{d}t}(t) &= -\beta \frac{I(t)}{N} S(t), \\
\frac{dE}{\mathrm{d}t}(t) &= \beta \frac{I(t)}{N} S(t) - \epsilon E(t), \\
\frac{dI}{\mathrm{d}t}(t) &= \epsilon E(t) - \gamma I(t), \\
\frac{dR}{\mathrm{d}t}(t) &= \gamma I(t).
\end{aligned}$$

Figure 4.6 shows a typical solution to the system of differential equations, where we start with a high number $S(0)$, a small number $I(0)$, and $E(0) = R(0) = 0$.

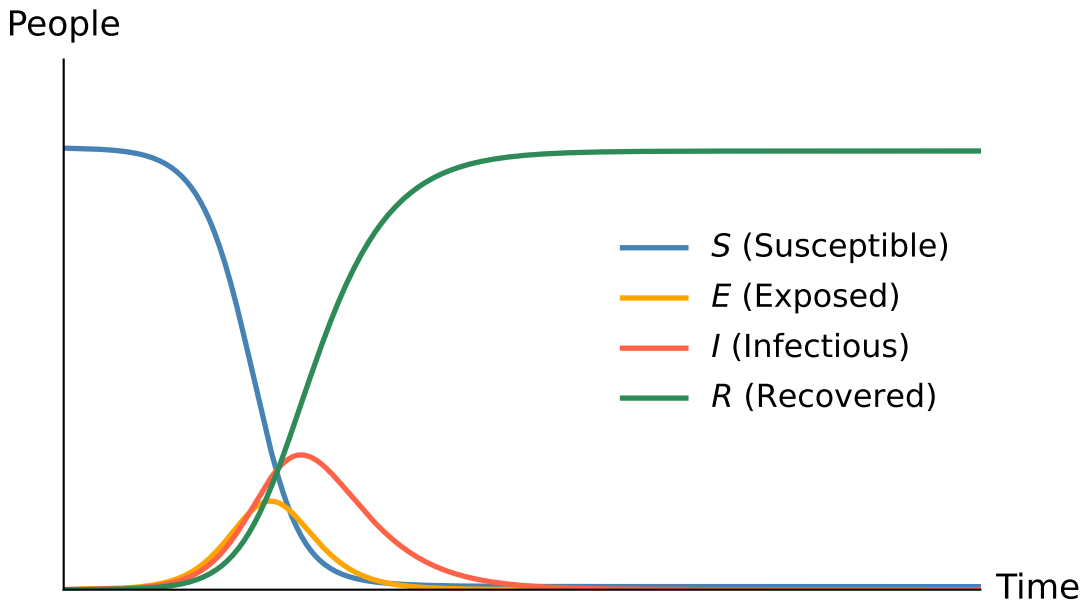


Figure 4.6: Example of the type of solutions the SEIR model produces. The number of infected individuals $I(t)$ increases to a maximum before decaying again.

The curve for I looks very promising, this is effectively the behaviour that our numbers display and that we expect!

Getting back to our prediction problem, we choose parameters β, γ, ϵ and initial values such that fairly good fit of the $I(t)$ values with the infections in the first 30 days is obtained.[12] The result is shown in Figure 4.7, where we can see that the SEIR model is able to capture the trend of the data for the first 30 days.

[12] In practice this is a very hard problem and one cannot just go and fit all parameters, one needs to use some prior knowledge about the disease to reduce the number of free parameters.

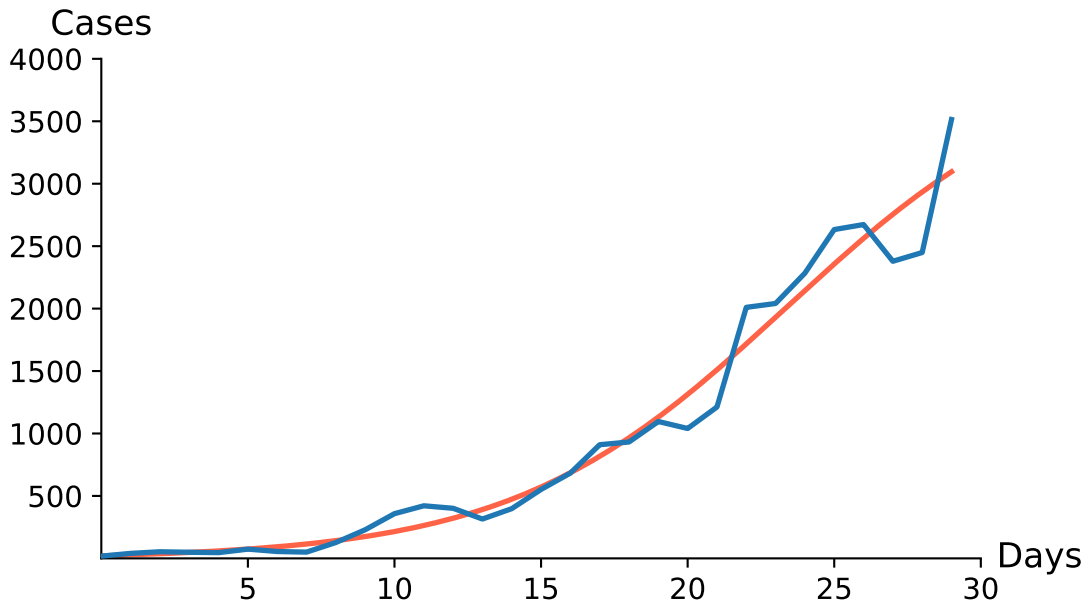


Figure 4.7: SEIR model for the first 30 days, where we have fitted the model to data to determine the parameters.

Let us now use these parameters or, respectively, the solution to the corresponding system of differential equations to make a prediction. Below we plot the infection data and the numerically obtained $I(t)$ values for the first 120 days.

This looks quite good, the solution captures the trend in the data points with the increase and the decay, even the maximum looks at a good position. So the mess of data is effectively fairly well described by a couple of equations with very few parameters. Now, can we further extend the prediction? Let us look into a year, as shown in Figure 4.8. Whilst the model can describe the first wave, the computed values for $I(t)$ stay close to zero after and the model fails to pick up the later waves, particularly the one early in 2021 caused by a new variant of the virus. For this purpose, more sophisticated models would be needed.

In this example we have seen the modelling cycle in action. In fact, we have done multiple cycles: we first tried to fit simple curves to the data, which did not work. We then formulated a model based on assumptions about the disease spreading and fitted it to the data, which worked well for the first wave. However, we also saw that this model is not able to predict later waves, and we would need to refine it further in order to capture them.

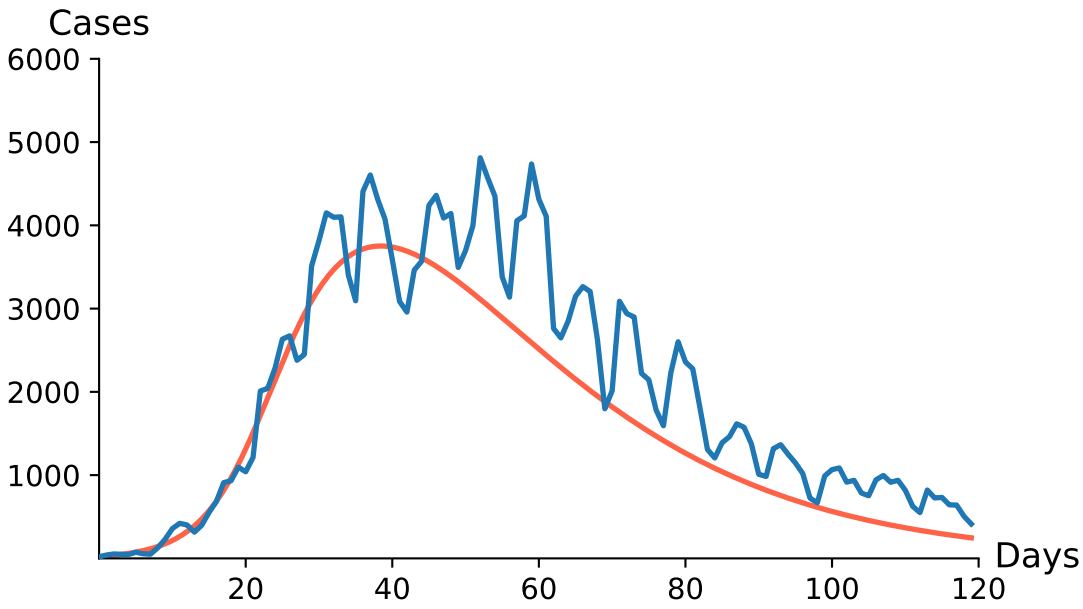


Figure 4.8: SEIR model for the first 30 days, where we have fitted the model to data to determine the parameters.

4.4 Word of caution: all models are wrong!

Before concluding this section, a word of caution. We have seen that it is critical to validate the model against reality, and that it is important to be aware of the model limitations. This is because, in general, models are not perfect representations of reality. They are simplifications and abstractions that capture certain aspects of the phenomenon we are studying, but they may not capture all the complexities and nuances of the real world. Therefore, it is important to be critical and cautious when interpreting the results of a model, and to always keep in mind that:

> **!** Important
>
> **Mathematical models always have a limited range of validity!**

Many scientists have expressed views about the limitations of models. Some notable examples are:

- Mark Kac explained *Models are, for the most part, caricatures of reality, but if they are good, they portray some features of the real world.*[13]

[13] M. Kac: *Some mathematical models in science.* Science 166, 695–699 (1969).

- In one of his papers on developmental biology[14] Alan Turing wrote *This model will be a simplification and an idealisation, and consequently a falsification. It is to be hoped that the features retained for discussion are those of greatest importance ...*
- Or, the most famous quote about modelling: *... all models are wrong, but some are useful.*[15]

[14] A.M. Turing: *The chemical basis of morphogenesis.* Phil. Trans. R. Soc. Lond., Series B – Biol. Sci. 237(641), 37–72 (1952).

[15] G.E.P. Box, N.R. Draper: *Empirical Model-Building and Response Surfaces.* Wiley, New York (1987).

5 Dimensional Analysis

Aims of the chapter

By the end of this chapter, you should be able to:

- Derive the nondimensional form of a given mathematical model.
- Apply the principles of dimensional analysis to determine the parameter dependence of a variable.

In the last chapter we focused our attention on mathematical modelling, and on how the differential and difference equations we studied in Chapters 2 and 3, respectively, are at the core of most models. In this chapter, we introduce another key modelling tool: dimensional analysis, which leverages the basic principles governing how dimensions of physical quantities work. As we will see, these principles allow us to (a) simplify model equations, and (b) infer which parameter combinations dictate the behaviour of a given variable, even without the need to write down a full mathematical model.

From an abstract point of view, a mathematical model is simply a relationship between variables of interest. Within this relationship we distinguish *indepedent variables*, denoted by $i = (i_1, \dots i_M)$, which are the variables that drive the model, and *dependent variables* $d = (d_1, \dots, d_N)$, which depend on them. The end goal of modelling is to find a set unknown function u_i that tell us how we can calculate the dependent variables from the independent ones:

$$\begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_N \end{pmatrix} = \begin{pmatrix} u_1(i_1, \dots i_M) \\ u_2(i_1, \dots i_M) \\ \vdots \\ u_N(i_1, \dots i_M) \end{pmatrix}. \tag{5.1}$$

These functions $u_1, \dots, u_N$ might be obtained in multiple ways, for example they could be solutions to a differential equation, like the mass-spring-damper system (Section 2.8.5). In that case, we have a single dependent variable (the position of the mass $x(t)$) and we have five independent variables: t, m, c, k and x_0. Note that, for the purpose of dimensional analysis, model parameters are counted as independent variables even though from an equation point of view they are not.[1] The rationale is that the parameters

[1] Do not forget initial conditions, they can have parameters too!

dictate the behaviour of the dependent variable in the same way as the independent variable of the equation, t, does. Indeed, as we saw in Section 2.8.5, the solution to our model allowed us to write

$$x = u(t, m, c, k, x_0).$$

Dimensional analysis is a complementary tool that allows us to reduce the effort required to solve such a problem, in some cases allowing us to draw conclusions even without knowing the equations for our model. It uses some fundamental ideas related to dimensions. Defining dimension is tricky, so one way to think about it is that a dimension is a dimension we can measure (even if in an indirect way). For example, time, length and mass are dimensions. Dimensions are measured in units,[2] but they are fundamentally different: units are arbitrary (e.g. seconds, minutes, hours...) while the dimension is the abstract concept (e.g. time). Dimensional analysis then uses the two following principles:

1. All quantities and variables have *dimensions* attached to them.[3]
2. The relations or laws between them do not alter when changing the units in which we measure them. For example, Newton's second law states that force equals mass times acceleration, and this is true independently of how you measure mass and acceleration.[4]

The main goal of dimensional analysis is to transform model equations into a *nondimensional form* (or dimensionless form), which means that every single term in the equation has no dimensions. This is a kind of reference or fundamental form of the model, which is usually much simpler than the original one. Any analysis or solution to the nondimensional form can then be converted back to the dimensional form. This sounds a bit pointless, but there are two essential benefits to this procedure:

1. The number of independent variables in the problem often is reduced in nondimensional form. This makes it easier to find the solution.
2. Many specific problems reduce to the same nondimensional problem, which means that we can just study that single nondimensional problem. All dimensional problems are then automatically covered too.
3. In the process of nondimensionalising, we usually end up with most variables taking values of the same order of magnitude (as opposed to some being huge and others tiny). This means that if we solve them numerically, the computer will be much

[2] Note that the distinction between dimension and units sometimes is not (or not properly) made in the literature. I will aim to be consistent throughout the module, but I may unconsciously mix them up.
[3] Not having dimensions is a dimension on itself, in a similar way that the empty set is an actual set.
[4] You will need to measure force in a consistent way, but that's totally fine: if you measured your mass in pounds and your acceleration in yards per year, you would have to measure force in pounds times yards per year. This is an incredibly weird unit, but from a mathematical point of view it is not any different to the unit we call Newton. In science, we usually stick to the International System of Units (SI units for short).

happier about it as having all quantities of similar magnitude reduces floating-point errors.

5.1 Dimensions

! Important

As you can imagine there are a lot of dimensions, but it turns out that any quantity you can think of can be written as a unique combination (products and powers) of seven *fundamental dimensions*.[5] For example, the dimension of volume is the dimension of length to the cube. In this module we will only use six of them,[6] which are shown in Table 5.1.

Table 5.1: Fundamental dimensions used in this module.

notation	dimension	possible units
T	time	seconds, days
L	length	nanometers, feet, lightyears
M	mass	micrograms, pounds
A	quantity	head count, mol
Θ	temperature	kelvin, degrees fahrenheit
Q	charge	coulomb

Note

Throughout the module I will stick to the notation in the table, and I encourage you to do so. If for whatever reason you do not follow it in the exam or assignment (for example, because you forgot of the specific symbol) please state clearly which dimension your symbol represents.

Proposition 5.1. *Let $[v_j]$ denote the dimension of a variable v_j. Give two variables v_1 and v_2, the dimension of their product is the product of their dimensions, that is*

$$[v_1 v_2] = [v_1][v_2].$$

Example 5.1. Let us illustrate this proposition with a couple of examples.

[6] They form a basis, analogous to a basis of a vector space.

[6] The seventh one, that we will not use, is the luminous intensity, usually measured in candelas. Note that the canonical set of fundamental variables uses electric current instead of electric charge, but current is just charge per unit time.

1. A velocity u is defined as a change in position Δx in a given time interval Δt (which could both be infinitely small). The (change in) position has dimension L while the time (interval) has dimension T. Then

$$[u] = \left[\frac{\Delta x}{\Delta t}\right] = \frac{[\Delta x]}{[\Delta t]} = \frac{L}{T},$$

as expected. A way to sanity check this, is that the SI units of meters per second (i.e. length per time)...

2. Newton's second law states that $F = ma$. We know that mass has dimension M while, by a similar argument as in point 1, acceleration has dimension LT^{-2}. Then, we deduce that the dimension of force is

$$[F] = [ma] = [m][a] = \frac{ML}{T^{-2}}.$$

This is not as obvious to check but, if we do the unit sanity check and look at the definition of the Newton, we will see that in SI units[7] is kg m s^{-2}.

Let us now introduce the fundamental principle in dimensional analysis.

! Important

Proposition 5.2 (Principle of dimensional homogeneity)**.** *All terms in an equation must have the same dimensions, i.e.*

$$v_1 = v_2 \quad \Longleftrightarrow \quad [v_1] = [v_2].$$

Remark 5.1. This sounds kind of trivial (and not too useful).[8] However, a consequence of that is that we can only add (or substract) variables if they have the same dimensions, i.e. $v_1 + v_2$ is only defined if $[v_1] = [v_2]$ and, naturally, $[v_1 + v_2] = [v_1] = [v_2]$.

We can now combine these results (and some that we will deduce from them) in order to provide a list on how we will manipulate dimensions.

Proposition 5.3. *For two given variables v_1 and v_2, their dimensions satisfy:*

1. $[v_1 v_2] = [v_1][v_2]$ *(Proposition 5.1).*
2. $v_1 = v_2 \quad \Longleftrightarrow \quad [v_1] = [v_2]$ *(Proposition 5.2).*
3. *Derivative rule:*

[7] That is why SI units are very useful, they allow us to build a whole system of consistent units, so when we perform calculations we do not have to worry about tracking them.

$$\left[\frac{\mathrm{d}v_1}{\mathrm{d}v_2}\right] = \frac{[v_1]}{[v_2]}.$$

4. *Integral rule:*

$$\left[\int v_1 \, \mathrm{d}v_2\right] = [v_1][v_2].$$

5. *All arguments of more complex functions (e.g.* exp *or* sin*) must be nondimensional.*[9] *That is, for a given function $f(v_1)$, we must have $[v_1] = 1$.*

Proof. The first two points are the propositions we stated above and we are not going to prove them.

Consider the definition of the derivative. For a function $v_1(v_2)$, we have

$$\frac{\mathrm{d}v_1}{\mathrm{d}v_2} = \lim_{\Delta v_2 \to 0} \frac{v_1(v_2 + \Delta v_2) - v_1(v_2)}{\Delta v_2}.$$

Using the product rule we obtain

$$\begin{aligned}
\left[\frac{\mathrm{d}v_1}{\mathrm{d}v_2}\right] &= \lim_{\Delta v_2 \to 0} \left[\frac{v_1(v_2 + \Delta v_2) - v_1(v_2)}{\Delta v_2}\right] \\
&= \lim_{\Delta v_2 \to 0} \frac{[v_1(v_2 + \Delta v_2) - v_1(v_2)]}{[\Delta v_2]} \\
&= \lim_{\Delta v_2 \to 0} \frac{[v_1]}{[v_2]} \\
&= \frac{[v_1]}{[v_2]}.
\end{aligned}$$

We can now use the Fundamental Theorem of Calculus (Theorem 2.1) to prove point 4:

$$\left[\int v_1 \mathrm{d}v_2\right] = [V_1]\,,$$

such that

$$\frac{\mathrm{d}V_1}{\mathrm{d}v_2} = v_1(v_2).$$

[9] Still, you would be surprise of the number of times people write equations in research articles that do not satisfy the principle (and thus are wrong).

[9] We denote a variable being nondimensional as $[v_1] = 1$. This makes it consistent with the product rule Proposition 5.1, which you can think of as the identity element of the group of all dimensions.

From the derivative rule we have that $[V_1] = [v_1][v_2]$, hence

$$\left[\int v_1 \mathrm{d}v_2\right] = [v_1][v_2].$$

Finally, let us consider rule 5. We will not give a general proof but prove it for a specific example, but one would proceed similarly for other functions. Consider the series representation of the exponential:

$$\exp(v_1) = \sum_{n=0}^{\infty} \frac{1}{n!} v_1^n.$$

By the principle of dimensional homogeneity all terms in the series must have the same dimension. Consider two consecutive terms:

$$\left[\frac{1}{n!} v_1^n\right] = \left[\frac{1}{(n+1)!} v_1^(n+1)\right].$$

Applying the product rule we find that this only holds if $[v_1]^n = [v_1]^(n+1)$, which means that we must have $[v_1] = 1$. □

These rules are useful to determine the dimensions of all variables and parameters in an equation, as long as we know a few of them.

Example 5.2 (Mass-spring-damper: dimensions)**.** Consider, yet again, the mass-spring-damper equation:

$$m\ddot{x}(t) = -kx(t) - c\dot{x}(t), \quad \text{with} \quad x(0) = x_0, \quad \dot{x}(0) = 0.$$

We want to deduce the dimensions of all parameters involved. First let us write down what we know. We know that time has dimension T, that m is a mass, so it has dimension M, and x is a position, so it has dimension L.

What is the dimension of x_0? One could argue that it is a position, so it must have dimension of L but we can also show it by the principle of dimensional homogeneity:

$$x(0) = x_0 \quad \Longleftrightarrow \quad [x(0)] = [x_0].$$

We can proceed similarly for c and k, just with some extra manipulation. The dimensions of the left-hand side term are

$$[m\ddot{x}(t)] = [m][\ddot{x}(t)] = MLT^{-2}.$$

By the principle of dimensional homogeneity we conclude[10]

$$[kx(t)] = [m\ddot{x}(t)], \quad \Longrightarrow \quad [k] = \frac{[m][\ddot{(x)}]}{[x(t)]} = MT^{-2},$$

and

$$[c\dot{x}(t)] = [m\ddot{x}(t)], \quad \Longrightarrow \quad [c] = \frac{[m][\ddot{(x)}]}{[\dot{x}(t)]} = MT^{-1}.$$

5.2 Nondimensionalisation

We now want to use the principles we presented in the previous section to simplify differential equations.[11] In order to do so, we will rescale the variables in the equation. Rescaling means that we change how we measure them. By choosing these rescalings cleverly, we will be able to simplify the equations. A way to think about this is as a simple change of units, just that we use some unusual and problem-dependent units. Let us first introduce the recipe to nondimensionalise, and we will then illustrate it with some examples.

! Important

Algorithm 5.1 (Nondimensionalisation recipe)**.** *In order to nondimensionalise a differential equation, we follow these steps:*

1. *Rescale the dependent and independent variables in the equation (not the parameters). This means that we redefine each variable v_j as the product of a nondimensional variable $\tilde{v}_j$ and a constant scaling Δv_j:*

   $$v_j := \Delta v_j \tilde{v}_j.$$

   The nondimensional variable $\tilde{v}_j$, as the name suggests, is nondimensional, so the dimensions will be carried by the scaling Δv_j.

2. *Substitute the rescaling in step 1 into the differential equations, applying the chain rule in the derivative as needed, e.g.*

   $$\frac{\mathrm{d}v_1}{\mathrm{d}v_2} = \frac{\Delta v_1}{\Delta v_2}\frac{\mathrm{d}\tilde{v}_1}{\mathrm{d}\tilde{v}_2}.$$

3. *Divide the equation by a suitable factor to make it nondimensional. Essentially, the factor is chosen to be the prefactor of one of the terms (i.e. the bit that ends up carrying the dimensions).*

[10] Note that minus signs play no role in the dimensional analysis, so we can just ignore them.
[11] This method works on all types of equations, e.g. algebraic equations, partial differential equations...

4. *Choose the suitable values for the rescaling from the differential equation. This step is a bit of an art, an will be better illustrated through the examples below, but the goal is to make as many of the factors in the equation equal to one. We usually rename the remaining factors for simplicity.*
5. *Solve or analyse the equation.*
6. *Convert the solutions or the conclusions of the analysis back to dimensional form.*

Remark 5.2. In step 2 we glossed very quickly over the rescaling of a derivative, let us now go through it in a bit more detail. Effectively, we are doing a change of variable:

$$\frac{\mathrm{d}v_1}{\mathrm{d}v_2} = \frac{\mathrm{d}v_1}{\mathrm{d}\tilde{v}_2}\frac{\mathrm{d}\tilde{v}_2}{\mathrm{d}v_2}.$$

The first factor in the right-hand side is close to what we want (i.e. a derivative with respect to $\tilde{v}_2$) and the second factor can be computed from the rescaling:

$$\tilde{v}_2 = \frac{v_2}{\Delta v_2}, \quad \Longrightarrow \quad \frac{\mathrm{d}\tilde{v}_2}{\mathrm{d}v_2} = \frac{1}{\Delta v_2}.$$

Then, combining both results we get

$$\frac{\mathrm{d}v_1}{\mathrm{d}v_2} = \frac{1}{\Delta v_2}\frac{\mathrm{d}v_1}{\mathrm{d}\tilde{v}_2}.$$

This is not exactly what we want, as we want the derivative of $\tilde{v}_1$ rather than v_1. However, this has a simple fix. Given that the rescaling is a constant we know that

$$\frac{\mathrm{d}v_1}{\mathrm{d}\tilde{v}_2} = \frac{\mathrm{d}\left(\Delta v_1 \tilde{v}_1\right)}{\mathrm{d}\tilde{v}_2} = \Delta v_1 \frac{\mathrm{d}\tilde{v}_1}{\mathrm{d}\tilde{v}_2},$$

and combining everything we obtain the result we wanted in step 2 of alg. 5.1.

Example 5.3 (Mass-spring-damper: nondimensionalisation)**.** As usual, let us illustrate these new concepts using the mass-spring-damper equation:

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + c\frac{\mathrm{d}x}{\mathrm{d}t} + kx(t) = 0, \quad \text{with} \quad x(0) = x_0, \quad \dot{x}(0) = 0.$$

Let us apply the algorithm.

1. We rescale the variables in the equation (not the parameters), which in this case are x and t:
$$x = \Delta x\, \tilde{x}, \quad \text{and} \quad t = \Delta t\, \tilde{t}.$$

2. Substituting into the equation we obtain
$$\frac{m\Delta x}{\Delta t^2}\frac{\mathrm{d}^2\tilde{x}}{\mathrm{d}\tilde{t}^2} + \frac{c\Delta x}{\Delta t}\frac{\mathrm{d}\tilde{x}}{\mathrm{d}\tilde{t}} + k\Delta x\, \tilde{x}(\tilde{t}) = 0, \quad \Delta x\, \tilde{x}(0) = x_0, \quad \frac{\Delta x}{\Delta t}\dot{\tilde{x}}(0) = 0.$$
A couple of remarks here. First, note that for each time derivative we get an additional Δt^{-1} factor, i.e. the second derivative has a Δt^{-2} in front (but only a Δx). This can be shown applying the argument in Remark 5.2 twice. Second, note that arguments of the dependent variables can just be switched to nondimensional form (i.e. we just changed t into $\tilde{t}$ in the $kx(t)$ term), as any rescalings would be absorbed by the functional form of the dependent variable once the equation has been solved.

3. Now we divide the equation (and the initial condition) by a suitable factor. For the initial conditions it is straightforward:
$$\Delta x\, \tilde{x}(0) = x_0, \quad \Longrightarrow \quad \tilde{x}(0) = \frac{x_0}{\Delta x},$$
and
$$\frac{\Delta x}{\Delta t}\dot{\tilde{x}}(0) = 0, \quad \Longrightarrow \quad \dot{\tilde{x}}(0) = 0.$$
For the equation we have various options: we could divide by $m\Delta x/\Delta t^2$, $c\Delta x/\Delta t$ or $k\Delta x$. It turns out that nondimensionalisations are often non-unique and depending on the choice we get a different nondimensional form. However, they are all equivalent (and they all give the same result once the solution is brought back to dimensional form) so it does not really matter which one you choose.[12] In this case, let us choose to divide everything by $m\Delta x/\Delta t^2$:
$$\frac{\mathrm{d}^2\tilde{x}}{\mathrm{d}\tilde{t}^2} + \frac{c\Delta t}{m}\frac{\mathrm{d}\tilde{x}}{\mathrm{d}\tilde{t}} + \frac{k\Delta t^2}{m}\, \tilde{x}(\tilde{t}) = 0.$$
You can check that, indeed, now each of the terms in the equation is nondimensional.

4. At this point, we set the values of the rescalings Δx and Δt. As we said in the algorithm, the goal is to make as many parameters equal to one as possible. Δx only appears in the initial condition, so we choose $\Delta x = x_0$. For Δt we have two options, depending on which of the remaining parameter combinations we want to set to one. Again, it does not really matter which one you choose, as results are

[12] In fact, in real applications this choice is normally determined by the analysis you want to perform.

equivalent.[13] This means that we want

$$\frac{k\Delta t^2}{m} = 1, \quad \Longrightarrow \quad \Delta t = \sqrt{\frac{m}{k}},$$

and you can check that, indeed, this has dimensions of time. In fact, one can show that this is the period of the oscillations if the damper was removed. The resulting nondimensional equation is

$$\frac{\mathrm{d}^2\tilde{x}}{\mathrm{d}\tilde{t}^2} + \xi\frac{\mathrm{d}\tilde{x}}{\mathrm{d}\tilde{t}} + \tilde{x}(\tilde{t}) = 0, \quad \text{with} \quad \tilde{x}(0) = 1, \quad \dot{\tilde{x}}(0) = 0,$$

where we have relabelled the parameter grouping as $\xi := \frac{c}{\sqrt{mk}}$. Here we see the whole point of nondimensionalisation: the original problem had 4 parameters (m, c, k and x_0), while the nondimensional form only has one (ξ). This means, for example, that if we were to do a bifurcation analysis[14] we would only have to worry about one parameter.

5. We now solve the nondimensional equation. We omit the details here for conciseness, but you know the method from Chapter 2. The roots of the auxiliary equation are

$$\lambda_\pm = \frac{1}{2}\left(-\xi \pm \sqrt{\xi^2 - 4}\right),$$

and we have different behaviours depending on whether $\xi > 2$ or $0 < \xi < 2$. For example, in the latter case we have complex roots, so we have

$$p = -\frac{\xi}{2}, \quad \text{and} \quad q = \frac{1}{2}\sqrt{4 - \xi^2}.$$

Then, the solution is

$$\tilde{x}(\tilde{t}) = \mathrm{e}^{-\frac{\xi}{2}\tilde{t}}\left(\cos\left(\frac{1}{2}\sqrt{4-\xi^2}\tilde{t}\right) + \frac{\xi}{\sqrt{4-\xi^2}}\sin\left(\frac{1}{2}\sqrt{4-\xi^2}\tilde{t}\right)\right).$$

6. Finally, we bring the solution back to dimensional form. To do this, we plug the inverse of the rescalings back into the solution: $\tilde{x} = x/\Delta x$ and $\tilde{t} = t/\Delta t$:

$$\frac{1}{\Delta x}x(t) = \mathrm{e}^{-\frac{\xi}{2}\frac{t}{\Delta t}}\left(\cos\left(\frac{1}{2}\sqrt{4-\xi^2}\frac{t}{\Delta t}\right) + \frac{\xi}{\sqrt{4-\xi^2}}\sin\left(\frac{1}{2}\sqrt{4-\xi^2}\frac{t}{\Delta t}\right)\right).$$

Manipulating the expression and using that $\xi = \frac{c}{\sqrt{mk}}$, $\Delta x = x_0$ and $\Delta t = \sqrt{m/k}$ we obtain

$$x(t) = x_0 e^{-\frac{c}{2m}t}\left(\cos\left(\frac{1}{2m}\sqrt{4mk - c^2}t\right) + \frac{c}{\sqrt{4mk-c^2}}\sin\left(\frac{1}{2m}\sqrt{4mk-c^2}t\right)\right),$$

[13] In practice, once again, depending on the analysis you want to make, you might be inclined to make a certain choice. For example, one may want to study what happens if we remove the damper, which would make the middle term disappear. In that case, choosing that term to be one would result into a singularity.

[14] Not that the methods we explained work in this case.

which is the solution we determined in Section 2.8.5. Note that one could also bring back to dimensional form the condition to have oscillatory solutions:

$$0 < \xi = \frac{c}{\sqrt{mk}} < 2, \quad \iff \quad c^2 < 4mk,$$

as we already knew.

Remark 5.3. Remember than a key advantage of using nondimensional equations is that various dimensional equations can reduce to the same nondimensional one. For example, in the example above, any combination of parameters that yields the same value of ξ are equivalent. This is not too important when one can solve the equation analytically but, if we did it numerically, it would mean that we only need to run a single simulation and then just scale the solution back to the relevant problem.

5.3 Scaling arguments

As we stated earlier in this chapter, one could think of a model as a relation between dependent and independent variables, independently on how this relation is given. There is a very interesting application of dimensional analysis that allows us to determine, even without knowing the model equations, which combination of independent variables drives our dependent variables. For the purposes of this section, we count parameters as independent variables.

Before we go into details on how to do that, a theorem.

> **!** Important
>
> **Theorem 5.1** (Buckingham Pi (π) Theorem)**.** *A given problem with n independent variables (including parameters) that contains m independent dimensions (with $m < n$) can be reduced to a nondimensional problem completely determined by $(n - m)$ nondimensional independent variables.*

We skip the proof for this theorem, as it requires some linear algebra concepts. Let us apply this to our favourite example.

Example 5.4 (Mass-spring-damper: Buckingham Pi theorem)**.** Consider the mass-spring-damper equation again. The model has one dependent variable $x(t)$ and five independent variables: t, m, c, k and x_0. The dimensions of each variable (as determined in Example 5.2) are:

$$\begin{aligned} [x] &= L, & [t] &= T, & [m] &= M, \\ [c] &= MT^{-1}, & [k] &= MT^{-2}, & [x_0] &= L. \end{aligned}$$

Therefore, we have three independent dimensions: M, T and L. Then, from the Buckingham Pi theorem we have $n = 5$ and $m = 3$, therefore the nondimensional problem will depend on $n - m = 5 - 3 = 2$ independent variables. We already knew that from Example 5.3: the nondimensional solution depended only on $\tilde{t}$ and ξ.

Remark 5.4. A note on what we mean by independent dimensions: a dimension is independent of a set of other dimensions, if it cannot be constructed by combining dimensions in the set. In the Buckingham Pi theorem we are counting the minimal set of independent dimensions, not fundamental ones (as in Table 5.1). In most cases they will be equivalent, but let us see a simple example where that is not true.

Consider an exponential decay problem, where a concentration c (i.e. amount of particles per unit of volume) decays in time proportionally to itself for a given constant a:

$$\frac{\mathrm{d}c}{\mathrm{d}t} = -ac, \quad \text{with} \quad c(0) = c_0. \tag{5.2}$$

The dimensions of the variables of the equation are

$$= AL^{-3}, \quad [t] = T, \quad [a] = T^{-1}, \quad [c_0] = AL^{-3}.$$

In this case, we have three fundamental dimensions (A, L and T), but only two independent ones as A and L always appear together as AL^{-3} and, therefore, they count as a single dimension.

! Important

Algorithm 5.2. *The recipe to deduce the parameter dependency of a certain variable d is as follows:*

1. *Identify the independent variables $i_1, i_2, \ldots, i_N$ that the dependent variable might depend on. Determining how many variables and which ones is the key modelling step.*
2. *Write the dimensions of all variables (dependent and independent).*
3. *Express the dimension of the dependent variable as a product of the independent ones:*
$$[d] = [i_1^{n_1} i_2^{n_2} \cdots i_n^{n_N}] = [i_1]^{n_1} [i_2]^{n_2} \cdots [i_n]^{n_N}.$$
4. *Substitute in the dimensions of all variables and solve for the exponents $n_1, n_2, \ldots, n_N$. Depending on the number of independent dimensions, we will have a system of equations with a unique solution or with infinitely many solutions. In the latter case, we can express some of the exponents in terms of the others, such that*
$$[d] = [\Delta d \, \pi_1^{m_1} \pi_2^{m_2} \cdots \pi_M^{m_M}],$$

where Δd is a scaling for d, and $\pi_1, \ldots, \pi_M$ are nondimensional combinations of the independent variables.[15] *The number of nondimensional combinations is $M = N - D$, where D is the number of independent dimensions.*

5. *Then, we can write the dependency of d on the independent variables as*

$$d = \Delta d \, f(\pi_1, \ldots, \pi_M),$$

where f is an unknown function that we can determine by experiments or measurements. In the particular case where $M = 0$, we have

$$d = \Delta d \, f(1, \ldots, 1) = \Delta d \cdot C,$$

where C is a constant to be determined.

Example 5.5 (Exponential decay)**.** To illustrate the method, let us pick up the exponential decay problem we mentioned in Remark 5.4.

1. Imagine we did not know the equation. Even in that case, we could guess that the concentration would depend on the initial concentration, time and a decay constant.

2. As we saw in Remark 5.4, we then have the following variables and their corresponding dimensions:

$$= AL^{-3}, \quad [t] = T, \quad [a] = T^{-1}, \quad [c_0] = AL^{-3}.$$

3. Hence,

$$[c] = [c_0^{n_1} a^{n_2} t^{n_3}] = [c_0]^{n_1} [a]^{n_2} [t]^{n_3}.$$

4. Substituting the dimensions in, it becomes

$$AL^{-3} = (AL^{-3})^{n_1} (T^{-1})^{n_2} (T)^{n_3}.$$

Remember, as per Remark 5.4, that we need to count AL^{-3} as a single dimension, so we have two independent dimensions: AL^{-3} and T. We need to solve the system above for n_1, n_2 and n_3. This might look hard, but note that, as the dimensions are independent, we can solve for each of them separately (i.e. match the exponents of all T terms and AL^{-3} terms, respectively). We get

$$\begin{aligned} AL^{-3}: &\quad 1 = n_1, \\ T: &\quad 0 = -n_2 + n_3, \end{aligned}$$

hence

$$n_1 = 1, \quad n_2 = n_3.$$

[15] Using pi_j is the traditional notation, and that is why Theorem 5.1 is known as Buckingham Pi Theorem.

Therefore, we can express the concentration as

$$[c] = [c_0 a^{n_2} t^{n_2}] = [c_0 (at)^{n_2}].$$

We end up with a single nondimensional combination of the independent variables, $\pi_1 = at$, which we could have already guessed as we had 3 independent variables and 2 independent dimensions.

5. Thus, we can write the dependency of c on the independent variables as

$$c = c_0 f(at).$$

Let us sanity check the result. If we went and solved Equation 5.2 for the concentration, we would obtain

$$c(t) = c_0 \mathrm{e}^{-at},$$

which is of the form we predicted, with $f(at) = \mathrm{e}^{-at}$, but in our scaling argument we did not need to know or solve any equations. Of course, this is a simple case so solving the equation is not too hard, but you can imagine how this technique can be useful in more complex problems where the equations are not known or are too hard to solve.

This approach is extremely useful when one needs to rely on experiments that cannot be performed in the original setting of the problem because they would be too expensive or too dangerous. For example, imagine we want to design a skyscraper and we want to know how it will react to wind. We cannot build the skyscraper and then wait for a storm to see what happens, but we can build a small-scale model and test it in a wind tunnel. However, we need to make sure that the results we obtain from the model are meaningful for the real skyscraper. This is where dimensional analysis comes into play: it allows us to determine how we need to scale all the variables in the problem to ensure that the results from the experiment are applicable to the real skyscraper.

Example 5.6 (Mars rover)**.** In 2012, NASA's Mars rover Curiosity reached its destination. During the descent, a parachute was used to bring down its velocity. This was possible because Mars has an atmosphere, but that atmosphere is very different to the one in the Earth. As experiments on Mars were out of reach, engineers had to set them up on Earth. For this purpose they had to make sure that the experience and data they collected were meaningful to ensure that everything would work on Mars. One of the problems they had to solve was to determine the size of the parachute, and this is what we are going to do here.

We are interested in the diameter d of the (spherical) parachute.

1. First, we need to determine on which variables it depends. As we said earlier, this is a key modelling step and the right choice is not always obvious. In this case, it sounds reasonable to assume that it will depend on the desired terminal velocity v,

the mass of the rover m, the gravity acceleration g, and the atmospheric viscosity μ.[16]

2. The dimensions of the variable are

$$[d] = L, \quad [v] = \frac{L}{T}, \quad [m] = M, \quad [g] = \frac{L}{T^2}, \quad [\mu] = \frac{M}{LT}.$$

3. The dependency of the dimensions is given by

$$[d] = [v^{n_1} m^{n_2} g^{n_3} \mu^{n_4}] = [v]^{n_1} [m]^{n_2} [g]^{n_3} [\mu]^{n_4}.$$

4. Substituting the dimensions in we obtain

$$L = \left(\frac{L}{T}\right)^{n_1} M^{n_2} \left(\frac{L}{T^2}\right)^{n_3} \left(\frac{M}{LT}\right)^{n_4}.$$

Collecting the powers of each independent dimension we get

$$\begin{aligned} L: &\quad 1 = n_1 + n_3 - n_4, \\ T: &\quad 0 = -n_1 - 2n_3, \\ M: &\quad 0 = n_2 + n_4, \end{aligned}$$

which yields

$$n_1 = 2 + 3n_4, \quad n_2 = -n_4, \quad n_3 = -1 - 2n_4.$$

We choose to leave everything in terms of n_4, but you could choose otherwise. If you did so, you would possibly end up with a different nondimensional combination of the independent variables, but they are equivalent as all possible nondimensional combinations are the power of each other. Anyway, for our choice we obtain

$$[d] = \left[\frac{v^2}{g}\left(\frac{\mu v^3}{m g^2}\right)^{n_4}\right].$$

5. Therefore, the dependency is given by

$$d = \frac{v^2}{g} f\left(\frac{\mu v^3}{m g^2}\right),$$

and you can check that, indeed, the prefactor has dimensions of length and the argument of the function is nondimensional.

[16] The viscosity is a quantity that measures how "thick" a fluid is, and it has dimensions of M/LT (you do not need to remember that for the exam).

The function f is unknown, but that is precisely what the experiments on the Earth would determine. Note that some of the parameters we cannot really change: the mass of the rover, the gravity acceleration and the viscosity of the atmosphere are what they are. But we can change the desired terminal velocity and the diameter of the parachute. So our experiments on Earth would effectively determine the relation between d and v for the given values of m, g and μ. Then, we would decide on a safe value of v for the descent of the rover and plug in the values of g and μ on Mars to determine the size of the parachute.

Example 5.7 (Mass-spring-damper: scaling argument)**.** We cannot finish this chapter without revisiting our good old friend: the mass-spring-damper system. We have already seen how to nondimensionalise it, but let us now see how we can use scaling arguments to determine the dependency of the solution on the parameters.

1. We have one dependent variable x and five independent variables: t, m, c, k and x_0.
2. The dimensions of each variable are
$$\begin{aligned} [x] &= L, & [t] &= T, & [m] &= M, \\ [c] &= MT^{-1}, & [k] &= MT^{-2}, & [x_0] &= L. \end{aligned}$$
3. The relation between dimensions is
$$[x] = [t^{n_1} m^{n_2} c^{n_3} k^{n_4} x_0^{n_5}] = [t]^{n_1} [m]^{n_2} [c]^{n_3} [k]^{n_4} [x_0]^{n_5}.$$
4. Substituting the dimensions in we obtain
$$L = T^{n_1} M^{n_2} (MT^{-1})^{n_3} (MT^{-2})^{n_4} L^{n_5}.$$
The system of equations for the exponents is
$$\begin{aligned} L: &\quad 1 = n_5, \\ T: &\quad 0 = n_1 - n_3 - 2n_4, \\ M: &\quad 0 = n_2 + n_3 + n_4, \end{aligned}$$
which yields
$$n_2 = -\frac{1}{2}(n_1 + n_3), \quad n_4 = \frac{1}{2}(n_1 - n_3), \quad n_5 = 1.$$
I chose to leave everything in terms of n_1 and n_3 so the results have the same form as in Example 5.3. Therefore, we can express the dependency of x on the independent variables as
$$[x] = \left[x_0 \left(\frac{t}{\sqrt{m/k}} \right)^{n_1} \left(\frac{c}{\sqrt{mk}} \right)^{n_3} \right].$$

5. Therefore, we can write the dependency of x on the independent variables as

$$x = x_0 f\left(\frac{t}{\sqrt{m/k}}, \frac{c}{\sqrt{mk}}\right).$$

Note that, comparing with Example 5.3, the first nondimensional variable is the nondimensional time $\tilde{t}$ and the second one is the nondimensional parameter ξ we defined there. Therefore, the scaling argument gives us the same result as the nondimensionalisation.

6 Systems

Aims of the chapter

By the end of this chapter, you should be able to:

- Explain what a system of equations is and how it relates to higher order equations for both difference and differential equations.
- Solve 2×2 first order linear systems of difference equations with constant coefficients.
- Describe the general form of a solution of a inhomogeneous first order linear system of difference equations.
- Solve 2×2 first order linear systems of differential equations with constant coefficients.
- Analyse the stability and produce a phase portrait for a 2×2 first order autonomous systems of differential equations, both linear and nonlinear.

So far, we have studied differential and difference equations for a single dependent variable. But in the same way as you can have systems of algebraic equations,[1] you can also have systems of differential or difference equations. In fact, we already came across a system of differential equation in Example 4.5, where we saw the SEIR model for the spread of an infectious disease. In this chapter we will study systems of first order equations, that is systems composed of several first order equations. We will focus mostly on 2×2 systems, but most of the ideas and concepts generalise to all $n \times n$ systems, $n \in \mathbb{N}$.

Note

Unless stated otherwise for the whole of this chapter we assume that any functions involved in the systems are sufficiently "nice", such that there is a unique solution of the system of equations we are considering.

6.1 Systems of difference equations

We start by considering systems of difference equations.

[1] You might know them as coupled equations.

Definition 6.1. In their most general form, a first order system of difference equations can be written as

$$\begin{aligned} x_{1,k+1} &= f_1\left(x_{1,k}, x_{2,k}, \ldots, x_{n,k}, k\right), \\ x_{2,k+1} &= f_2\left(x_{1,k}, x_{2,k}, \ldots, x_{n,k}, k\right), \\ &\vdots \\ x_{n,k+1} &= f_n\left(x_{1,k}, x_{2,k}, \ldots, x_{n,k}, k\right), \end{aligned}$$

for $k \in \mathbb{N}$ and with some initial conditions $x_{1,0}, x_{2,0}, \ldots, x_{n,0}$. Here $x_{i,k}$ is the ith dependent variable at step k, $f_i : \mathbb{R}^n \times \mathbb{N} \to \mathbb{R}$ are given functions, and n is the number of equations and variables in the system. The system is called an $n \times n$ system of first order difference equations.[2]

It is often convenient to write the system in vector form.[3] This is simply a relabelling of the variables and functions. Let $\boldsymbol{x}_k = (x_{1,k}, x_{2,k}, \ldots, x_{n,k})^T$ and $\boldsymbol{f} = (f_1, f_2, \ldots, f_n)^T$. Then we can write the system as

$$\boldsymbol{x}_{k+1} = \boldsymbol{f}\left(\boldsymbol{x}_k, k\right),$$

where $\boldsymbol{x}_0$ is known and $\boldsymbol{f} : \mathbb{R}^n \times \mathbb{N} \to \mathbb{R}^n$ is a vector-valued function.

We do not give a formal definition, but note that the same classification of difference equations that we introduced in Definition 3.2 naturally generalises to systems. In particular, the order of the system is still the difference between the highest and lowest index of the dependent variables and the system is linear if $\boldsymbol{f}$ is linear.

> **!** Important
>
> For simplicity, in this module we will focus on 2×2 systems of first order difference equations of the form, and to simplify the notation we will usually write them in the form
>
> $$\begin{aligned} x_{k+1} &= f_1\left(x_k, y_k, k\right), \\ y_{k+1} &= f_2\left(x_k, y_k, k\right), \end{aligned} \tag{6.1}$$
>
> with $k \in \mathbb{N}$, some given functions $f_i : \mathbb{R}^2 \times \mathbb{N} \to \mathbb{R}$, $i = 1, 2$ and initial conditions x_0 and y_0. This is equivalent to the notation in Definition 6.1 but using x_k and y_k instead of $x_{1,k}$ and $x_{2,k}$ to simplify the notation.

[2] To be precise, the most general form would be something of the form $f(...) = 0$, as we did with differential equations, but the notation would get even messier so we will stick to the above form.

[3] In this module we will consider vectors to be column vectors, hence why we have all the transpose operations.

6.1.1 Relation to higher order equations

> ! Important
>
> **Lemma 6.1.** *Any (scalar) second order difference equation can be written as a* 2×2 *system of first order equations.*

Proof. Recalling the definition of a general scalar difference equation, let

$$x_{k+2} = f(x_{k+1}, x_k, k)$$

be any second order equation.

Now let $y_k = x_{k+1}$. Then $y_{k+1} = x_{k+2}$ and we can reformulate the equation in the form

$$\begin{aligned} y_{k+1} &= f(y_k, x_k, k)\,, \\ x_{k+1} &= y_k\,. \end{aligned}$$

This is a 2×2 system. □

Remark 6.1. The above concepts for 2×2 systems can be generalised to $n \times n$ systems for $n \in \mathbb{N}$ in a straightforward way.

Example 6.1. Consider the second order difference equation

$$x_{k+2} + e^k x_{k+1} + 10 x_k = \log(k)\,.$$

Writing $y_k = x_{k+1}$ we obtain the system

$$\begin{aligned} y_{k+1} + e^k y_k + 10 x_k &= \log(k)\,, \\ x_{k+1} &= y_k\,. \end{aligned}$$

As we can rewrite the second order equations as a system, this raises the question whether we can do so the other way round as well. In general, this is not possible, however sometimes we can do so.

Example 6.2. Consider the system

$$\begin{aligned} x_{k+1} &= \sin(k)x_k + y_k\,, \\ y_{k+1} &= x_k\,. \end{aligned}$$

Substituting $x_k = y_{k+1}$ and $x_{k+1} = y_{k+2}$ into the first equations, we get

$$y_{k+2} = \sin(k)y_{k+1} + y_k.$$

6.1.2 Linear systems with constant coefficients

Let us now briefly talk about how one solves 2×2 linear systems of difference equations with constant coefficients. This is systems of the form

$$\begin{aligned} x_{k+1} &= a_{1,1}x_k + a_{1,2}y_k, \\ y_{k+1} &= a_{2,1}x_k + a_{2,2}y_k, \end{aligned} \tag{6.2}$$

for $k \in \mathbb{N}$ and some initial conditions.

Note that, defining the matrix

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix},$$

the system can be written, in vector form, as

$$\boldsymbol{x}_{k+1} = A\boldsymbol{x}_k.$$

A simple way to solve the system would be to convert it into a second order difference equation (as it is linear, we will always be able to) and solve it as we did in Section 3.4. However, we can also solve the system directly using the eigenvalues and eigenvectors of the matrix A. This is a more general approach that can be used for any $n \times n$ linear system of difference equations with constant coefficients.

However, let us study it as a system. Similarly to what we did in Section 3.2.1.1, we can calculate the first few values of $\boldsymbol{x}_k$ to deduce a pattern:

$$\begin{aligned} \boldsymbol{x}_1 &= A\boldsymbol{x}_0, \\ \boldsymbol{x}_2 &= A\boldsymbol{x}_1 = A^2\boldsymbol{x}_0, \\ \boldsymbol{x}_3 &= A\boldsymbol{x}_2 = A^3\boldsymbol{x}_0, \\ &\vdots \\ \boldsymbol{x}_k &= A^k\boldsymbol{x}_0. \end{aligned} \tag{6.3}$$

Here, A^k is the kth power of the matrix A, i.e. $A^k = A \cdot A \cdots A$ (k times). Therefore, the solution to the system can be expressed as $\boldsymbol{x}_k = A^k \boldsymbol{x}_0$. Of course, one needs to compute A^k to get the solution. We will not do this in this module, but if you take a linear algebra module, you will learn how to do that using the eigenvalues and eigenvectors of A.

6.2 Systems of differential equations

Let us now consider first order systems of differential equations. In vector notation, the most general form of an $m \times m$ first order system of differential equations is

$$\boldsymbol{\mathcal{F}}\left(t, \boldsymbol{x}(t), \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}(t)\right) = \mathbf{0}, \quad \text{with} \quad \boldsymbol{x}(t_0) = \boldsymbol{x}_0, \tag{6.4}$$

where $\boldsymbol{\mathcal{F}} : (\alpha, \beta) \times (\mathbb{R}^m)^2 \to \mathbb{R}^m$ is a given function, $\boldsymbol{x}(t) = (x_1(t), x_2(t), \ldots, x_m(t))^T$ is the vector of dependent variables, and $\boldsymbol{x}_0$ is a given initial condition.

> **!** Important
>
> In this module we will constrain ourselves to studying 2×2 systems of first order equations, and we will assume we can always write them in the form
>
> $$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}x(t) &= f_1\left(x(t), y(t), t\right), \quad \text{with} \quad x(t_0) = x_0, \\ \frac{\mathrm{d}}{\mathrm{d}t}y(t) &= f_2\left(x(t), y(t), t\right), \quad \text{with} \quad y(t_0) = y_0, \end{aligned} \tag{6.5}$$
>
> for $t \in (\alpha, \beta)$ and with some given functions $f_i : \mathbb{R}^2 \times (\alpha, \beta) \to \mathbb{R}$, where $i = 1, 2$.

Remark 6.2. A couple of remarks. First, as it happened with systems of difference equations, the definitions of order, linearity, etc. that we introduced for scalar differential equations in Definition 2.1 generalise to systems of differential equations. Second, we will interchangeably use the notation $x(t)$, $y(t)$ and $x_1(t)$, $x_2(t)$ for the dependent variables, depending on which is more convenient in the context.

Note that we assumed that t belongs to the open interval (α, β). If $t_0 = \alpha$ or $t_0 = \beta$ then we implicitly assume that we can extend the solution to these boundary points.

6.2.1 Relation to higher order equations

> **!** Important
>
> **Lemma 6.2.** *Any (scalar) second order differential equation can be written as a* 2×2 *system of first order equations.*

Recalling the definition Equation 2.1 of a general scalar differential equation, let

$$\mathcal{F}(x, y(x), y'(x), y''(x)) = 0$$

be any second order equation. Let now $\eta(x) = y'(x)$. Then $\eta'(x) = y''(x)$ and we can reformulate the equation in the form

$$\begin{aligned} y'(x) - \eta(x) &= 0, \\ \mathcal{F}(x, y(x), \eta(x), \eta'(x)) &= 0. \end{aligned}$$

This is a 2×2 system of the most general form Equation 6.4. These concepts for 2×2 systems can be generalised to $n \times n$ systems for $n \in \mathbb{N}$ in a straightforward way, including rewriting higher order equations of nth order as a system.

Example 6.3. Consider the second order equation

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x + \cos(t)\frac{\mathrm{d}}{\mathrm{d}t}x - x^2 = 0.$$

Writing $x_1(t) = x(t)$ and $x_2(t) = \frac{\mathrm{d}}{\mathrm{d}t}x_1(t)$ we obtain the system

$$\frac{\mathrm{d}}{\mathrm{d}t}x_1 - x_2 = 0, \quad \frac{\mathrm{d}}{\mathrm{d}t}x_2 + \cos(t)x_2 - x_1^2 = 0.$$

As we can rewrite the second order equation as a system this raises the question whether we can do so the other way round as well. In general, this is not possible, however sometimes we can do so.

Example 6.4. Consider the system

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}x_1 &= x_1 + x_2, \\ \frac{\mathrm{d}}{\mathrm{d}t}x_2 &= 4x_1 - 2x_2 + 4e^{-2t}. \end{aligned}$$

Rearranging the first equation we see that

$$x_2 = \frac{\mathrm{d}}{\mathrm{d}t}x_1 - x_1 \quad \Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}x_2 = \frac{\mathrm{d}^2}{\mathrm{d}t^2}x_1 - \frac{\mathrm{d}}{\mathrm{d}t}x_1.$$

Therefore, using the second equation,

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x_1 - \frac{\mathrm{d}}{\mathrm{d}t}x_1 = \frac{\mathrm{d}}{\mathrm{d}t}x_2 = 4x_1 - 2x_2 + 4e^{-2t} = 4x_1 - 2\Big(\frac{\mathrm{d}}{\mathrm{d}t}x_1 - x_1\Big) + 4e^{-2t}.$$

Rearranging yields the second order equation

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x_1 + \frac{\mathrm{d}}{\mathrm{d}t}x_1 - 6x_1 = 4e^{-2t}.$$

6.3 Linear systems

We will first focus on linear systems of differential equations, particularly with constant coefficients. Before we analyse the latter in more detail in Section 6.4, we first establish some more general results for linear systems where the coefficients can depend on the independent variable. These are of the form

! Important

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}x_1(t) &= a_{1,1}(t)x_1(t) + a_{1,2}(t)x_2(t) + s_1(t), \\ \frac{\mathrm{d}}{\mathrm{d}t}x_2(t) &= a_{2,1}(t)x_1(t) + a_{2,2}(t)x_2(t) + s_2(t), \end{aligned} \tag{6.6}$$

where the $a_{i,j} : [\alpha, \beta] \to \mathbb{R}$ and the $s_i : [\alpha, \beta] \to \mathbb{R}$ are given functions, $i, j \in \{1, 2\}$. We can formulate this system in matrix-vector form as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}(t) = A(t)\boldsymbol{x}(t) + \boldsymbol{s}(t), \tag{6.7}$$

with

$$A(t) = \begin{pmatrix} a_{1,1}(t) & a_{1,2}(t) \\ a_{2,1}(t) & a_{2,2}(t) \end{pmatrix} \in \mathbb{R}^{2\times 2}, \quad \boldsymbol{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}, \quad \boldsymbol{s}(t) = \begin{pmatrix} s_1(t) \\ s_2(t) \end{pmatrix}.$$

This notation, in addition to being more compact, allows us to use the tools of linear algebra to study the system later on.

6.3.1 Homogeneous system, complementary function

! Important

We first study the homogeneous case $\boldsymbol{s}(t) = 0$ for all $t \in [\alpha, \beta]$ in Equation 6.7, i.e.

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}(t) = A(t)\boldsymbol{x}(t), \tag{6.8}$$

with some suitable initial condition.
Suppose that $\boldsymbol{x}^{(1)}(t)$ and $\boldsymbol{x}^{(2)}(t)$ are two solutions. Thanks to the linearity of the differential equation any *linear combination*

$$l_1\boldsymbol{x}^{(1)}(t) + l_2\boldsymbol{x}^{(2)}(t), \quad l_1, l_2 \in \mathbb{R},\ t \in (\alpha, \beta), \tag{6.9}$$

then is a solution, too.

Definition 6.2 (Linearly independent functions)**.** Two functions $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} : [\alpha, \beta] \to \mathbb{R}^2$ are *linearly independent*[4] if the only solution of

$$c_1\boldsymbol{x}^{(1)}(t) + c_2\boldsymbol{x}^{(2)}(t) = 0 \quad \text{for any } c_1, c_2 \in \mathbb{R} \tag{6.10}$$

at all times $t \in [\alpha, \beta]$ is $c_1 = c_2 = 0$.

Observe that in the this case where we have two functions only, linear dependence effectively means that the two functions are multiples of each other, because if $c_1, c_2 \neq 0$ exist then we can write $\boldsymbol{x}^{(1)}(t) = -\frac{c_2}{c_1}\boldsymbol{x}^{(2)}(t)$ and $\boldsymbol{x}^{(2)} = -\frac{c_1}{c_2}\boldsymbol{x}^{(1)}$. (We here omit a tedious discussion of trivial cases when any of the functions $\boldsymbol{x}^{(i)}$ or any of the coefficients c_i is zero.)

Example 6.5. Consider $\boldsymbol{x}^{(1)}(t) = (t, 1)$ and $\boldsymbol{x}^{(2)}(t) = (t^2, t)$ for $t \in [0, 2]$ and let $c_1, c_2 \in \mathbb{R}$ be such that

$$c_1\boldsymbol{x}^{(1)}(t) + c_2\boldsymbol{x}^{(2)}(t) = 0 \quad \Leftrightarrow \quad \begin{pmatrix} c_1 t + c_2 t^2 \\ c_1 + c_2 t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

at all times $t \in [0, 2]$. Choosing $t = 0$ we see from the second equation that necessarily $c_1 = 0$. If we then choose $t = 2$ then the second equation yields that $2c_2 = 0$, so necessarily $c_2 = 0$ as well.

Note that we are free to choose the time. In fact, if we only consider a specific time we might get a wrong answer. Say, if we only consider $t = 1$ then both equations read $c_1 + c_2 = 0$, and there are many solutions to this equation, for instance, $c_1 = 42$ and $c_2 = -42$.

[4] The concept of linear independence naturally extends from the scalar case to the vector case.

! Important

Theorem 6.1. *Assume that* $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} : (\alpha, \beta) \to \mathbb{R}^2$ *are two linearly independent solutions of Equation 6.8. Then all solutions are of the form*

$$\boldsymbol{x}(t) = l_1 \boldsymbol{x}^{(1)}(t) + l_2 \boldsymbol{x}^{(2)}(t), \quad l_1, l_2 \in \mathbb{R},\ t \in [\alpha, \beta].$$

Moreover, if an initial condition of the form

$$\boldsymbol{x}(t_0) = \boldsymbol{x_0} \in \mathbb{R}^2 \quad \text{for some } t_0 \in [\alpha, \beta] \tag{6.11}$$

is imposed then l_1, l_2 *are given as the unique solution of*

$$\boldsymbol{x}_0 = l_1 \boldsymbol{x}^{(1)}(t_0) + l_2 \boldsymbol{x}^{(2)}(t_0). \tag{6.12}$$

Proof. We omit the details of the proof, but it follows easily from substituting the linear combination Equation 6.9 into the system Equation 6.8 and using the linear independence of $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(2)}$ to show that any solution must be of this form. The uniqueness of l_1, l_2 follows from the linear independence as well. □

Definition 6.3 (Complementary function)**.** Given any two linearly independent solutions $\boldsymbol{x}^{(1)}$, $\boldsymbol{x}^{(2)}$ of the 2×2 linear system of homogeneous differential equation Equation 6.8, the set of functions

$$l_1 \boldsymbol{x}^{(1)} + l_2 \boldsymbol{x}^{(2)} : (\alpha, \beta) \to \mathbb{R}^2, \quad l_1, l_2 \in \mathbb{R},$$

is called *general solution* or *complementary function.*

In summary, Theorem 6.1 implies that this general solution is the set of *all* solutions to the differential equation, and if an initial condition of the form Equation 6.11 is imposed then the system Equation 6.12 has to be solved to identify the then unique coefficients l_1, l_2.

6.3.2 Inhomogeneous system, particular integral

Let us now consider the inhomogeneous case, i.e. Equation 6.7 where $\boldsymbol{s}(t) \neq \boldsymbol{0}$ for at least some $t \in (\alpha, \beta)$. It turns out that knowing only one solution to the differential equation is sufficient to find all solutions and the one for a specific initial condition.

! Important

Theorem 6.2 (Particular integral)**.** *Assume that* $\boldsymbol{x}_p : (\alpha, \beta) \to \mathbb{R}^2$ *is a solution to the inhomogeneous system Equation 6.7, called the* particular integral*, and that* $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} : (\alpha, \beta) \to \mathbb{R}^2$ *are two linearly independent solutions of the corresponding*

homogeneous system Equation 6.8. Then all solutions to the inhomogeneous system Equation 6.7 are of the form

$$\boldsymbol{x}(t) = \boldsymbol{x}_p(t) + l_1\boldsymbol{x}^{(1)}(t) + l_2\boldsymbol{x}^{(2)}(t), \quad l_1, l_2 \in \mathbb{R},\ t \in [\alpha, \beta].$$

Moreover, if we impose an initial condition of the form in Equation 6.11, then l_1, l_2 are given as the unique solution of

$$\boldsymbol{x}_0 - \boldsymbol{x}_p(t_0) = l_1\boldsymbol{x}^{(1)}(t_0) + l_2\boldsymbol{x}^{(2)}(t_0). \tag{6.13}$$

Proof. Assume that $\boldsymbol{x}$ is a solution of the inhomogeneous system Equation 6.7. Define now the function $\boldsymbol{x}_h(t) = \boldsymbol{x}(t) - \boldsymbol{x}_p(t)$, $t \in (\alpha, \beta)$. Observe that it satisfies

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}_h(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\boldsymbol{x}(t) - \boldsymbol{x}_p(t)\right) \\
&= \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}(t) - \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}_p(t) \\
&= A(t)\boldsymbol{x}(t) + s(t) - A(t)\boldsymbol{x}_p(t) - s(t) \\
&= A(t)\left(\boldsymbol{x}(t) - \boldsymbol{x}_p(t)\right) \\
&= A(t)\boldsymbol{x}_h(t),
\end{aligned}$$

so $\boldsymbol{x}_h$ is a solution to the homogeneous system Equation 6.8. From Theorem 6.1 we know that its solutions are of the form

$$\boldsymbol{x}_h(t) = l_1\boldsymbol{x}^{(1)}(t) + l_2\boldsymbol{x}^{(2)}(t), \quad l_1, l_2 \in \mathbb{R},\ t \in [\alpha, \beta].$$

Altogether, $\boldsymbol{x}(t) = \boldsymbol{x}_p(t) + \boldsymbol{x}_h(t) = \boldsymbol{x}_p(t) + l_1\boldsymbol{x}^{(1)}(t) + l_2\boldsymbol{x}^{(2)}(t)$ is of the form as claimed.

We may proceed as in the proof of Theorem 6.1 to show that Equation 6.13 has a unique solution l_1, l_2. Then

$$\boldsymbol{x}(t_0) = \boldsymbol{x}_p(t_0) + l_1\boldsymbol{x}^{(1)}(t_0) + l_2\boldsymbol{x}^{(2)}(t_0) = \boldsymbol{x}_p(t_0) + \boldsymbol{x}_0 - \boldsymbol{x}_p(t_0) = \boldsymbol{x}_0,$$

so $\boldsymbol{x}$ also satisfies the correct initial condition. As we assume that the system is well-posed, the solution to the initial value problem is unique, so there is no solution other than $\boldsymbol{x} = \boldsymbol{x}_p + l_1\boldsymbol{x}^{(1)} + l_2\boldsymbol{x}^{(2)}$. □

Such a solution $\boldsymbol{x}_p$ of the inhomogeneous system Equation 6.7 is known in the literature as a *particular integral*. We see in Theorem 6.2 how it can be used together with solutions to the corresponding homogeneous system to synthesize the solution to the initial value problem as a linear combination. This idea is called *superposition principle* and generally works for linear models, not only for differential equations. In practice, finding a particular integral can be quite challenging, though.

6.4 Homogeneous linear systems with constant coefficients

We learned in the previous section that two linearly independent solutions to the linear system (plus one particular integral in the inhomogeneous case) are sufficient to solve any initial value problem. In the case of constant coefficients it is actually possible to find such two linearly independent solutions.

! Important

Let us look into linear systems of the form

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}x_1(t) &= a_{1,1}x_1(t) + a_{1,2}x_2(t), \\ \frac{\mathrm{d}}{\mathrm{d}t}x_2(t) &= a_{2,1}x_1(t) + a_{2,2}x_2(t), \end{aligned} \qquad t \in \mathbb{R},$$

where the $a_{i,j} \in \mathbb{R}$ now are independent of t. These can be formulated in the matrix-vector form

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}(t) = A\boldsymbol{x}(t), \qquad A = \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} \in \mathbb{R}^{2\times 2}, \quad \boldsymbol{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}. \tag{6.14}$$

We found that exponential functions have been working well with the linear equations that we saw so far, so let us try to find a solution in the form

$$\boldsymbol{x}(t) = e^{\lambda t}\boldsymbol{v},$$

where $\lambda \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^2$ are to be determined from the system.

Substituting this approach in the system yields

$$\lambda e^{\lambda t}\boldsymbol{v} = \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}(t) = A\boldsymbol{x}(t) = e^{\lambda t}A\boldsymbol{v}.$$

Using that $e^{\lambda t} \neq 0$ we obtain that

$$\lambda\boldsymbol{v} = A\boldsymbol{v}. \tag{6.15}$$

So $\boldsymbol{x}(t) = e^{\lambda t}\boldsymbol{v}$ is a solution if λ and $\boldsymbol{v}$ solve Equation 6.15. It turns out that there are at most two distinct pairs of λ and $\boldsymbol{v}$ that solve this equation. For a given matrix A, values of λ and $\boldsymbol{v}$ that satisfy Equation 6.15 are called the eigenvalues and eigenvectors of the matrix. We will discuss them in more detail in Section 6.4.1.

6.4.1 Eigenvalues and eigenvectors

In general, the 2×2 matrix A has at most two distinct eigenvalues (similarly, an $n \times n$ matrix would have at most n distinct eigenvalues) and each eigenvalue is paired with a corresponding eigenvector solving Equation 6.15. Assuming this is the case we denote them by λ_1, λ_2 and corresponding eigenvectors by $\boldsymbol{v}^{(1)}, \boldsymbol{v}^{(2)}$. These eigenvectors are always linearly independent.[5]

! Important

Definition 6.4. The eigenvalues of a matrix $A \in \mathbb{R}^{2\times 2}$ are the roots to its *characteristic polynomial*, $\det(\lambda I - A) = 0$.

This gives the polynomial

$$\lambda^2 - (a_{1,1} + a_{2,2})\lambda + a_{1,1}a_{2,2} - a_{2,1}a_{1,2}\,,$$

with roots

$$\lambda_{1,2} = \frac{a_{1,1} + a_{2,2}}{2} \pm \frac{1}{2}\sqrt{(a_{1,1} - a_{2,2})^2 + 4a_{2,1}a_{1,2}}. \tag{6.16}$$

Remark 6.3 (Complex eigenvalues). From Equation 6.16 we see that the eigenvalues could be complex. In this case, the corresponding eigenvectors are also complex, so from now on we will allow $\boldsymbol{v}^{(i)} \in \mathbb{C}^2$ and $\lambda_i \in \mathbb{C}$, $i = 1, 2$.

Definition 6.5. The corresponding eigenvectors $\boldsymbol{v}^{(1)}$ and $\boldsymbol{v}^{(2)}$ to eigenvalues λ_1 and λ_2 are the solutions $\boldsymbol{v}^{(i)} \in \mathbb{C}^2$, $\boldsymbol{v}^{(i)} \neq (0, 0)$, of

$$(\lambda_i I - A)\boldsymbol{v}^{(i)} = 0, \quad i = 1, 2.$$

There are two equations, and if $\boldsymbol{v}^{(i)}$ is an eigenvector then also every multiple $c\boldsymbol{v}^{(i)}$, $c \in \mathbb{C}$, is an eigenvector again. In practice, we just choose an arbitrary instance of $c\boldsymbol{v}^{(i)}$ to represent the eigenvector, for instance, by setting one of the components to 1 and solving for the other component.

Example 6.6. Consider the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}.$$

In this case $a_{1,1} = 1$, $a_{1,2} = 2$, $a_{2,1} = 1$ and $a_{2,2} = 0$ so that

$$\lambda_{1,2} = \frac{1}{2} \pm \frac{1}{2}\sqrt{1 - 4(-2)} = \frac{1}{2} \pm \frac{3}{2},$$

[5] This will be proved in MA149 Linear Algebra

therefore $\lambda_1 = 2$ and $\lambda_2 = -1$. An eigenvector $\boldsymbol{v}^{(1)} = (v_1^{(1)}, v_2^{(1)})$ corresponding to λ_1 satisfies

$$0 = (\lambda_1 I - A)\boldsymbol{v} = \begin{pmatrix} 2-1 & -2 \\ -1 & 2-0 \end{pmatrix} \begin{pmatrix} v_1^{(1)} \\ v_2^{(1)} \end{pmatrix} = \begin{pmatrix} v_1^{(1)} - 2v_2^{(1)} \\ -v_1^{(1)} + 2v_2^{(1)} \end{pmatrix}.$$

We see that a solution is given by $\boldsymbol{v}^{(1)} = (2, 1)$. Let us briefly check that this is an eigenvalue for λ_1:

$$A\boldsymbol{v}^{(1)} = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \lambda_1 \boldsymbol{v}^{(1)}.$$

Regarding λ_2, a corresponding eigenvector $\boldsymbol{v}^{(2)} = (v_1^{(2)}, v_2^{(2)})$ satisfies

$$0 = (\lambda_2 I - A)\boldsymbol{v} = \begin{pmatrix} -1-1 & -2 \\ -1 & -1-0 \end{pmatrix} \begin{pmatrix} v_1^{(2)} \\ v_2^{(2)} \end{pmatrix} = \begin{pmatrix} -2v_1^{(2)} - 2v_2^{(2)} \\ -v_1^{(2)} - v_2^{(2)} \end{pmatrix}.$$

A solution to these equations is $\boldsymbol{v}^{(2)} = (1, -1)$, and it can be checked similarly.

6.4.2 Solving homogeneous linear systems with constant coefficients

We now have that for eigenvalues λ_1 and λ_2 with corresponding eigenvectors $\boldsymbol{v}^{(1)}$ and $\boldsymbol{v}^{(2)}$, the functions $\boldsymbol{x}^{(1)}(t) = e^{\lambda_1 t}\boldsymbol{v}^{(1)}$ and $\boldsymbol{x}^{(2)}(t) = e^{\lambda_1 t}\boldsymbol{v}^{(2)}$ then are solutions to the homogenenous system of linear equations with constant coefficients, Equation 6.14. Moreover, these solutions are linearly independent.[6] The previous results on linear systems apply, in particular Theorem 6.1.[7]

! Important

Hence, all solutions of Equation 6.14 are of the form of the complementary function,

$$\boldsymbol{x}(t) = l_1\boldsymbol{x}^{(1)}(t) + l_2\boldsymbol{x}^{(2)}(t) = l_1 e^{\lambda_1 t}\boldsymbol{v}^{(1)} + l_2 e^{\lambda_2 t}\boldsymbol{v}^{(2)}, \quad l_1, l_2 \in \mathbb{C},$$

and the unique solution to the initial value problem consisting of Equation 6.14 and Equation 6.11 is obtained by substituting $\boldsymbol{x}_0$ and solving for l_1, l_2.

[6] Just evaluate them at $t = 0$ to obtain the $\boldsymbol{v}^{(i)}$, which are linearly independent.

[7] Actually, we have studied the real-valued case only, but the steps of the proof still are valid if the functions $\boldsymbol{x}^{(i)} : (\alpha, \beta) \to \mathbb{C}^2$ are complex-valued and if the l_i are complex.

Like we did with second order linear equations, we will now consider three separate cases for the eigenvalues of A and discuss the corresponding solutions to the system of differential equations.[8]

6.4.3 Real eigenvalues

> **!** Important
>
> If $(a_{1,1}-a_{2,2})^2+4a_{2,1}a_{1,2}>0$ then the two eigenvalues are real and distinct and the corresponding eigenvectors $\boldsymbol{v}^{(1)},\boldsymbol{v}^{(2)}\in\mathbb{R}^2$ are linearly independent. The functions
>
> $$\boldsymbol{x}^{(1)}(t)=e^{\lambda_1 t}\boldsymbol{v}^{(1)},\quad \boldsymbol{x}^{(2)}(t)=e^{\lambda_2 t}\boldsymbol{v}^{(2)}$$
>
> are then are linearly independent solutions, with general solution
>
> $$\boldsymbol{x}(t)=l_1e^{\lambda_1 t}\boldsymbol{v}^{(1)}+l_2e^{\lambda_2 t}\boldsymbol{v}^{(2)}\,.$$
>
> Then the initial conditions $\boldsymbol{x}(t_0)=\boldsymbol{x}_0$ can be used to determine $l_1,l_2\in\mathbb{R}$.

Example 6.7. Consider Equation 6.14 with an initial condition of the form Equation 6.11 where $t_0=0$ and

$$A=\begin{pmatrix}2&6\\-2&-5\end{pmatrix},\quad \boldsymbol{x}_0=\begin{pmatrix}1\\0\end{pmatrix}.$$

The eigenvalues of A satisfy $\lambda^2+3\lambda+2=0$ and are

$$\lambda_{1,2}=-\frac{3}{2}\pm\frac{1}{2}\sqrt{3^2-4\cdot 2}\quad\Longrightarrow\quad \lambda_1=-1,\quad \lambda_2=-2.$$

An eigenvector $\boldsymbol{v}^{(1)}$ has to satisfy

$$0=(\lambda_1 I-A)\boldsymbol{v}^{(1)}=\begin{pmatrix}-3&-6\\2&4\end{pmatrix}\boldsymbol{v}^{(1)}.$$

A possible choice is $\boldsymbol{v}^{(1)}=(-2,1)$. Proceeding similarly for λ_2 we find that a corresponding eigenvector is given by $\boldsymbol{v}^{(2)}=(-3,2)$.

A general solution to the differential equations is thus given by

[8] At this point you probably have noticed the similiarities between 2×2 linear systems and second order linear equations. You can dig deeper in this similarity by converting a second order linear equation into a 2×2 linear system and vice versa, as we did in Example 6.4, and solving both.

$$\boldsymbol{x}(t) = l_1 e^{\lambda_1 t}\boldsymbol{v}^{(1)} + l_2 e^{\lambda_2 t}\boldsymbol{v}^{(2)} = l_1 e^{-t}\begin{pmatrix}-2\\1\end{pmatrix} + l_2 e^{-2t}\begin{pmatrix}-3\\2\end{pmatrix}, \quad l_1, l_2 \in \mathbb{R}.$$

The initial condition is satisfied if

$$\boldsymbol{x}_0 = \boldsymbol{x}(0) = l_1\boldsymbol{v}^{(1)} + l_2\boldsymbol{v}^{(2)} \quad \Leftrightarrow \quad \begin{pmatrix}1\\0\end{pmatrix} = l_1\begin{pmatrix}-2\\1\end{pmatrix} + l_2\begin{pmatrix}-3\\2\end{pmatrix}.$$

The solution to this 2×2 systems is $l_1 = -2$, $l_2 = 1$, so the unique solution of Equation 6.14, Equation 6.11 with the above data is

$$\boldsymbol{x}(t) = -2e^{-t}\begin{pmatrix}-2\\1\end{pmatrix} + e^{-2t}\begin{pmatrix}-3\\2\end{pmatrix}.$$

Figure 6.1 shows the solution in the x_1-x_2 plane. To facilitate the understanding of the solution, the arrows indicate the direction of the solution at each point $\boldsymbol{x}$, which is given by $A\boldsymbol{x}$.[9]

As $\lambda_2 < \lambda_1 < 0$ the solution converges to $(0,0)$ as $t \to \infty$ and, you can imagine, that if either of the eigenvalues were positive then the solution would diverge to infinity. We will discuss this in more detail in Section 6.5.

6.4.4 Complex eigenvalues

> **!** Important
>
> If $(a_{1,1} - a_{2,2})^2 + 4a_{2,1}a_{1,2} < 0$ then the two eigenvalues are complex and satisfy
>
> $$\lambda_{1,2} = \frac{a_{1,1} + a_{2,2}}{2} \pm \frac{i}{2}\sqrt{-4a_{2,1}a_{1,2} - (a_{1,1} - a_{2,2})^2},$$
>
> where $\lambda_2 = \overline{\lambda_1}$ (complex conjugate).

Moreover $\boldsymbol{v}^{(1)} \in \mathbb{C}^2$ is an eigenvector corresponding to λ_1 if and only if its conjugate $\overline{\boldsymbol{v}^{(1)}}$ is an eigenvector corresponding to $\lambda_2 = \overline{\lambda_1}$. We do not prove it here, but combining both results we obtain

$$\overline{e^{\lambda_1 t}\boldsymbol{v}^{(1)}} = e^{\lambda_2 t}\boldsymbol{v}^{(2)}.$$

Recall that any linear combinations

[9] I have normalised the arrows to have the same length, so it is easier to read.

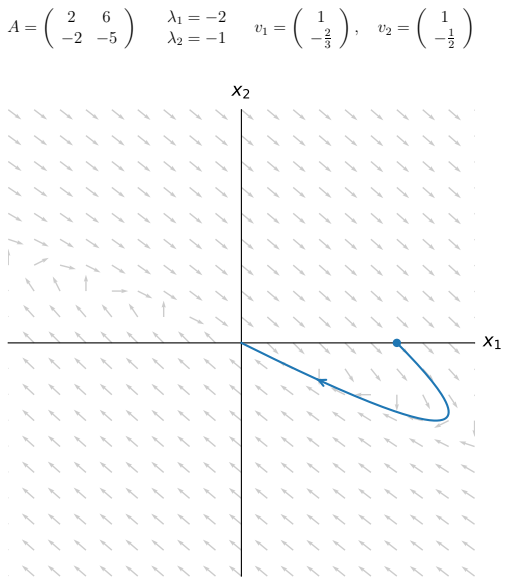


Figure 6.1: Plot of the solution of Example 6.7.

$$l_1 e^{\lambda_1 t}\boldsymbol{v}^{(1)} + l_2 e^{\lambda_2 t}\boldsymbol{v}^{(2)}, \quad l_1, l_2 \in \mathbb{C},$$

are solutions to the system of differential equations. But note that the real and imaginary parts of each independent solution can be written as a linear combination of the two independent solutions:

$$\boldsymbol{x}^{(1)}(t) = \operatorname{Re}\left[e^{\lambda_1 t}\boldsymbol{v}^{(1)}\right] = \frac{1}{2}\left(e^{\lambda_1 t}\boldsymbol{v}^{(1)} + \overline{e^{\lambda_1 t}\boldsymbol{v}^{(1)}}\right) = \frac{1}{2}e^{\lambda_1 t}\boldsymbol{v}^{(1)} + \frac{1}{2}e^{\lambda_2 t}\boldsymbol{v}^{(2)} = \operatorname{Re}\left[e^{\lambda_2 t}\boldsymbol{v}^{(2)}\right],$$
$$\boldsymbol{x}^{(2)}(t) = \operatorname{Im}\left[e^{\lambda_1 t}\boldsymbol{v}^{(1)}\right] = \frac{1}{2i}\left(e^{\lambda_1 t}\boldsymbol{v}^{(1)} - \overline{e^{\lambda_1 t}\boldsymbol{v}^{(1)}}\right) = \frac{1}{2i}e^{\lambda_1 t}\boldsymbol{v}^{(1)} - \frac{1}{2i}e^{\lambda_2 t}\boldsymbol{v}^{(2)} = -\operatorname{Im}\left[e^{\lambda_2 t}\boldsymbol{v}^{(2)}\right],$$

where Re represents the real component, p, of a complex number $p+iq$ and Im represents the imaginary component, q.

To identify and further work out real and imaginary parts let us write

$$\lambda_1 = p + i\omega \text{ with } p, \omega \in \mathbb{R}, \quad \text{and } \boldsymbol{v}^{(1)} = \boldsymbol{r} + i\boldsymbol{b} \text{ with } \boldsymbol{r}, \boldsymbol{b} \in \mathbb{R}^2.$$

Note that $\omega \neq 0$, and one can show that $\boldsymbol{r} \neq (0,0)$ and $\boldsymbol{b} \neq (0,0)$. Then

$$e^{\lambda_1 t} = e^{pt}(\cos(\omega t) + i\sin(\omega t)),$$

and consequently

$$\begin{aligned} e^{\lambda_1 t}\boldsymbol{v}^{(1)} &= e^{pt}\left(\cos(\omega t) + i\sin(\omega t)\right)(\boldsymbol{r} + i\boldsymbol{b}) \\ &= e^{pt}\left(\cos(\omega t)\boldsymbol{r} - \sin(\omega t)\boldsymbol{b}\right) + ie^{pt}\left(\sin(\omega t)\boldsymbol{r} + \cos(\omega t)\boldsymbol{b}\right). \end{aligned}$$

! Important

Two linearly independent real-valued solutions therefore are

$$\boldsymbol{x}^{(1)}(t) = e^{pt}\left(\cos(\omega t)\boldsymbol{r} - \sin(\omega t)\boldsymbol{b}\right), \quad \boldsymbol{x}^{(2)}(t) = e^{pt}\left(\sin(\omega t)\boldsymbol{r} + \cos(\omega t)\boldsymbol{b}\right).$$

We obtain a general (real-valued) solution of the form

$$\begin{aligned} \boldsymbol{x}(t) &= l_1\boldsymbol{x}^{(1)}(t) + l_2\boldsymbol{x}^{(2)}(t) \\ &= l_1 e^{pt}\left(\cos(\omega t)\boldsymbol{r} - \sin(\omega t)\boldsymbol{b}\right) + l_2 e^{pt}\left(\sin(\omega t)\boldsymbol{r} + \cos(\omega t)\boldsymbol{b}\right), \end{aligned} \tag{6.17}$$

for $l_1, l_2 \in \mathbb{R}$.
Any initial condition of the form Equation 6.11 with $t_0 = 0$ (this is only for in the notation simplicity) then becomes

$$\boldsymbol{x}_0 = \boldsymbol{x}(0) = l_1\boldsymbol{r} + l_2\boldsymbol{b}.$$

Remark 6.4. Note that cos and sin will give rise to oscillations, which in the two-dimensional case manifest as spiralling around. For $p = 0$ this will become a circle, maintaining a fixed distance away from $(0, 0)$.

Example 6.8. Consider Equation 6.14 with an initial condition of the form Equation 6.11 where $t_0 = 0$ and

$$A = \begin{pmatrix} 1 & 5 \\ -1 & -3 \end{pmatrix}, \quad \boldsymbol{x}_0 = \begin{pmatrix} 20 \\ 0 \end{pmatrix}.$$

The eigenvalues of A satisfy $\lambda^2 + 2\lambda + 2 = 0$ and are

$$\lambda_{1,2} = -\frac{2}{2} \pm \frac{1}{2}\sqrt{2^2 - 4 \cdot 2} = -1 \pm i, \quad \text{so } p = -1,\ \omega = 1.$$

An eigenvector corresponding to λ_1 can be worked out to be

$$\boldsymbol{v}^{(1)} = \begin{pmatrix} 5 \\ -2 + i \end{pmatrix}, \quad \text{so } \boldsymbol{r} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}, \boldsymbol{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A general solution to the differential equations is thus given by

$$\begin{aligned}\boldsymbol{x}(t) &= l_1 e^{pt}\left(\cos(\omega t)\boldsymbol{r} - \sin(\omega t)\boldsymbol{b}\right) + l_2 e^{pt}\left(\sin(\omega t)\boldsymbol{r} + \cos(\omega t)\boldsymbol{b}\right)\\ &= l_1 e^{-t}\cos(t)\begin{pmatrix}5\\-2\end{pmatrix} - l_1 e^{-t}\sin(t)\begin{pmatrix}0\\1\end{pmatrix} + l_2 e^{-t}\sin(t)\begin{pmatrix}5\\-2\end{pmatrix} + l_2 e^{-t}\cos(t)\begin{pmatrix}0\\1\end{pmatrix}\\ &= l_1 e^{-t}\begin{pmatrix}5\cos(t)\\-2\cos(t)-\sin(t)\end{pmatrix} + l_2 e^{-t}\begin{pmatrix}5\sin(t)\\-2\sin(t)+\cos(t)\end{pmatrix}.\end{aligned}$$

The initial condition is satisfied if

$$\boldsymbol{x}_0 = l_1\boldsymbol{r} + l_2\boldsymbol{b} \quad \Leftrightarrow \quad \begin{pmatrix}20\\0\end{pmatrix} = l_1\begin{pmatrix}5\\-2\end{pmatrix} + l_2\begin{pmatrix}0\\1\end{pmatrix}.$$

Hence $l_1 = 4$ and $l_2 = 8$, and the unique solution of Equation 6.14, Equation 6.11 with the above data is

$$\begin{aligned}\boldsymbol{x}(t) &= e^{-t}\begin{pmatrix}20\cos(t)\\-8\cos(t)-4\sin(t)\end{pmatrix} + e^{-t}\begin{pmatrix}40\sin(t)\\-16\sin(t)+8\cos(t)\end{pmatrix}\\ &= e^{-t}\begin{pmatrix}20\cos(t)+40\sin(t)\\-20\sin(t)\end{pmatrix}.\end{aligned}$$

The plot of the solution is shown in Figure 6.2. Because $e^{-t} \to 0$ as $t \to \infty$, the solution converges to $(0,0)$ as $t \to \infty$. Notice the difference from Figure 6.1: now the solutions "spiral" into the origin rather than "hitting" it directly.

6.4.5 Repeated real eigenvalue

It can happen that the two eigenvalues of A coincide. If $(a_{1,1} - a_{2,2})^2 + 4a_{2,1}a_{1,2} = 0$, then we have $\lambda = \lambda_1 = \lambda_2 \in \mathbb{R}$. In this instance we also only get one eigenvector, $\boldsymbol{v}$, and we have that $\boldsymbol{x}^{(1)}(t) = e^{\lambda t}\boldsymbol{v}$ is still a solution to our first order 2×2 system. So we require a second, linearly independent, solution.

For scalar second order differential equations with repeated real roots of the auxiliary equation we saw that we could take $x(t) = te^{\lambda t}$ as a solution, so we try the ansatz

$$\boldsymbol{x}^{(2)}(t) = te^{\lambda t}\boldsymbol{v} + e^{\lambda t}\boldsymbol{v}^{(2)}$$

for some vector $\boldsymbol{v}^{(2)}$ that is linearly independent to our eigenvector $\boldsymbol{v}$. We want to check that this is a solution and find a formula for $\boldsymbol{v}^{(2)}$.

We start by differentiating $\boldsymbol{x}^{(2)}$ and setting equal to $\dot{\boldsymbol{x}}^{(2)} = A\boldsymbol{x}^{(2)}$ to get

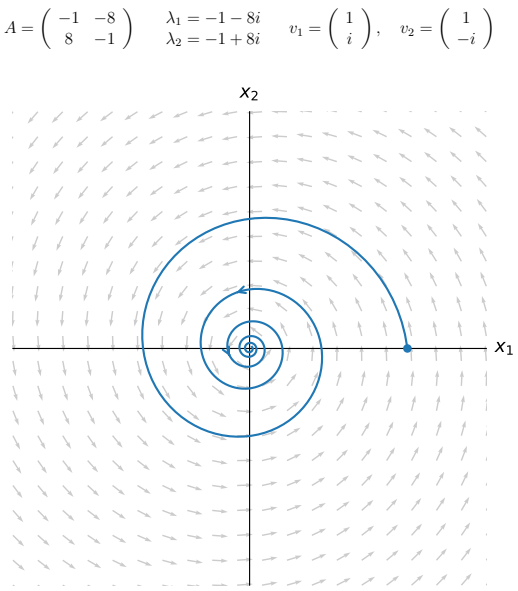


Figure 6.2: Plot of the solution of Example 6.8.

$$\dot{\boldsymbol{x}}^{(2)}(t) = e^{\lambda t}\boldsymbol{v} + t\lambda e^{\lambda t}\boldsymbol{v} + \lambda e^{\lambda t}\boldsymbol{v}^{(2)} = te^{\lambda t}A\boldsymbol{v} + e^{\lambda t}A\boldsymbol{v}^{(2)}\,.$$

Now we divide through by $e^{\lambda t}$ to get

$$\boldsymbol{v} + t\lambda\boldsymbol{v} + \lambda\boldsymbol{v}^{(2)} = tA\boldsymbol{v} + e^{\lambda t}A\boldsymbol{v}^{(2)} = t\lambda\boldsymbol{v} + e^{\lambda t}A\boldsymbol{v}^{(2)}$$

since $A\boldsymbol{v} = \lambda\boldsymbol{v}$, which cancels and rearranges to give

$$\boldsymbol{v} = (A - \lambda I)\boldsymbol{v}^{(2)}\,.$$

! Important

So now we have that $\boldsymbol{x}^{(2)}(t)$ solves our linear 2×2 system if $\boldsymbol{v}^{(2)}$ satisfies $\boldsymbol{v} = (A - \lambda I)\boldsymbol{v}^{(2)}$, and our full general solution is given by

$$\boldsymbol{x}(t) = (l_1 e^{\lambda t} + l_2 te^{\lambda t})\boldsymbol{v} + l_2 e^{\lambda t}\boldsymbol{v}^{(2)}\,.$$

Example 6.9. Now consider Equation 6.14 with an initial condition of the form Equation 6.11 where $t_0 = 0$ and

$$A = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}, \quad \boldsymbol{x}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The eigenvalues satisfy $\lambda^2 + 2\lambda + 1 = 0$, therefore

$$\lambda = -1,$$

and we have a repeated eigenvalue. The corresponding eigenvector $\boldsymbol{v}^{(1)}$ (also repeated) is

$$\boldsymbol{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

We now need to find a vector $\boldsymbol{v}^{(2)}$ such that

$$\boldsymbol{v} = (A - \lambda I)\boldsymbol{v}^{(2)}, \quad \Longrightarrow \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \boldsymbol{v}^{(2)}.$$

A solution to this equation is $\boldsymbol{v}^{(2)} = (0, 1)$.

Then, a general solution to the differential equations is thus given by

$$\boldsymbol{x}(t) = \left(l_1 e^{-t} + l_2 t e^{-t}\right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + l_2 e^{-t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} l_1 e^{-t} + l_2 t e^{-t} \\ l_2 e^{-t} \end{pmatrix}.$$

The initial condition is satisfied if

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = l_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + l_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} l_1 \\ l_2 \end{pmatrix},$$

hence $l_1 = l_2 = 1$, and the unique solution of Equation 6.14, Equation 6.11 with the above data is

$$\boldsymbol{x}(t) = e^{-t} \begin{pmatrix} 1 + t \\ 1 \end{pmatrix}.$$

The plot of the solution is shown in Figure 6.3. Because $e^{-t} \to 0$ as $t \to \infty$, the solution converges to $(0, 0)$ as $t \to \infty$. However, the solutions approach the origin in a more complex manner, not quite directly as in Figure 6.1, and not quite spiralling as in Figure 6.2. This is the contribution of the t term in the solution, and we will discuss this in more detail in Section 6.5.

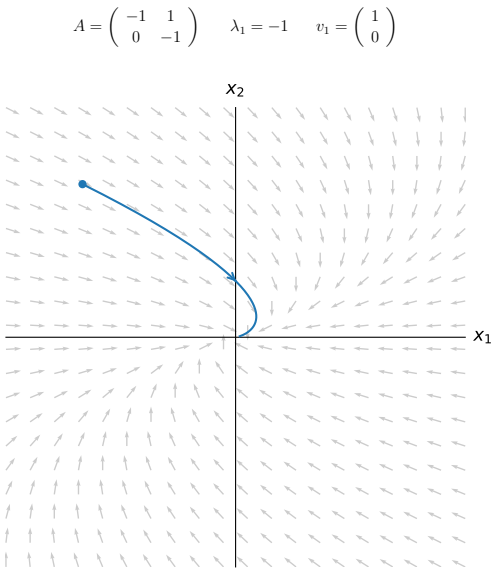


Figure 6.3: Plot of the solution of Example 6.9.

6.5 Phase portraits

In this section we generalise the idea of a phase line that we used in Chapter 2 to systems of differential equations. This generalisation is called a *phase portrait* and is a very useful tool to visualise solutions to systems of differential equations, including those that we cannot solve analytically. To introduce the key ideas though, we will focus on 2×2 systems of linear constant-coefficient first order autonomous differential equations.[10]

> **!** Important
>
> **Definition 6.6.** A *phase portrait* (also called *phase diagram*) of Equation 6.14 is the set of curves in the x_1-x_2-plane that are parametrised by solutions to Equation 6.14.

Note that Figure 6.1, Figure 6.2 and Figure 6.3 are technically phase portraits, but they only show a single solution. Typically, in phase portraits, we represent a few solutions to showcase the overall behaviour of the system.

Before going into detail on how to draw phase portraits, let us talk stability. We will give a formal definition of stationary points in Section 6.6, but for now we will just say that the definition we gave for differential equations naturally generalises. A stationary point is a point $\boldsymbol{x}^*$ that, if the solution starts there, it will stay there forever. Alternatively, we can

[10] In Section 6.6 we will see how we can link nonlinear systems to the results discussed in this section.

picture it as a point for which the right-hand side of the system of differential equations is zero. For a homogeneous linear system with constant coefficients, this means

$$A\boldsymbol{x}^* = \mathbf{0}, \quad \Longrightarrow \quad \boldsymbol{x}^* = \mathbf{0}, \tag{6.18}$$

so for the rest of the section we will focus on the stationary point at the origin.[11] We can consider the stability of this stationary point but, as we will see, a lot more complex behaviour can occur in systems of differential equations than in the one-dimensional case. We will showcase these behaviours through examples as we discuss how to draw phase portraits.

! Important

Algorithm 6.1 (Drawing phase portraits)**.** *Consider a* 2×2 *system of linear constant-coefficient first order autonomous differential equations*

$$\dot{\boldsymbol{x}} = A\boldsymbol{x}.$$

Before any drawing, we need to compute the eigenvalues and eigenvectors of A *(or, if the eigenvalue is repeated, the vector* $\boldsymbol{v}^{(2)}$*). Then we can follow the steps below to draw a phase portrait:*

0. *Draw the axes:* x_1 *and* x_2*, with the stationary point at the origin.*

Now we need to consider three cases based on the eigenvalues of A*.*
Case 1: real (distinct) eigenvalues:

1. *Draw a line through the stationary point along each eigenvalue.*
2. *Draw arrows on these lines:*
   - *Pointing towards the stationary point if the corresponding eigenvalue is negative.*
   - *Pointing away from the stationary point if the corresponding eigenvalue is positive.*
3. *Using the lines as a guide, add in additional lines demonstrating the evolution of various solutions as* t *increases, including arrows to determine the direction of travel. As a guideline, solutions start moving parallel to the eigenvector with smallest eigenvalue (in absolute value), and then bend towards the eigenvector with largest eigenvalue (in absolute value) as* t *increases, with the direction of travel determined by the arrows from Step 2.*

Case 2: complex eigenvalues:

[11] If Equation 6.18 is underdetermined, then we will have infinitely many stationary points (a line, or maybe even the whole plane). For now let us not worry about that.

1. *Work out the direction in which solutions cross the axes. To do so, evaluate the right-hand side of the system of differential equations at a point along a given axis. The resulting vector will point in the direction of travel, i.e. the direction in which the solution spirals.*[12]
2. *Draw the solutions:*
   - *If $p = 0$ then draw circles.*
   - *If $p < 0$ then draw inward spirals in the direction of travel.*
   - *If $p > 0$ then draw outward spirals in the direction of travel.*

In all cases, draw arrows on the spirals/circles to indicate the direction of travel.

Case 3: repeated eigenvalue:

1. *Draw a line through the stationary point along the eigenvalue.*
2. *Draw arrows on the line based on the sign of the eigenvalue (like in Case 1).*
3. *Draw a second line through the stationary point along $\boldsymbol{v}^{(2)}$ (this time without arrows).*
4. *Determine the direction in which solutions cross the second line by evaluating the right-hand side of the system at points along the line. You want, at least, to evaluate one point at each side of the stationary point.*
5. *Using the lines as a guide, add in additional lines demonstrating the evolution of various solutions as t increases, including arrows to determine the direction of travel. As a guideline, solutions start moving parallel to the eigenvector, then bend around crossing the second line in the direction determined in Step 4, and then bend back to be parallel to the eigenvector again, but in the opposite direction.*

Remark 6.5. Phase portraits are a qualitative tool to visualise the behaviour of solutions to systems of differential equations, not a quantititative one (at least not in this module). Therefore, when you draw phase portraits, you should focus on the overall structure and behaviour of the solutions (qualitative shape, direction of travel, etc.) rather than the specific shape of the solutions or the exact points it goes through.

With this information, we now draw phase portraits for various systems to showcase all the different behaviours that can occur. In each case, the details of the system (the matrix A, eigenvalues and eigenvectors) are shown in the plot so you can work out the results yourself.[13]

[12] It is actually sufficient to calculate the direction on one axis only, but it is safer to do it on both axes in practice.

[13] In the figures here we also show the grey arrows that indicate the direction of travel of the solution at each point, but you do not have to do that. Similarly, we use different colours for the different solutions for clarity, but you do not have to do that either.

! Important

First we discuss the case of real eigenvalues. We need to distinguish three separate cases, based on the signs of the eigenvalues. If both eigenvalues are positive (Figure 6.4a), then the stationary point is unstable and is called a *source node*. If both eigenvalues are negative (Figure 6.4b), then the stationary point is stable and is called a *sink node*. The source/sink node terminology comes from the fact that solutions "flow" away from the stationary point in the case of a source node, and "flow" towards the stationary point in the case of a sink node. If one eigenvalue is positive and the other is negative (Figure 6.4c), then the stationary point is unstable and is called a *saddle node*.

! Important

Next we consider the case of complex eigenvalues. If the real part of the eigenvalues is positive, then the stationary point is unstable and is called a *source spiral*. If the real part of the eigenvalues is negative, then the stationary point is stable and is called a *sink spiral*. If the real part of the eigenvalues is zero, it is called a *center*. In this case, the solution does neither converge nor diverge from the stationary point, but rather oscillates around it. As the solution does not get further and further from the stationary point as time increases, the solution is stable. Sometimes, we may refer it as marginally stable, because the solution does not converge to the stationary point either.

! Important

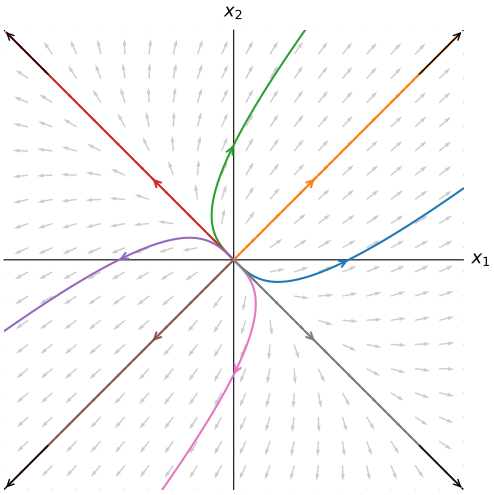
Finally, we consider the case for repeated eigenvalues. If the eigenvalue is positive, then the stationary point is unstable and is called an *improper source node*. If the eigenvalue is negative, then the stationary point is stable and is called an *improper sink node*. Notation is similar to the source/sink node case in Figure 6.4, but the adjective "improper" is added to indicate that the eigenvalue is repeated.

6.5.1 Degenerate cases

! Important

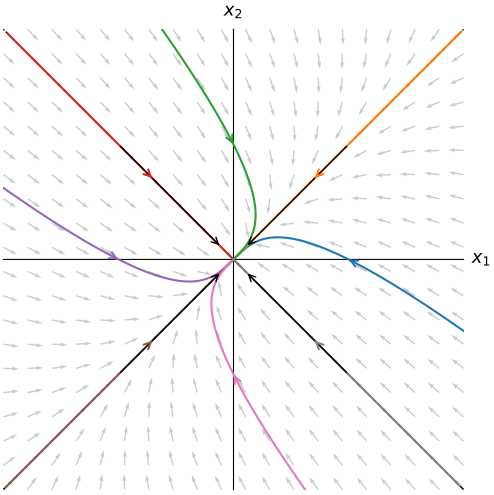
In the classification above, we have assumed that the eigenvalues are nonzero, but they could be. In that case, the behaviour of the system is a bit different, and they are called degenerate cases. For a 2×2 system, there are two degenerate cases: either one eigenvalue is zero and the other is nonzero (and real) or both eigenvalues are zero. The latter is quite simple: if both eigenvalues are zero, then the matrix A

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ $\lambda_1 = 1$ $\lambda_2 = 3$ $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$



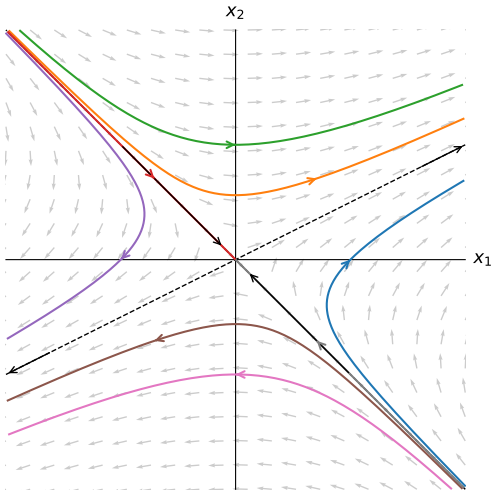
(a) Source node: $\lambda_1, \lambda_2 > 0$.

$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$ $\lambda_1 = -3$ $\lambda_2 = -1$ $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$



(b) Sink node: $\lambda_1, \lambda_2 < 0$.

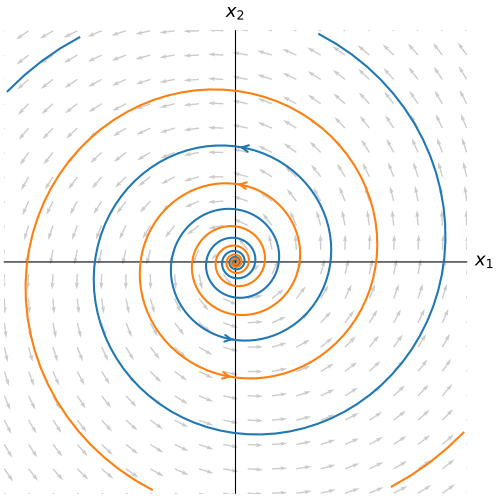
$A = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$ $\lambda_1 = -1$ $\lambda_2 = 2$ $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}$



(c) Saddle node: $\lambda_1 < 0 < \lambda_2$.

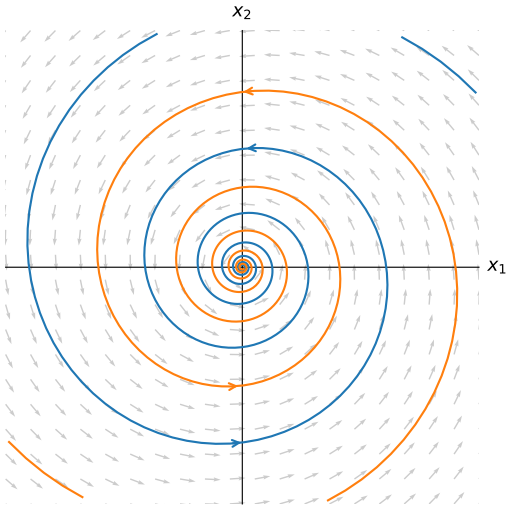
Figure 6.4: Phase portraits for real eigenvalues.

$A = \begin{pmatrix} 1 & -8 \\ 8 & 1 \end{pmatrix}$ $\lambda_1 = 1 - 8i$ $\lambda_2 = 1 + 8i$ $v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$



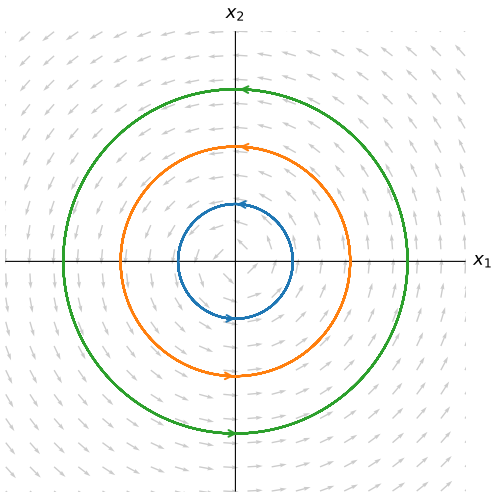
(a) Source spiral: $p > 0$.

$A = \begin{pmatrix} -1 & -8 \\ 8 & -1 \end{pmatrix}$ $\lambda_1 = -1 - 8i$ $\lambda_2 = -1 + 8i$ $v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$



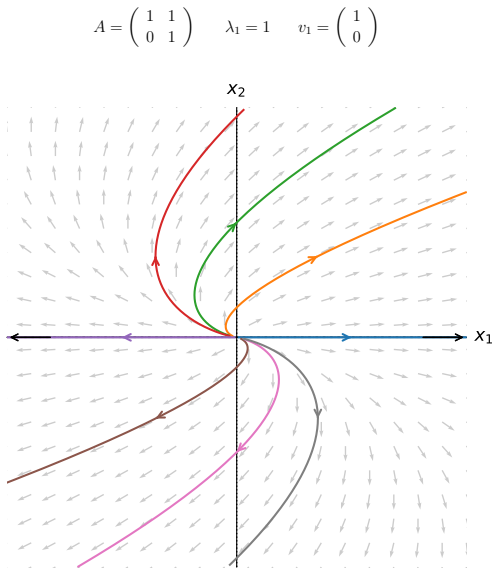
(b) Sink spiral: $p < 0$.

$A = \begin{pmatrix} 0 & -8 \\ 8 & 0 \end{pmatrix}$ $\lambda_1 = -8i$ $\lambda_2 = 8i$ $v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$

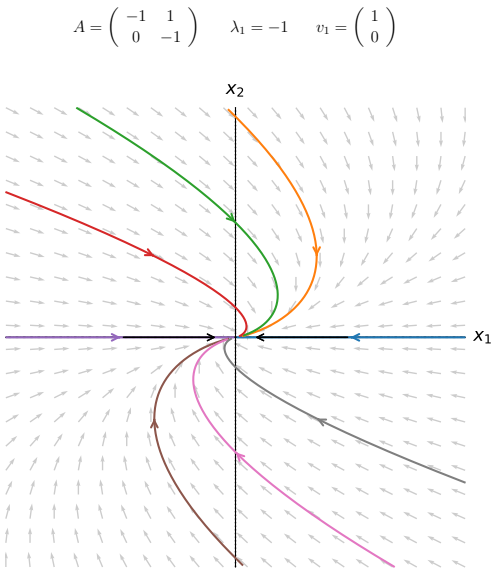


(c) Centre: $p = 0$.

Figure 6.5: Phase portraits for complex eigenvalues.



(a) Improper source node: $\lambda > 0$.



(b) Improper sink node: $\lambda < 0$.

Figure 6.6: Phase portraits for repeated eigenvalues.

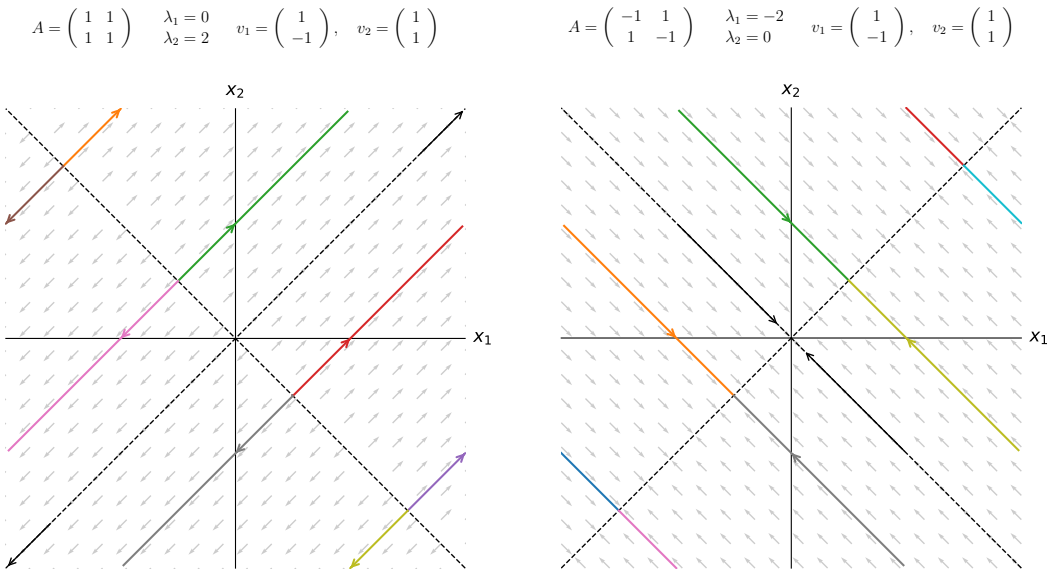
is the zero matrix, therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}(t) = A\boldsymbol{x}(t) = \mathbf{0}, \quad \Longrightarrow \quad \boldsymbol{x}(t) = \boldsymbol{x}_0,$$

which implies that all points $\boldsymbol{x}_0 \in \mathbb{R}^2$ are stationary points. As nothing happens, there is nothing we can draw in a phase portrait.
The former case is a bit more interesting. If one eigenvalue is zero and the other is nonzero, then the line through the stationary point along the eigenvector corresponding to the zero eigenvalue is a line of stationary points. To see that, consider the definition of the eigenvalue and eigenvector for $\lambda = 0$:

$$(\lambda I - A)\boldsymbol{v} = \mathbf{0} \quad \Longrightarrow \quad A\boldsymbol{v} = \mathbf{0},$$

but this means that the eigenvector (and any multiple of it) satisfies the definition of a stationary point, i.e. $A\boldsymbol{x}^* = \mathbf{0}$. The other eigenvector (corresponding to the nonzero eigenvalue) is not a stationary point, but it is a line of solutions that either converge to or diverge from the line of stationary points, depending on the sign of the nonzero eigenvalue. In that case, the phase portrait looks as shown in Figure 6.7.



(a) Degenerate case: $\lambda_1 = 0, \lambda_2 > 0$.

(b) Degenerate case: $\lambda_1 < 0, \lambda_2 = 0$.

Figure 6.7: Phase portraits for the degenerate case.

6.6 Autonomous systems

We conclude this section with a discussion of autonomous nonlinear systems. The idea is to leverage the concepts we have just introduced for phase portraits of linear systems to understand the behaviour of nonlinear systems.

We constrain ourseles to 2×2 autonomous systems, that is systimes of the form

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} x_1(t) &= f_1\left(x_1(t), x_2(t)\right), \\ \frac{\mathrm{d}}{\mathrm{d}t} x_2(t) &= f_2\left(x_1(t), x_2(t)\right), \end{aligned} \qquad t \in (\alpha, \beta) \tag{6.19}$$

with some given functions $f_1, f_2 : \mathbb{R}^2 \to \mathbb{R}$. Remember that, as the system is autonomous, these functions f_i do not depend on the independent variable t.

We can also use the vector notation and write Equation 6.19 as

$$\frac{\mathrm{d}}{\mathrm{d}t} \boldsymbol{x}(t) = \boldsymbol{f}(\boldsymbol{x}(t)), \qquad t \in (\alpha, \beta), \tag{6.20}$$

where $\boldsymbol{x}(t) = (x_1(t), x_2(t))^T$ and $\boldsymbol{f}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), f_2(\boldsymbol{x}))^T$. This is analogous to the vector notation we used for linear systems in Equation 6.7.

Let us now give a more formal definition of a stationary point and its stability.

! Important

Definition 6.7 (Stationary point)**.** A *stationary point* of Equation 6.20 is a point $\boldsymbol{x}^* \in \mathbb{R}^2$ such that $\boldsymbol{f}(\boldsymbol{x}^*) = \mathbf{0}$, i.e. $f_1(\boldsymbol{x}^*) = 0$ and $f_2(\boldsymbol{x}^*) = 0$.

Definition 6.8 (Stability)**.** The stationary solution (or point) of Equation 6.20 is called *stable* if solutions $(x_1(t), x_2(t))$ that start nearby remain close as time t increases.[14] It is called *unstable* if solutions that start nearby do not remain close as time t increases.

Example 6.10 (Lotka-Volterra predator-prey model)**.** Let us consider a model that describes the interaction between two species in a predator-prey relationship, which is known as the Lotka-Volterra model. Let $x(t)$ be the population of prey and $y(t)$ the population of predators at time t. The model is given by the following system of differential equations:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}x(t) &= \alpha x(t) - \beta x(t)y(t), \\ \frac{\mathrm{d}}{\mathrm{d}t}y(t) &= -\gamma y(t) + \delta x(t)y(t), \end{aligned} \qquad t \in \mathbb{R}, \tag{6.21}$$

with given parameters $\alpha, \beta, \gamma, \delta > 0$. The first equation describes the growth of the prey population, which grows at a rate α in the absence of predators, but is reduced by the presence of predators at a rate β. The second equation describes the growth of the predator population, which decreases at a rate γ in the absence of prey, but increases at a rate δ due to the presence of prey. Note that the interaction term between the two species is proportional to the product of their populations, which reflects the idea that more prey leads to more food for predators, and more predators leads to more predation on prey.

Let us now determine the stationary points of this system x^* and y^*. They have to satisfy

$$\begin{aligned} \alpha x^* - \beta x^* y^* &= 0, \\ -\gamma y^* + \delta x^* y^* &= 0. \end{aligned}$$

Solving the two equations simultaneously, we find that the stationary points are $(x^*, y^*) = (0, 0)$ and $(x^*, y^*) = (\gamma/\delta, \alpha/\beta)$. The first stationary point corresponds to the extinction of both species, while the second stationary point corresponds to a stable coexistence of both species.

[14] A precise definition of stability will be given in follow-up modules once measuring distances and limits will have been discussed in greater depth.

We have now seen how to calculate stationary points of a nonlinear system, but how do we determine their stability? In order to do so, we can linearise the system around the stationary point and study the resulting linear system using the tools in Section 6.5.

Let (x_1^*, x_2^*) denote a stationary solution (or point) of Equation 6.19. In order to get some more insight into whether (x_1^*, x_2^*) is stable we study solutions of the form

$$(x_1(t), x_2(t)) = (x_1^* + \xi_1(t), x_2^* + \xi_2(t))$$

where $(\xi_1(t), \xi_2(t))$ is a "small" deviation from the stationary point.[15] If we take the time derivative of both sides, we get

$$\frac{\mathrm{d}}{\mathrm{d}t}(x_1(t), x_2(t)) = \tfrac{\mathrm{d}}{\mathrm{d}t}(\xi_1(t), \xi_2(t)),$$

so that the system for $(\xi_1(t), \xi_2(t))$ becomes

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\xi_1(t) &= f_1\left(x_1^* + \xi_1(t), x_2^* + \xi_2(t)\right), \\ \frac{\mathrm{d}}{\mathrm{d}t}\xi_2(t) &= f_2\left(x_1^* + \xi_1(t), x_2^* + \xi_2(t)\right), \end{aligned} \qquad t \in (\alpha, \beta).$$

Now, we take a Taylor expansion[16] of the right-hand-side around the stationary point (x_1^*, x_2^*). Let us work first with the first equation. We have that

$$f_1\left(x_1^* + \xi_1(t), x_2^* + \xi_2(t)\right) \approx f_1(x_1^*, x_2^*) + \frac{\partial}{\partial x_1} f_1(x_1^*, x_2^*)\xi_1(t) + \frac{\partial}{\partial x_2} f_1(x_1^*, x_2^*)\xi_2(t),$$

where we have dropped the higher order terms, hence the $\approx$. Here, the *partial derivatives* $\frac{\partial}{\partial x_1}$ and $\frac{\partial}{\partial x_2}$ with respect to the first and second argument of the functions f_1 and f_2 appear. There are defined by differentiating with respect to one argument whilst keeping the other fixed, for instance,

$$\frac{\partial}{\partial x_2} f_1(x_1, x_2) = \lim_{\Delta x_2 \to 0} \frac{f_1(x_1, x_2 + \Delta x_2) - f_1(x_1, x_2)}{\Delta x_2}.$$

Then, because (x_1^*, x_2^*) is a stationary point, we have that $f_1(x_1^*, x_2^*) = 0$, so that the first term on the right-hand-side vanishes. A similar expansion can be done for the second equation, and we obtain

[15] In order to properly formulate what "small" means, one needs to nondimensionalise the problem.

[16] You are probably familiar with Taylor expansions for functions of a single variable: $f(x) \approx f(a) + f'(a)(x - a)$. For functions of multiple variables, the idea is similar, but one needs to take partial derivatives with respect to each variable while keeping the others fixed.

$$\begin{aligned} f_1\left(x_1^* + \xi_1(t), x_2^* + \xi_2(t)\right) &\approx \frac{\partial}{\partial x_1} f_1(x_1^*, x_2^*)\xi_1(t) + \frac{\partial}{\partial x_2} f_1(x_1^*, x_2^*)\xi_2(t), \\ f_2\left(x_1^* + \xi_1(t), x_2^* + \xi_2(t)\right) &\approx \frac{\partial}{\partial x_1} f_2(x_1^*, x_2^*)\xi_1(t) + \frac{\partial}{\partial x_2} f_2(x_1^*, x_2^*)\xi_2(t). \end{aligned} \tag{6.22}$$

Important

With this approximation we obtain the following system for $(\xi_1(t), \xi_2(t))$:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\xi_1(t) &= \frac{\partial}{\partial x_1} f_1(x_1^*, x_2^*)\xi_1(t) + \frac{\partial}{\partial x_2} f_1(x_1^*, x_2^*)\xi_2(t), \\ \frac{\mathrm{d}}{\mathrm{d}t}\xi_2(t) &= \frac{\partial}{\partial x_1} f_2(x_1^*, x_2^*)\xi_1(t) + \frac{\partial}{\partial x_2} f_2(x_1^*, x_2^*)\xi_2(t), \end{aligned} \qquad t \in (\alpha, \beta). \tag{6.23}$$

This system is *linear* in $(\xi_1(t), \xi_2(t))$. We therefore talk about a *linearisation* of Equation 6.19 around the stationary solution (x_1^*, x_2^*). In particular, the coefficients $\frac{\partial}{\partial x_2} f_1(x_1^*, x_2^*)$, etc. are constant, and in that case we can find explicit solutions.
The *principle of linearised stability* now relates the stability properties of the original system Equation 6.19 around the stationary point (x_1^*, x_2^*) to the stability properties of the linearised system Equation 6.23 around $(0, 0)$. Roughly, the stationary point is stable if and only if the linearised system is stable around the origin.

Remark 6.6. As we have seen, it is often more convenient to write linear systems in matrix form. In that case, the linearised system Equation 6.23 can be written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\xi_1(t)\\ \xi_2(t)\end{pmatrix} = \begin{pmatrix}\frac{\partial}{\partial x_1} f_1(x_1^*, x_2^*) & \frac{\partial}{\partial x_2} f_1(x_1^*, x_2^*)\\ \frac{\partial}{\partial x_1} f_2(x_1^*, x_2^*) & \frac{\partial}{\partial x_2} f_2(x_1^*, x_2^*)\end{pmatrix}\begin{pmatrix}\xi_1(t)\\ \xi_2(t)\end{pmatrix}.$$

The matrix on the right-hand-side is called the *Jacobian* of the system Equation 6.19 at the stationary point (x_1^*, x_2^*), and is denoted by $J(x_1^*, x_2^*)$. The Jacobian is a 2×2 matrix that contains all the partial derivatives of the functions f_1 and f_2 evaluated at the stationary point.[17]

Example 6.11 (Lotka-Volterra predator-prey model (continued))**.** Let us linearise the predator-prey model Equation 6.21 around the stationary points. We will compute the Jacobian matrix first, and then evaluate it at the stationary points. The Jacobian matrix is given by

$$J(x, y) = \begin{pmatrix}\frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y}\\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y}\end{pmatrix} = \begin{pmatrix}\alpha - \beta y & -\beta x\\ \delta y & -\gamma + \delta x\end{pmatrix}.$$

[17] You will learn more about Jacobians in the follow-up module MA145 Mathematical Methods and Modelling 2.

First we study the stationary point $(x^*, y^*) = (0, 0)$. Evaluating the Jacobian at this point gives

$$J(0,0) = \begin{pmatrix} \alpha & 0 \\ 0 & -\gamma \end{pmatrix}.$$

This is already a diagonal matrix, and we can read off the eigenvalues as $\lambda_1 = \alpha > 0$ and $\lambda_2 = -\gamma < 0$. Therefore, the stationary point $(0, 0)$ is a saddle point, and hence unstable. A side note, if there is only predators present in the system, then they will die out, as there is no food for them. This is consistent with the definition of a saddle point, because having only predators corresponds to starting on the y-axis, which is the direction of the eigenvector corresponding to the negative eigenvalue.

Next we study the stationary point $(x^*, y^*) = (\gamma/\delta, \alpha/\beta)$. Evaluating the Jacobian at this point gives

$$J\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right) = \begin{pmatrix} 0 & -\beta\frac{\gamma}{\delta} \\ \delta\frac{\alpha}{\beta} & 0 \end{pmatrix}.$$

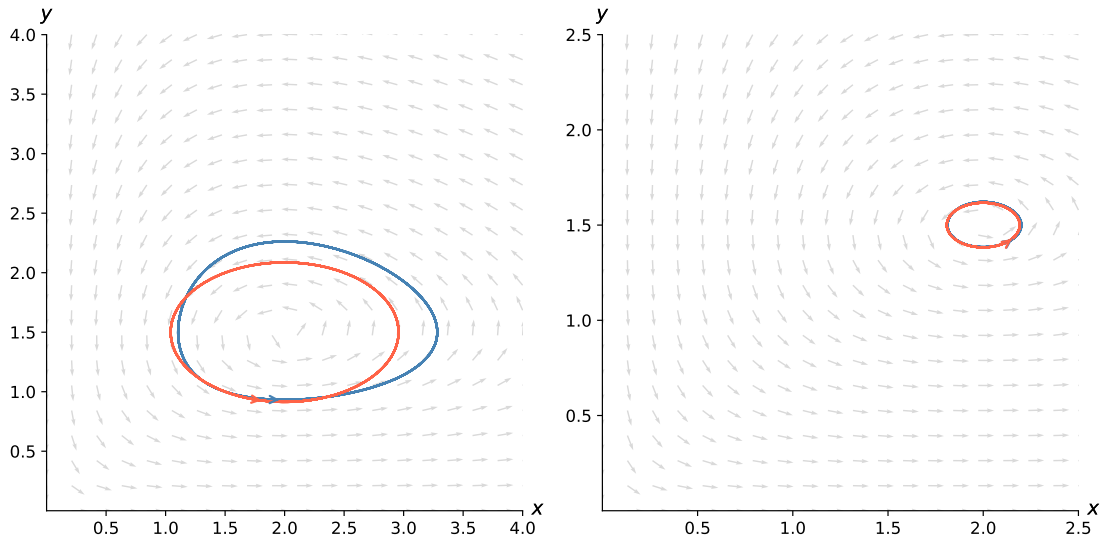
Now the eigenvalues of the matrix satisfy $\lambda^2 + \alpha\gamma = 0$, which gives

$$\lambda_{1,2} = \pm i\sqrt{\alpha\gamma}.$$

Therefore, the eigenvalues are purely imaginary, and the stationary point $(\gamma/\delta, \alpha/\beta)$ is a centre. This means that the populations of both species will oscillate around this stationary point, and the system is stable in the sense that the populations will not diverge to infinity.

Let us now visualise the phase portrait of the Lotka-Volterra system around the non-zero stationary point, Figure 6.8. We take the parameters $\alpha = 3$, $\beta = 2$, $\gamma = 2$, and $\delta = 1$ and compare the true solution (obtained numerically) with the linearise solution for a starting point $(x, y) = (1.5, 1)$ (Figure 6.8a). We observe that, even though the quantitative agreement is not perfect, the qualitative behaviour is captured well by the linearised system. In particular, the linearised system captures the oscillatory behaviour of the populations around the stationary point. Note that if instead we start much closer to the stationary point (Figure 6.8b), then the agreement between the true solution and the linearised solution is much better, as expected.[18]

[18] Note that Figure 6.8b has a narrower plotting range than Figure 6.8a, which magnifies the discrepancy between solutions.



(a) Starting point $(x, y) = (1.5, 1)$.

(b) Starting point $(x, y) = (2.1, 1.4)$.

Figure 6.8: Phase portraits for the Lotka-Volterra system, taking $\alpha = 3$, $\beta = 2$, $\gamma = 2$, and $\delta = 1$, for two different initial conditions. The exact solution is shown in blue, while the linearised system is shown in red.